

Resources

- ▶ Won't be following any one particular book.
- ▶ Lecture slides will have material from variety of sources.
- ▶ Some popular books
 1. Introduction to probability by Bertsekas and Tsitsiklis (Athena Scientific)
 2. Intro. to Probability and Statistics for Engineers and Scientists by Sheldon Ross (Elsevier)
 3. A first course in probability by Sheldon Ross (Prentice Hall)
- ▶ Some urls
 1. <https://www.probabilitycourse.com/>
 2. <https://www.statlect.com/>
 3. <https://www.randomservices.org/>

Course Outline

- ▶ Module 1 (3 Lectures)
Motivation & Probability basics
- ▶ Module 2 (10 Lectures)
All about random variables!
- ▶ Module 3 (4 Lectures)
Convergence of random variables, Stochastic Simulation
- ▶ Module 4 (5 Lectures)
All about Statistics
- ▶ Module 5 (4 lectures)
Random vectors and Random Processes

} Finance

[3Blue1Brown]

→ Random Experiments:

Experiments involving randomness

• Sample Space (Ω):

Set of all possible outcomes of the random experiment.
Can be finite or infinite.

e.g. $\Omega_c = \{H, T\}$
 $\Omega_{2c} = \{H, T\} \times \{H, T\}$
 $= \{HH, HT, TH, TT\}$

• Sample point / Possible outcome: $\omega \in \Omega$

• Event: $A \subseteq \Omega$

$C_1 = \{T\}$ i.e. $C_1 \rightarrow$ Event

• Probability of event A : $\mathbb{P}(A)$

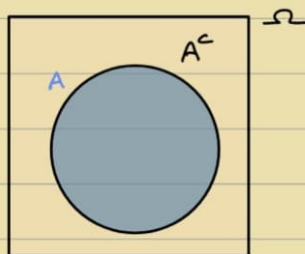
i.e. $\mathbb{P}(C_1) = 1/2$

→ Probability Theory:

Probability is a measure.
→ Set function

Set Theory :

• → H.W



Cardinality of A : $|A|$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

(for n sets)

$\emptyset$: Empty set , $\emptyset \subseteq A, \forall A$

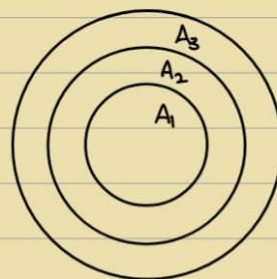
Union : $A \cup B$

Intersection : $A \cap B$

Difference : $A \setminus B \equiv A - B \equiv A \cap B^c$

Symmetric difference : $A \Delta B \equiv (A \setminus B) \cup (B \setminus A)$

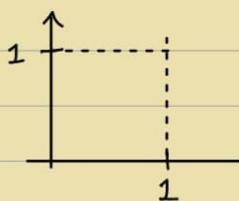
Disjoint events $\equiv$ Mutually exclusive



Increasing Sequence
 $A_1 \subseteq A_2 \subseteq A_3$

Cartesian Product : $A \times B$
 $\hookrightarrow (a, b) \forall (a \in A \text{ \& } b \in B)$

$$[0, 1] \times [0, 1] = \{(a, b) : a \in \underbrace{[0, 1]}_{\Omega_1} \text{ \& } b \in \underbrace{[0, 1]}_{\Omega_2}\}$$



• Powerset of A : $\mathcal{P}(A)$

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

x	x	x	x	x	x	000000
x	x	x	x	x	✓	000001
x	x	x	x	✓	x	000010

} Binary Representation

$$\text{i.e. } 2^6 \rightarrow |\mathcal{P}(\Omega)| \text{ i.e. } 2^n$$

$$\mathcal{P}([0, 1]) = ? = \{(a, b) : a \leq b ; a, b \in [0, 1]\} \leftarrow \text{COMES LATER (Sigma Algebra)}$$

Cardinality is uncountable infinite.

$\mathcal{B}(\mathbb{R})$??

→ Functions:

Domain D
Range R $\left\{ f: D \rightarrow R \right.$

(Injection
Surjection
Bijection)

e.g. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x$

• Set Functions:

Functions which act on sets

D : Collection of sets

→ Probability: P
 ↳ Set function

Axioms:

(1) $P(\emptyset) = 0$

$P(\Omega) = 1$

(2) For a set $A \subseteq \Omega$, we have $0 \leq P(A) \leq 1$

(3) For a disjoint collection of events $A_1, A_2, A_3, \dots$, where $A_i \subseteq \Omega$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad [\text{Union} \rightarrow \text{Countable sets}]$$

Domain of P in general $\rightarrow$ Powerset of A , i.e. $P(A)$

$P(\Omega) = 1$

$P(\Omega) = \{A: A \subseteq \Omega\}$

6/8 Counter-example:

$\Omega = \mathbb{R} \Rightarrow P(\mathbb{R}) = 1$

Domain: $P(\mathbb{R}) \rightarrow \text{Complex}$

$P: P(\mathbb{R}) \rightarrow [0, 1]$

P has the property that sets of equal 'length' have equal probability.

$$\mathbb{R} = \bigcup_{n=-\infty}^{\infty} [n, n+1), \text{ where } [n, n+1) \in P(\mathbb{R})$$

$$P(\mathbb{R}) = 1 = \sum P[n, n+1)$$

$\therefore$ Restrict domain to measurable sets

• Towards Sigma Algebra:

$$\mathcal{F} = \{\emptyset, \Omega, \mathbb{R}_-, \mathbb{R}_+\}$$

Event space or Sigma-Algebra $\mathcal{F}$ associated with a set Ω is a collection of subsets of Ω that satisfy

$$(1) \emptyset \in \mathcal{F}$$

$$\Omega \in \mathcal{F}$$

$$(2) A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$$

$$(3) A_1, A_2, A_3, \dots, A_n \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

σ -Algebra is said to be closed under formation of complements & countable unions.

$$\begin{array}{c|c} A_1 \in \mathcal{F} & A_1^c \in \mathcal{F} \\ A_2 \in \mathcal{F} & A_2^c \in \mathcal{F} \end{array}$$

$(A_1^c \cup A_2^c)^c \Rightarrow$ Closed under the formation of countable intersections.

Note: When Ω is countable and finite $\Rightarrow P(\Omega)$ is the domain
 $\hookrightarrow$ Sample space $\hookrightarrow$ Sigma Algebra

$(\Omega, \mathcal{F}, P)$: Probability Space

$\hookrightarrow$ Probability measure on $(\Omega, \mathcal{F})$

$$P: \mathcal{F} \rightarrow [0, 1]$$

S.T. Axioms hold true

$$|\Omega| < \infty, \mathcal{F} = 2^n$$

• Probability space for $U[0, 1]$:

$$\Omega = [0, 1]$$

$$\text{Suppose, } \mathcal{F} = \{\emptyset, [0, 1], [0, .5), [.5, 1]\}$$

$$P([.25, .75]) = .5$$

$$\mathcal{F}^+ = \{\emptyset, [0, 1], [0, .5), [.5, 1], [.25, .75]\}$$

but can we show?

$\hookrightarrow$ Not a sigma algebra

$\mathcal{F}^{++} \rightarrow$ Not a sigma-algebra

Borel-Sigma Algebra $\mathcal{B}[0, 1]$

$\mathcal{B}[0,1] \rightarrow \sigma\text{-Algebra generated by closed sets of form } [a, b]$
 $a \leq b \text{ \& } a, b \in [0,1]$

$$(a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}]$$

Can also be of form
 $(a, b]$ or $[a, b)$

$$(a, b] = \bigcap_{n=1}^{\infty} (a, b + \frac{1}{n})$$

$$[a, b) = \bigcap_{n=1}^{\infty} (a + \frac{1}{n}, b)$$

- $\mathcal{B}(\mathbb{R})$:

$$\Omega = \mathbb{R}$$

$$\text{form } (a, b) \mid a \leq b; a, b \in \mathbb{R}$$

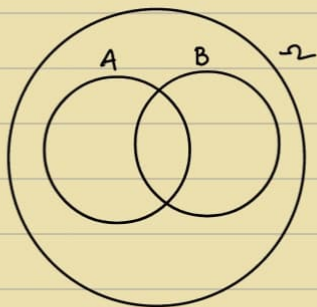
$\mathcal{B}(\mathbb{R})$ contains intervals of the form

$$[a, b], [a, b), [a, \infty), (a, \infty), (-\infty, b], (-\infty, b], \{a\}$$

• Consequences of Probability Axioms:

$$- P(A^c) = 1 - P(A)$$

$$- P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$A = (A \cap B) \cup (A \setminus B)$$

$$P(A) = P(A \cap B) + P(A \setminus B)$$

$\rightarrow$ H.W

- $A \subseteq B$, Then prove that $P(A) \leq P(B)$

$$(A \subseteq B \Rightarrow \text{Event } A \text{ implies event } B)$$

$$- P(A \cup B \cup C) = ?$$

Hint: 3rd Axiom

H.W: Inclusion-exclusion principle

- Impossible event v/s Zero prob. event :

In $U[0,1]$, $P(\omega = 0.5) = ?$

$$P(\omega = 0.5) = 0$$

$$P(\omega \in [a, b]) = b - a$$

$$\text{Then, } P([.5, .5]) = P(\{.5\}) = 0$$

$$P(\emptyset) = 0 \Rightarrow \emptyset \text{ is impossible}$$

$$\text{e.g. } C = [0, .5] \cap [.8, 1]$$

→ Null Set

$$\Omega = \bigcup_{\omega \in \Omega} \{\omega\}$$

$$P(\Omega) = P\left(\bigcup_{\omega \in \Omega} \{\omega\}\right) = \sum_{\omega \in \Omega} P(\{\omega\}) = 0 \quad \times$$

9/8

$$a_1, a_2, \dots, a_n$$

$$n > N_\varepsilon$$

$$|a_n - L| \leq \varepsilon$$

$$\text{Continuous : } x \rightarrow c \Rightarrow f(x) \rightarrow f(c)$$

$$\text{Continuous set function } S, A_n \rightarrow A,$$

$$S(A_n) \rightarrow S(A)$$

Lemma :

$$A_n \uparrow A \text{ (or) } A_n \downarrow A \Rightarrow \lim_{n \rightarrow \infty} P(A_n) = P(A)$$

$$P(A) = \sum_{i=1}^{\infty} P(F_i)$$

$$F_n = A_n - A_{n-1}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n P(F_i)$$

$$= \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n F_i\right)$$

$$= \lim_{n \rightarrow \infty} P(A_n)$$

• Conditional Probability :

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

Property :

$$P(A/B) \cdot P(B) = P(A \cap B)$$

H.W : $P(A/(B \cap C))$

$$P(A \cap B/C) = P(A/B/C) \cdot P(B/C)$$

H.W : $P(A \cap B \cap C) = P(A) \cdot P(B/A) \cdot P(C/AB)$

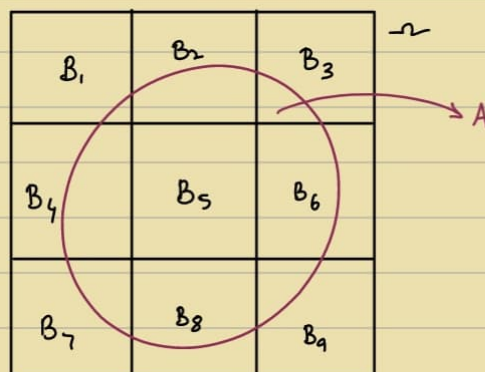
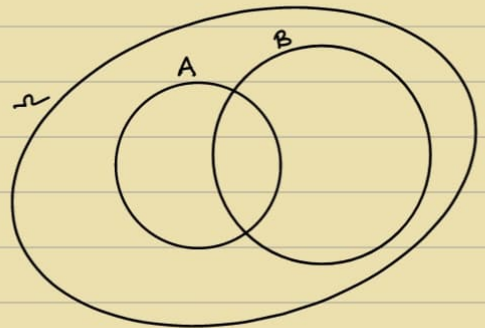
H.W : $P(A_1 \cap A_2 \cap A_3 \dots \cap A_n) = P(A_1) \cdot P(A_2/A_1) \cdot \dots$

Ex. $A = (A \cap B) \cup (A \cap B^c)$

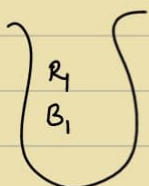
$$P(A) = P(A \cap B) + P(A \cap B^c)$$

$$= P(A/B) \cdot P(B) + P(A/B^c) \cdot P(B^c)$$

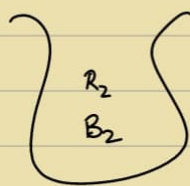
$$P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A/B_i) \cdot P(B_i)$$



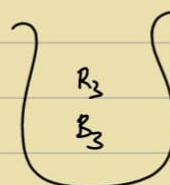
Q)



$$R_1 + B_1 = M$$



$$R_2 + B_2 = M$$



$$R_3 + B_3 = M$$

$$\frac{1}{3} \times \frac{R_1}{M} + \frac{1}{3} \times \frac{R_2}{M} + \frac{1}{3} \times \frac{R_3}{M}$$

- Independence:

Consider tossing a coin & rolling a dice simultaneously

$$\Omega = \{\{H, 1\}, \{H, 2\}, \dots, \{T, 1\}, \dots\}$$

$$P(\{H, 6\}) = 1/12$$

$$\begin{aligned} P(\{T, \text{odd}\}) &= P(\{ \bigcup_{i=1,3,5} \{T, i\} \}) \\ &= 1/12 + 1/12 + 1/12 \\ &= 3/12 \\ &= 1/4 \end{aligned}$$

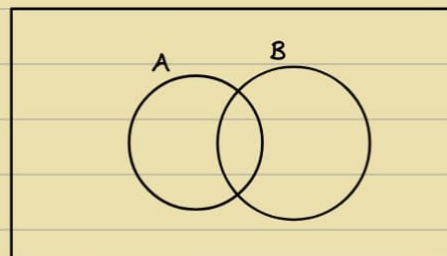
$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(A/B) = P(A)$$

To prove:

$$P(A^c \cap B^c) = P(A^c) \cdot P(B^c)$$

$$\begin{aligned} P(A^c) \cdot P(B^c) &= (1 - P(A))(1 - P(B)) \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) \\ &= 1 - P(A) - P(B) + P(A \cap B) \\ &= P(A^c \cap B^c) \end{aligned}$$



H.W: Are A and B^c independent?

$$P\left(\bigcup_{i=1}^n A_i\right) = 1 - \prod_{i=1}^n [1 - P(A_i)] \quad \text{if } A_1, A_2, \dots, A_n \text{ are independent}$$

- Mutual Independence:

$A_i, i \in I \leftarrow$ collection of events

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j) \quad \text{for any subset } J \text{ of } I$$

- Pairwise Independence

MI $\Rightarrow$ PI but vice-versa not always

All pairs are independent

H.W: find examples

9) $\Omega = \{1, 2, 3, \dots, 10\}$

Event A : number $< 7 \Rightarrow P(A) = 3/5$

Event B : number $< 8 \Rightarrow P(B) = 7/10$

Event C : number is even $\Rightarrow P(C) = 1/2$

$A \subseteq B \Rightarrow P(A \cap B) = P(A) \Rightarrow A$ and B are not independent

$P(A \cap C) = 3/10 \Rightarrow A$ and C are independent

$\therefore$ No mutual independence

A and B are positively correlated iff $P(A/B) > P(A)$

A and B are negatively correlated iff $P(A/B) < P(A)$

A and B have same correlation as A^c and B^c

A and B have opp. correlation as A and B^c

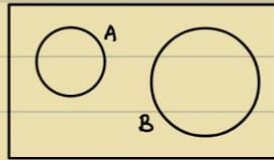
• Mutually exclusive:

One can happen $\Rightarrow$ Other event cannot occur

$P(A \cap B) = \emptyset$

$P(A/B) = 0$

$P(A/B^c) = \frac{P(A)}{P(B^c)}$



If $A \subseteq B$, then they are neither mutually exclusive nor independent.

Note: Zero probability events are always independent

Let $E \rightarrow$ Zero probability event, $P(E) = 0$

For any set, $P(E \cap F) = 0$

$(E \cap F) \subseteq E$

$P(E \cap F) \leq P(E)$

• Conditional Independence:

$P(A/B) = \frac{P(AB)}{P(B)}$

$P(A/BC) = \frac{P(ABC)}{P(BC)}$

$$P((AB)/C) = P(A/C) \cdot P(B/C), \quad P(C) > 0$$

$\therefore A$ and B are conditionally independent

$$\Rightarrow P(A/BC) = P(A/C)$$

Ex. 2 coins - 1 fair & 1 fake (Both heads)

Experiment $\rightarrow$ Choose a coin uniformly & toss twice

Event A : First coin toss results in H

$$\begin{aligned} \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 &= \frac{1}{4} + \frac{1}{2} \\ &= \frac{3}{4} \end{aligned}$$

Event B : Second coin toss results in H

$$= \frac{3}{4}$$

Event C : Coin 1 is chosen

$$P(A \cap B)$$

$\hookrightarrow$ Both coin tosses give heads

$$= P(A \cap B / C_F) \cdot P(C_F) + P(A \cap B / C_B) \cdot P(C_B)$$

$$= \frac{1}{8} + \frac{1}{2}$$

$$P(A/C) = \frac{1}{2}$$

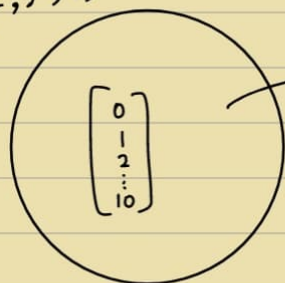
$$P(B/C) = \frac{1}{2}$$

$$P(A \cap B / C) = \frac{1}{4}$$

$\rightarrow$ Random variable:

$(\Omega, \mathcal{F}, P)$ to a simpler $(\Omega', \mathcal{F}', P_X)$

$(\Omega, \mathcal{F}, P)$



$X(w)$

$$\Omega' = \{0, 1, 2, \dots, 10\}$$

$$P \rightarrow P_X$$

$\mathcal{F}'$

Induced Probability
measure on Ω'

X is a function, $X: \Omega \rightarrow \Omega'$ that transforms the probability space $(\Omega, \mathcal{F}, P)$ to $(\Omega', \mathcal{F}', P_X)$ and is $(\mathcal{F}, \mathcal{F}')$ -measurable

$$X(w) \in \Omega' \quad \forall w \in \Omega$$

$$\forall B \in \mathcal{F}' \Rightarrow X^{-1}(B) := \{w \in \Omega : X(w) \in B\}, \text{ i.e. } X^{-1}(B) \in \mathcal{F}$$

16/8 & 20/8

Absent - Not well

23/8

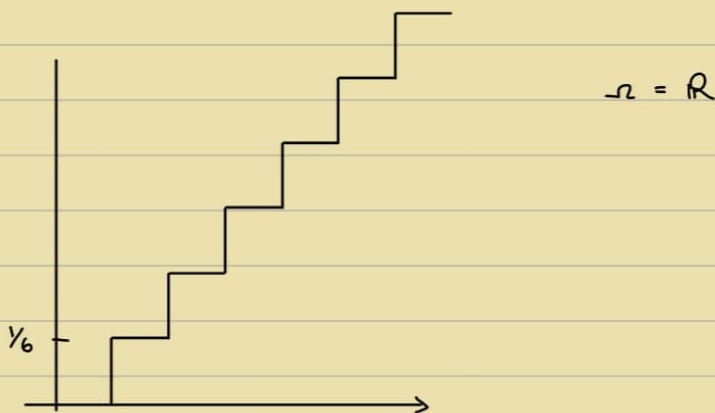
$$F_X(x) = \sum_{x \leq x_i} P(x_i)$$

$$\hookrightarrow P(X \leq x_i)$$

$$P(\omega \in \Omega, X(\omega) \leq x_i)$$

$$\bullet X : (\Omega, \mathcal{F}, P) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$$

$$\text{CDF} : F_X(x) = P\{\omega \in \Omega : X(\omega) \leq x\}$$



CDF : Continuous $\longrightarrow$ Continuous variable

CDF : Discontinuous $\longrightarrow$ Discrete variable

$$\text{PDF} : f_X(x) \quad \forall x \in \mathbb{R}$$

$$f_X(x) = \lim_{\Delta \rightarrow 0^+} \frac{P(x < X \leq x + \Delta)}{\Delta}$$

$$= \lim_{\Delta \rightarrow 0^+} \frac{P(X \leq x + \Delta) - P(X \leq x)}{(x + \Delta) - x}$$

$$= \lim_{\Delta \rightarrow 0^+} \frac{F_X(x + \Delta) - F_X(x)}{\Delta}$$

$$\left(F_X(x_1) = P(X \leq x_1) \right)$$

$$= \frac{d}{dx} F_X(x)$$

$$F_X(x) = \int_{u=-\infty}^x f_X(u) du$$

$$P(A \cap B^c) = P(A) - P(B)$$

$$A : X \leq x_1 + \Delta$$

$$B : X \leq x_1$$

$$\hookrightarrow B^c : X > x_1$$

$$P_X(\mathbb{R}) = \int_{u=-\infty}^{\infty} f_X(u) du = 1$$

$$P_X([a, b]) = F_X(b) - F_X(a) \\ = \int_a^b f_X(u) du$$

$$\text{For any } B \subseteq \mathbb{R}, P_X(B) = \int_{u \in B} f_X(u) du$$

$$P_X(\{a\}) = 0$$

$$P_X([a, b]) = P_X((a, b]) = P_X([a, b)) = P_X((a, b))$$

$$\bullet E[X] = \int_{-\infty}^{\infty} u f_X(u) du$$

$$E[X^n] = \int_{-\infty}^{\infty} u^n \cdot f_X(u) du$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(u) \cdot f_X(u) du$$

$$\text{Var}[X] = E[g(X)] \quad g(X) = (X - E[X])^2$$

$$\text{For } Y = aX + b,$$

$$E[Y] = a \cdot E[X] + b$$

$$\underbrace{F_Y(y)}_{\substack{\rightarrow P(Y \leq y) \\ P(aX + b \leq y) \\ P(aX \leq y - b) \\ P(X \leq \frac{y-b}{a}) \\ F_X(\frac{y-b}{a})}} = F_X\left(\frac{y-b}{a}\right) \quad \left| \quad F_Y(y) = 1 - F_X\left(\frac{y-b}{a}\right) \quad \text{For } a < 0$$

$$\left. \begin{array}{l} P(Y \leq y) \\ P(aX + b \leq y) \\ P(aX \leq y - b) \\ P(X \leq \frac{y-b}{a}) \\ F_X(\frac{y-b}{a}) \end{array} \right\} \underline{a > 0}$$

$$\frac{d}{du} F_X(u) \cdot \frac{du}{dy} \quad \left[u = \frac{y-b}{a} \right]$$

$$\frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

- Uniform Random variable:

$$f_X(x) = \frac{1}{b-a} \quad \forall x \in [a, b]$$

$$\text{CDF : } F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & x \in [a, b] \\ 1, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{a+b}{2}$$

$$\text{Var}[X] = \frac{(b-a)^2}{12}$$

- Exponential Random variable:

Non-negative R.V with parameter λ

$$f_X(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

$$F_X(x) = 1 - e^{-\lambda x} \quad \text{for } x \geq 0$$

$$E[X] = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless property

$$P(X > a+h \mid X > a) = \frac{e^{-\lambda(a+h)}}{e^{-\lambda(a)}} = e^{-\lambda h} = P(X > h)$$

Prep for Quiz - 1

Sample Space : Ω

Possible outcome : $\omega \in \Omega$

Event : $A \subseteq \Omega$

Probability : $P(A)$

Powerset : $P(A)$

$$(1) P(\emptyset) = 0, P(\Omega) = 1$$

$$(2) A \subseteq \Omega, 0 \leq P(A) \leq 1$$

$$(3) P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i), A_i \rightarrow \text{Disjoint events}$$

→ Sigma Algebra?

$$\mathcal{F} = \{\emptyset, \Omega, R_-, R_+\}$$

$$(i) \emptyset \in \mathcal{F} \text{ \& } \Omega \in \mathcal{F}$$

$$(ii) B \in \mathcal{F} \Rightarrow B^c \in \mathcal{F}$$

$$(iii) A_1 \in \mathcal{F} \text{ \& } A_2 \in \mathcal{F} \Rightarrow A_1 \cup A_2 \in \mathcal{F} \text{ \& } A_1 \cap A_2 \in \mathcal{F}$$

Probability space : $(\Omega, \mathcal{F}, P)$

Measurable space : $(\Omega, \mathcal{F})$

→ Borel - Sigma Algebra?

$$B[0,1]$$

→ LCD:

$$\lim_{n \rightarrow \infty} a_n = L$$

$$x \rightarrow c \Rightarrow f(x) \rightarrow f(c)$$



Q) Defective $\rightarrow 98\%$ Acc.

Not defective $\rightarrow 99\%$ Acc.

0.1% $\rightarrow$ Total Defective

$$P(\text{Robot says defective} \mid \text{Defective}) = 0.98$$

$$P(\text{Robot says not def.} \mid \text{Not Def.}) = 0.99$$

$$\Rightarrow P(\text{Robot says def.} \mid \text{Not def.}) = 0.01$$

$$P(\text{Defective}) = 0.01$$

$$\Rightarrow P(\text{Not def.}) = 0.99$$

$P(A|B)$
 $\hookrightarrow$ occurred

$$P(\text{def.} \mid \text{Robot says def.}) = \frac{P(\text{robot says def.} \mid \text{def.}) \cdot P(\text{def.})}{P(\text{Robot says def.})}$$

$$= \frac{0.98 \times 0.01}{\frac{98}{100} \times \frac{0.1}{100} + \frac{99.9}{100} \times \frac{1}{100}}$$

$$= 8.9\%$$

Independence: $P(A \cap B) = P(A) \cdot P(B)$

+ve correlation: $P(A|B) > P(A)$

-ve correlation: $P(A|B) < P(A)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B \cap C) = \frac{P(A \cap B \cap C)}{P(B \cap C)}$$

$$P(A \cap B \cap C) = P(A|B \cap C) \cdot P(B \cap C)$$

$\rightarrow$ Binomial:

$$P(i) = P(X=i) = \binom{n}{i} p^i \cdot (1-p)^{n-i}$$

$$E[X] = \sum x P(x)$$

$$= \sum_{i=0}^n i \cdot P(i)$$

$$= \sum_{i=0}^n (i) \cdot \binom{n}{i} \cdot p^i \cdot (1-p)^{n-i}$$

$$= \sum_{i=1}^n i \cdot \frac{n!}{(n-i)! i!} \cdot p^i \cdot \frac{(1-p)^n}{(1-p)^i}$$

$$= n \cdot \sum_{i=1}^n \frac{(n-1)!}{(n-i)! (i-1)!} \cdot p^i \cdot (1-p)^{n-i}$$

$$= np \cdot \sum_{i=1}^n \binom{n-1}{i-1} p^{i-1} \cdot (1-p)^{(n-1)-(i-1)}$$

$$= np \cdot \sum_{j=0}^{n-1} \binom{n-1}{j} p^j \cdot (1-p)^{n-1-j} = np$$

→ Geometric:

$$P(X=i) = (1-p)^{i-1} p$$

$$E[X] = \sum_{i=0}^n x \cdot P(x)$$

$$= \sum_{i=0}^n i \cdot (1-p)^{i-1} \cdot p$$

$$= \sum_{i=0}^n i \cdot (1-p)^i \cdot \frac{p}{1-p}$$

$$= \frac{p}{1-p} \cdot \sum_{i=1}^n i(1-p)^i$$

$$= \frac{p}{1-p} \cdot \frac{1-p}{p^2} \cdot (1-(1-p)^n)$$

$$= \frac{1}{p}$$

$$\begin{aligned} S &= 1(1-p) + 2(1-p)^2 + 3(1-p)^3 + \dots \\ (1-p)S &= 1(1-p)^2 + 2(1-p)^3 + \dots \\ S \cdot p &= (1-p) + (1-p)^2 + (1-p)^3 + \dots \infty \\ &= (1-p) \cdot \frac{1-(1-p)^n}{p} \end{aligned}$$

$$\Rightarrow S = \frac{1-p}{p^2} \cdot (1-(1-p)^n)$$

→ Uniform:

$$P(X=x) = \frac{1}{b-a} \quad \forall x \in [a, b]$$

$$E[X] = \int_{-\infty}^{\infty} x \cdot P(x) dx$$

$$= \int_a^b x \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \cdot \left(\frac{x^2}{2} \right)_a^b$$

$$= \frac{a+b}{2}$$

→ Exponential:

$$P(X=x) = \lambda e^{-\lambda x} \quad \forall x \geq 0$$

$$E[X] = \int_0^{\infty} \lambda e^{-\lambda x} = e^{-\lambda x} \Big|_0^{\infty} \cdot \frac{1}{\lambda}$$

$$= \frac{1}{\lambda} (0+1) = \frac{1}{\lambda}$$

Ø) 10 min break

$$P(\text{rain}) = \frac{3}{10}$$

$$P(\text{crowd}) = \frac{1}{2}$$

$$P(\text{late} | \text{rain}) = \frac{1}{2}$$

$$P(\text{late} | \text{crowded}) = \frac{3}{10}$$

$$\begin{aligned} \text{(a)} \quad P(\text{late}) &= P(\text{late} | \text{rain}) \cdot P(\text{rain}) + P(\text{late} | \text{crowded}) \cdot P(\text{crowded}) \\ &= \frac{1}{2} \cdot \frac{3}{10} + \frac{3}{10} \cdot \frac{1}{2} \\ &= \frac{3}{10} \end{aligned}$$

$$\text{(b)} \quad P(\text{rain} | \text{late}) = \frac{\frac{1}{2} \cdot \frac{3}{10}}{\frac{3}{10}} = \frac{1}{2}$$

$$P(\text{crowded} | \text{late}) = \frac{\frac{1}{2} \cdot \frac{3}{10}}{\frac{3}{10}} = \frac{1}{2}$$

$$\begin{aligned} \text{(c)} \quad P(\text{rain} \cap \text{crowded}) &= P(\text{rain}) \cdot P(\text{crowded}) \\ &= \frac{3}{10} \cdot \frac{1}{2} \\ &= \frac{3}{20} \end{aligned}$$

Ø) $E_{11} E_{12} E_{21} E_{22}$

$$(E_{11} E_{21} + E_{11} E_{22} + E_{12} E_{21} + E_{12} E_{22} + \dots)$$

$$\begin{aligned} P(\text{Crash}) &= \overline{E_{11}} \overline{E_{12}} + \overline{E_{21}} \overline{E_{22}} - \overline{E_{11}} \overline{E_{12}} \overline{E_{21}} \overline{E_{22}} \\ &= \frac{2}{10} \cdot \frac{2}{10} + \frac{2}{10} \cdot \frac{2}{10} - \frac{2}{10} \cdot \frac{2}{10} \cdot \frac{2}{10} \cdot \frac{2}{10} \\ &= \frac{8}{100} - \frac{16}{10000} = \frac{784}{10000} \end{aligned}$$

$$\begin{aligned} \overline{P}(\text{Crash}) &= 1 - \frac{784}{10000} = \frac{9216}{10000} \\ &= \underline{\underline{0.9216}} \end{aligned}$$

$$Q) P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} \quad (\lambda > 0), \quad k=0,1,2,\dots$$

$$E[X] = \sum_{x=0}^{\infty} x P(x)$$

$$= \sum_{i=0}^{\infty} i \cdot \frac{e^{-\lambda} \cdot \lambda^i}{i!}$$

$$= e^{-\lambda} \cdot \sum_{i=0}^{\infty} i \cdot \frac{\lambda^i}{i!}$$

$$= e^{-\lambda} \cdot \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} \cdot \lambda$$

$$= \lambda e^{-\lambda} \cdot \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

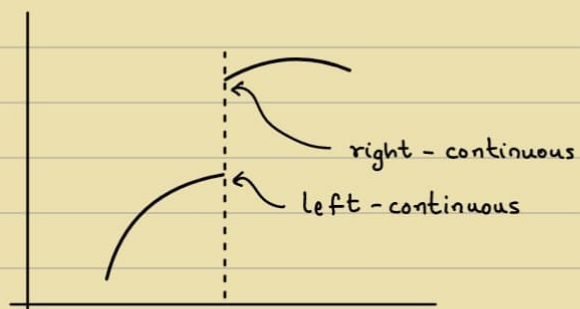
$$= \lambda \cdot e^{-\lambda} \cdot e^{\lambda} = \underline{\underline{\lambda}}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^{\lambda} = 1 + \lambda + \frac{\lambda^2}{2!} + \dots$$

$$\text{Var}[X] = \lambda (1 - \lambda/n)$$

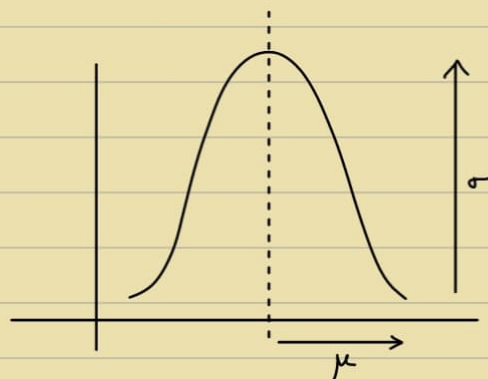
$$\curvearrowright E[X^2] - (E[X])^2$$



→ Gaussian Random variable $(N(\mu, \sigma^2))$:
 - Continuous RV
 mean variance

$$\text{PDF : } f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \forall x \in \mathbb{R}$$

$$\begin{aligned} \text{H.W : } \int_{-\infty}^{\infty} f_X(x) dx &= 1 \\ E[X] &= \mu \\ \text{Var}[X] &= \sigma^2 \end{aligned}$$



• Standard Normal R.V : $N(0, 1)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$F_X(x) : \Phi(x) \leftarrow \text{Table}$$

- Normality preserved under Linear Transformations :

$$X \sim N(\mu, \sigma^2)$$

$$Y = AX + B \leftarrow N(A\mu + B, A^2\sigma^2)$$

$$E[Y] = A\mu + B$$

$$\text{Var}[Y] = A^2\sigma^2$$

• Central Limit Theorem :

$$\frac{1}{n} \sum_{i=1}^n X_i \approx N(\mu, \sigma^2/n) \quad [X : \text{Any RV with mean } \mu \text{ \& \; variance } \sigma^2]$$

• Multinomial Gaussian vector $\leftarrow$ Countably finite

Gaussian Process $\leftarrow$ Countably infinite

Used in Bayesian Optimization (Black-box optimization)
 Brownian Motion

Note: Beta, Gamma, Erlang, Logistic

- $Y = aX + b$

$$F_Y(y) = P(Y \leq y) = P(aX + b \leq y) \\ = F_X\left(\frac{y-b}{a}\right) \text{ if } a > 0$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) \text{ when } a > 0$$

$$F_Y(y) = 1 - F_X\left(\frac{y-b}{a}\right) \text{ if } a < 0$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{-1}{a} f_X\left(\frac{y-b}{a}\right) \text{ when } a < 0$$

$$\therefore f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

Generalising: [Intuition]

$$Y = aX + b \Rightarrow X = \frac{Y-b}{a} \} \Rightarrow f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

$\Downarrow$

$$Y = g(X) \Rightarrow X = h(Y) \} \Rightarrow f_Y(y) = \left| \frac{dh(y)}{dy} \right| f_X(h(y))$$

$\hookrightarrow$ Monotone function, continuous and differentiable
 $\therefore h(x) = g^{-1}(x)$

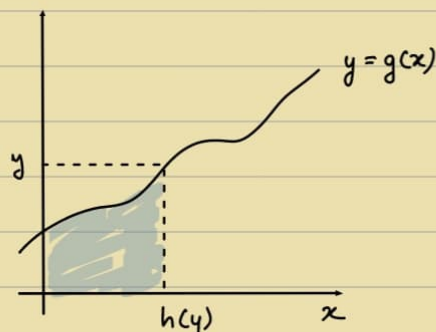
Proof:

$$Y = g(X) \Rightarrow X = h(Y)$$

$\hookrightarrow$ Monotone, continuous and differentiable function

$$F_Y(y) = P(g(X) \leq y) \\ = P(X \leq h(y))$$

$$\left[\begin{array}{l} g(X) \leq y : A = \{\omega \in \Omega \mid g(X(\omega)) \leq y\} \\ \Rightarrow h(g(X)) \leq h(y) \\ \Rightarrow X \leq h(y) : A = \{\omega \in \Omega \mid X(\omega) \leq h(y)\} \end{array} \right]$$



$\xrightarrow{\hspace{1cm}}$
 Event $\{X \leq h(y)\}$

As $g(x) \uparrow \Rightarrow h(x) \downarrow$

case - I :

$g(x) \leftarrow$ Non-decreasing

$$F_Y(y) = P(g(X) \leq y)$$

$$= P(X \leq h(y))$$

$$= F_X(h(y))$$

$$f_Y(y) = \frac{d}{dy} F_X(h(y))$$

$$= f_X(h(y)) \cdot \frac{dh}{dy}(y)$$

non-dec

≥ 0



$$f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy}(y) \right|$$

case - II :

$g(x) \leftarrow$ Non-increasing

$$F_Y(y) = P(g(X) \leq y)$$

$$= P(X \geq h(y))$$

$$= 1 - F_X(h(y))$$

$$f_Y(y) = \frac{d}{dy} (1 - F_X(h(y)))$$

$$= -f_X(h(y)) \cdot \frac{dh}{dy}(y)$$

non-inc.

≤ 0



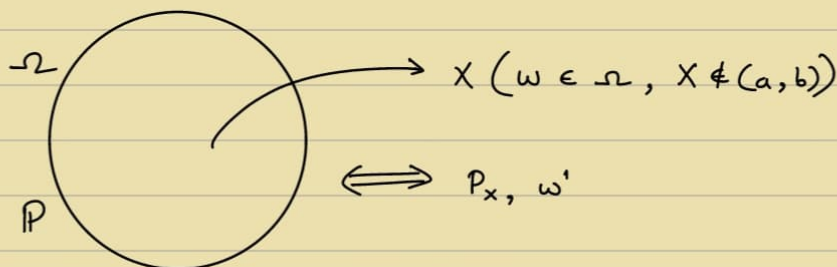
$$f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy}(y) \right|$$

Q) $Y = X^2$, $f_Y(y) = ?$ in terms of $f_X(x)$

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P : Probability measure

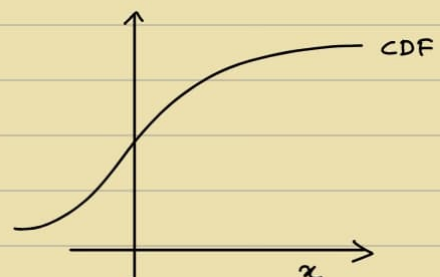
P_X : Induced Probability measure



$$F_X(a) = P_X(-\infty, a]$$

$$= P(X \leq a)$$

Point mass $\rightarrow$ Discontinuity



• CDF:

(i) $F_X(\infty) = 1$

$F_X(-\infty) = 0$, when $P(-\infty < X < \infty) = 1$

(ii) $F_X: \mathbb{R} \rightarrow [0, 1] \leftarrow$ Non-decreasing & right continuous

(iii) @ Points of discontinuity :

$$F_X(x^+) := \lim_{\epsilon \rightarrow 0} F_X(x + \epsilon)$$

$$F_X(x^-) := \lim_{\epsilon \rightarrow 0} F_X(x - \epsilon)$$

} ϵ decreases to 0, i.e. $\epsilon \downarrow 0$
 [$\because \epsilon$ is positive]

$$F_X(x^+) \neq F_X(x^-)$$

Right continuous : $F_X(x) = F_X(x^+)$

if $F_X(x)$ is continuous $\Rightarrow F_X(x^+) = F_X(x^-)$

Th :

$F_X: \mathbb{R} \rightarrow [0, 1]$ is non-decreasing and right continuous

(i) non-decreasing :

$$\forall x_1, x_2 : x_1 \leq x_2 \Rightarrow F_X(x_1) \leq F_X(x_2) \quad [x_1 \& x_2 \text{ are arbitrary}]$$

$$[\text{From } a \leq b \Rightarrow P(A) \leq P(B)]$$

$$\text{Define } A := \{ \omega \in \Omega : X(\omega) \leq x_1 \}$$

$$\text{Define } B := \{ \omega \in \Omega : X(\omega) \leq x_2 \}$$

$$\therefore A \subseteq B$$

$$\therefore P(A) \leq P(B)$$

$$F_X(x_1) = P_X((-\infty, x_1]) = P(A) < P(B) = F_X(x_2)$$

(ii) right-continuous :

Consider seq. of no. $\{x_n\}$ decreasing to x

$$F_X(x^+) = \lim_{x_n \rightarrow x} F_X(x_n)$$

$$\text{Define } A_n := \{ \omega : X(\omega) \leq x_n \}$$

$$\text{Define } A := \{ \omega : X(\omega) \leq x \}$$

$$[\text{When } A_n \uparrow A \text{ or } A_n \downarrow A \Rightarrow \lim_{n \rightarrow \infty} P(A_n) = P(A)]$$

$$\left[\begin{array}{c} (-\infty, x_n] = A_n \\ \downarrow \\ (-\infty, x] = A \end{array} \right\} \Rightarrow P_X$$

$$F_X(x) = P_X(-\infty, x]$$

$$\therefore \text{RHS} = F_X(x)$$

$$\text{LHS} = F_X(x_n) = F_X(x)$$

If $x_n \uparrow x \Rightarrow (-\infty, x_n]$

$$\lim_{x_n \uparrow x} (-\infty, x_n] = (-\infty, x) \quad [\because \text{Going to } x]$$

$$\underbrace{\quad}_{\downarrow} \& P_X(-\infty, x) \neq F_X(x) \quad [\text{By definition}]$$

$$\downarrow U_n(-\infty, x_n]$$

$\therefore$ There could be point mass at x .

→ Mixed RV :

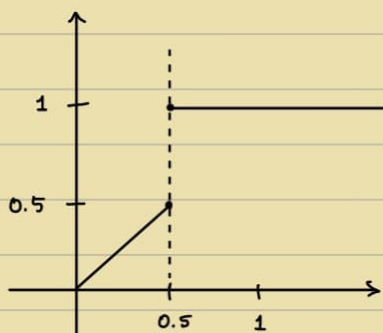
Neither discrete nor continuous, Partly both

CDF → Piece-wise continuous

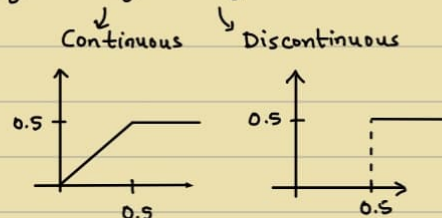
Ref : 4.3.1
probabilitycourse.com

e.g. $X = U[0,1]$ and $Y = X$ if $X \leq 0.5$
 $Y = 0.5$ if $X \geq 0.5$

CDF :



$$\therefore F_Y(y) = C(y) + D(y)$$

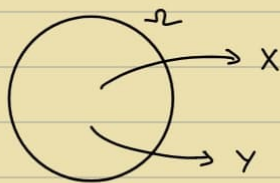


$$E[Y] = \int_{-\infty}^{\infty} x \cdot C(x) dx + \sum_{y_k} y_k P(Y = y_k)$$

where $\{y_1, y_2, \dots, y_n\}$ are jump points of $D(y)$

where $P(Y = y_k) > 0$

• Multiple R.V :



$$\Omega = \{0,1\} \times \{1,2,3,4,5,6\}, \mathcal{F} = 2^{\Omega}, P(\omega) = \frac{1}{12}$$

$$X_{\omega} \in \{0,1\} \leftarrow \text{Coin}$$

$$Y_{\omega} \in \{1,2,3,4,5,6\} \leftarrow \text{Dice}$$

$$\text{for } \omega = (1,5) \Rightarrow X(\omega) = 1 \text{ and } Y(\omega) = 5$$

Joint PMF : $P_{X,Y}(x,y)$

Joint CDF : $F_{X,Y}(x,y)$

$$P_{X,Y}(x,y) := P\{\omega \in \Omega : X(\omega) = x \text{ and } Y(\omega) = y\}$$

$$F_{X,Y}(x,y) := P\{\omega \in \Omega : X(\omega) \leq x \text{ and } Y(\omega) \leq y\}$$

$$P((X,Y) \in A) = P\{\omega \in \Omega : (X(\omega), Y(\omega)) \in A\}$$

$$= \sum_{(x,y) \in A} P_{X,Y}(x,y)$$

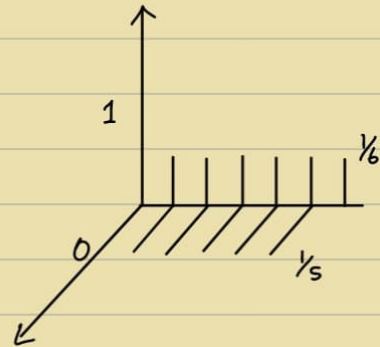
e.g. $P((X,Y) \in A)$; Given, A : coin $\rightarrow$ head & dice $\rightarrow 6$

if $H \rightarrow \{1, 2, 3, 4, 5, 6\}$

$T \rightarrow \{1, 2, 3, 4, 5\}$

$\{ \overset{1/12}{(1,1)}, \overset{1/12}{(1,2)}, \dots, \overset{1/12}{(1,6)}, \underset{1/10}{(0,1)}, \underset{1/10}{(0,2)}, \dots, \underset{1/10}{(0,5)} \}$

$$\begin{cases} 1/12 = 1/2 \times 1/6 \\ 1/10 = 1/2 \times 1/5 \end{cases}$$



IMP

• Marginals:

$$P(Y=3) = \frac{1}{10} + \frac{1}{12}$$

$$P(Y=6) = \frac{1}{12}$$

$$P(X=1) = \frac{1}{2}$$

}

$$P_{X,Y}(1,i) = 1/12$$

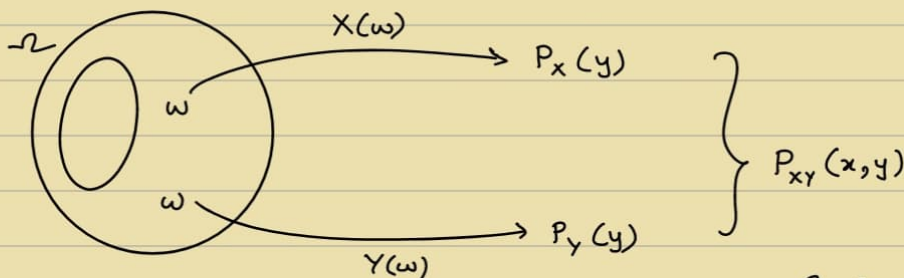
$$\sum_i P_{X,Y}(1,i) = P\{X \in \Omega, X(\omega) = 1\} = 1/2 = P_X(x)$$

$$P_{X,Y}(1,i) + P_{X,Y}(0,i) = 1/6 = P_Y(i)$$

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

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$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

$$\begin{aligned} \sum_{\substack{x \leq a \\ y \leq b}} P_{X,Y}(x,y) &= F_{X,Y}(a,b) \\ &= P(X \leq a, Y \leq b) \\ &= P(\omega \in \Omega, X(\omega) \leq a, Y(\omega) \leq b) \end{aligned}$$

$$\sum_y P_{X,Y}(x,y) = P_X(x) = P(\omega \in \Omega, X(\omega) = x)$$

$$P_{X,Y}(x,y) = P(\omega \in \Omega, X(\omega) = x \text{ \& } Y(\omega) = y)$$

$$\begin{aligned}\sum_y P_{XY}(x, y) &= P\left(\bigcup_y (\omega \in \Omega, X(\omega) = x \text{ \& } Y(\omega) = y)\right) \\ &= \sum_y P((\omega \in \Omega, X(\omega) = x \text{ \& } Y(\omega) = y))\end{aligned}$$

Note: $F_X(x) = \sum_y F_{XY}(x, y)$

- Independent:

$$P_{XY}(x, y) = P_X(x) \cdot P_Y(y)$$

$$F_{XY}(x, y) = F_X(x) \cdot F_Y(y)$$

$$P_{XY}(x, y) = P(\omega \in \Omega, X(\omega) = x \text{ \& } Y(\omega) = y)$$

$$A = \{\omega \in \Omega, X(\omega) = x\}$$

$$B = \{\omega \in \Omega, Y(\omega) = y\}$$

$$\Rightarrow A \cap B = \{\omega \in \Omega, X(\omega) = x \text{ \& } Y(\omega) = y\}$$

$$P_X(x) = P(A)$$

$$P_Y(y) = P(B)$$

$$\begin{aligned}F_{XY}(x, y) &= P(X \leq x, Y \leq y) \\ &= P(\omega \in \Omega, X(\omega) \leq x \text{ \& } Y(\omega) \leq y)\end{aligned}$$

- Expectation: $E[XY]$

$$\begin{aligned}E[X] &= \sum_x x \cdot P_X(x) \\ &= \sum_x \sum_y x \cdot P_{XY}(x, y)\end{aligned}$$

$$\begin{aligned}E[Y] &= \sum_y y \cdot P_Y(y) \\ &= \sum_y \sum_x y \cdot P_{XY}(x, y)\end{aligned}$$

$$E[XY] = \sum_x \sum_y xy \cdot P_{XY}(x, y)$$

$$E[XY] = E[X] \cdot E[Y] \quad \text{for independent events, } X \text{ and } Y \text{ given}$$

(Converse may not be true)

e.g. When X and Y are dependent.

Roll a dice &

$$X = \begin{cases} 1 & \text{if outcome is odd} \\ 0 & \text{, otherwise} \end{cases} \quad \Bigg| \quad Y = \begin{cases} 1 & \text{if outcome is even} \\ 0 & \text{, otherwise} \end{cases}$$

$$P_{XY}(0,0) = 0 \quad P_{XY}(0,1) = \frac{1}{2}$$

$$P_{XY}(1,0) = \frac{1}{2} \quad P_{XY}(1,1) = 0$$

$$P_Y(0) = P_X(0) = \sum_y P_{XY}(0,y) = \frac{1}{2}$$

$$P_Y(1) = P_X(1) = \sum_y P_{XY}(1,y) = \frac{1}{2}$$

$$F_{XY}(x,y) \neq F_X(x) \cdot F_Y(y)$$

$$E[XY] \neq E[X] \cdot E[Y]$$

- Consistency Properties :

$$(i) \sum_{x,y} P_{XY}(x,y) = 1$$

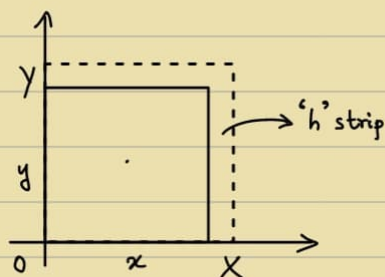
$$(ii) F_{XY}(\infty, \infty) = 1$$

$$(iii) F_{XY}(-\infty, -\infty) = 0$$

$$(iv) F_{XY}(-\infty, \infty) = F_{XY}(\infty, -\infty) = 0$$

$$(v) \left. \begin{aligned} F_{XY}(x, \infty) &= F_X(x) \\ F_{XY}(\infty, y) &= F_Y(y) \end{aligned} \right\} \leftarrow \text{Marginal}$$

$F_{XY}(x,y) :$



$$F_{XY}(x,y) = xy \quad \& \quad f_{XY}(x,y) = 1$$

$$F_{XY}(x+h, y) - F_{XY}(x, y)$$

$$\frac{\partial^2 F_{XY}(x,y)}{\partial x \partial y} \leftarrow \begin{cases} \frac{\partial F_{XY}(x,y)}{\partial x} = \lim_{h \rightarrow 0} \frac{F_{XY}(x+h, y) - F_{XY}(x, y)}{h} \\ \frac{\partial F_{XY}(x,y)}{\partial y} = \lim_{h \rightarrow 0} \frac{F_{XY}(x, y+h) - F_{XY}(x, y)}{h} \end{cases}$$

Joint PDF
 $f_{X,Y}(x,y)$

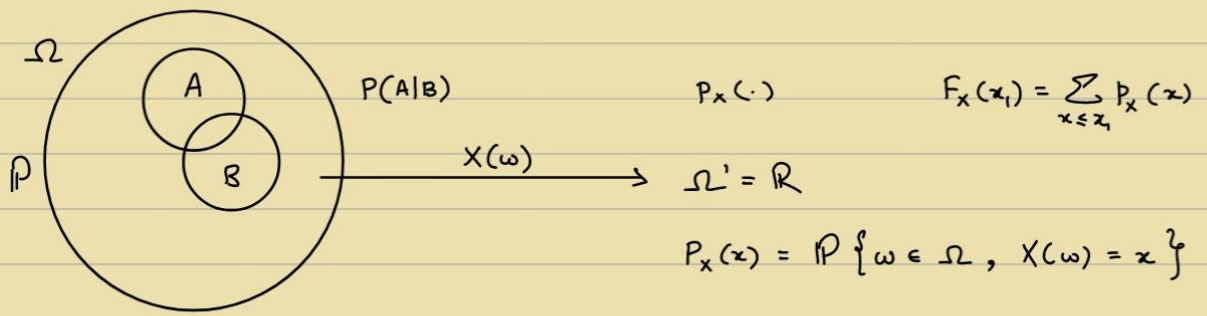
$$F_{XY}(x,y) := \int_{-\infty}^x \int_{-\infty}^y f_{XY}(a,b) \, da \cdot db$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y) \, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y) \, dx$$

Apply for ≥ 2 R.V.s

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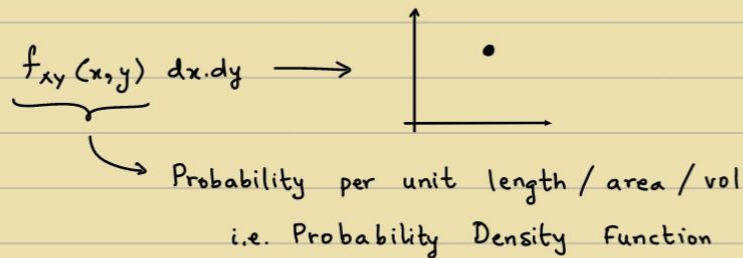


$$f_X(x) = \frac{d}{dx} F_X(x)$$

$P_{XY}(x, y) \leftarrow$ Joint PDF

$$F_{XY}(x_1, y_1) = \sum_{x \leq x_1} \sum_{y \leq y_1} P_{XY}(x, y)$$

$$f_{XY}(x, y) = \frac{\partial}{\partial x} \frac{\partial}{\partial y} F_{XY}(x, y)$$



• $X \rightarrow P_X(z)$

$$E[g(x)] = \sum_x g(x) P_X(x)$$

$\downarrow$
 z

$$P_Z(z) = \sum_{\substack{x \\ z=g(x)}} P_X(x)$$

If $z = g(x, y)$

$$P_Z(z) = ?$$

$$P_Z(z) = \sum_{\substack{x, y \\ g(x, y) = z}} P_{XY}(x, y)$$

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

Proof: $\sum_{xy} g(x, y) \cdot P_{XY}(x, y)$

$$\left\{ \sum_{xy} (ax + by + c) P_{XY}(x, y) \right\}$$

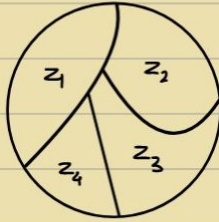
$$\sum_x \sum_y P_{XY}(x, y) = 1$$

$$E[Z] = \sum_z \sum_{x,y} z \cdot p_{xy}(x,y) \\ g(x,y) = z$$

$$\therefore p_z(z) = \sum_{x,y} p_{xy}(x,y) \\ g(x,y) = z$$

$$E[g(x,y)] = \sum_{x,y} g(x,y) p_{xy}(x,y) \\ = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) dx \cdot dy$$

$$E[Z] = \sum_z z p_z(z)$$



- \$X\$ and \$Y \rightarrow\$ continuous independent R.V

$$\left. \begin{array}{l} W = \max(X, Y) \\ Z = \min(X, Y) \end{array} \right\} \begin{array}{l} \text{i.e. } P_{XY}(x, y) = P_X(x) \cdot P_Y(y) \\ F_{XY}(x, y) = F_X(x) \cdot F_Y(y) \\ E[XY] = E[X] \cdot E[Y] \end{array}$$

$$(i) F_W(w_1) = P(W \leq w_1)$$

$$= P(X \leq w_1 \text{ \& } Y \leq w_1)$$

$$= P(X \leq w_1) \cdot P(Y \leq w_1)$$

$$= F_X(w_1) \cdot F_Y(w_1)$$

$$F_W(w_1) = F_{XY}(w_1, w_1) \text{ , i.e. } W \leq w_1 \iff X, Y \leq w_1$$

$$p_W(w) = \frac{d}{dw} F_W(w)$$

$$= F_X(w) \cdot f_Y(w) + f_X(w) \cdot F_Y(w)$$

$$(ii) F_Z(z_1) = P(Z \leq z_1)$$

$$Z \geq z_1 \iff X \geq z_1 \text{ and } Y \geq z_1$$

$$\overline{F_Z(z_1)} = P(Z > z_1)$$

$$= \overline{F_{XY}(z_1, z_1)}$$

$$= 1 - F_{XY}(z_1, z_1)$$

$$= 1 - F_X(z_1) \cdot F_Y(z_1)$$

e.g. if \$X\$ & \$Y \rightarrow\$ exponential , Then \$Z \rightarrow\$ exponential
 $(\lambda_1) \quad (\lambda_2) \quad (\lambda_1 + \lambda_2)$

$$\left\{ \text{Hint: } e^{-\lambda_1 w} \cdot e^{-\lambda_2 w} = e^{-(\lambda_1 + \lambda_2) w} \right\}$$

$$\Phi) Z = X + Y$$

$$P_Z(z_1) = \sum_{\substack{x,y \\ x+y=z_1}} P_{X,Y}(x,y)$$

$$= \sum_{x,y} P_X(x) \cdot P_Y(y)$$

$$= \sum_x P_X(x) \cdot \sum_y P_Y(y)$$

$$= \sum_x P_X(x) \cdot P_Y(z_1 - x)$$

$$\begin{cases} x + y = z \\ \Rightarrow y = z - x \end{cases}$$

for satisfying (x,y)

- Covariance of X and Y :
(Joint Variation)

$$\begin{aligned} \text{Cov}(X,Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X] \cdot E[Y] \end{aligned}$$

$$\left\{ \text{What is } \begin{bmatrix} \text{cov}(X,X) & \text{cov}(X,Y) \\ \text{cov}(Y,X) & \text{cov}(Y,Y) \end{bmatrix} ? \right\} \leftarrow \begin{array}{l} \text{Covariance Matrix} \\ \text{cov}[X,Y] \\ \text{Gives correlation} \end{array}$$

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$g(x,y) = (x - E[X]) \cdot (y - E[Y])$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

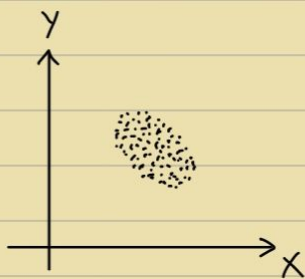
$$\underbrace{E[g(X,Y)]}_{E_{P_{X,Y}(x,y)}} = \sum_{x,y} g(x,y) \cdot P_{X,Y}(x,y)$$

$$E_{P_{X,Y}(x,y)} [(X - E_{P_X(x)}[X])(Y - E_{P_Y(y)}[Y])] = \sum_{x,y} g(x,y) \cdot P_{X,Y}(x,y)$$

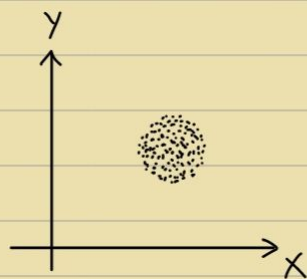
Note: $E[XY] = E[X] \cdot E[Y]$ only when X and Y are uncorrelated & doesn't necessarily imply independence

Independence $\Rightarrow$ Uncorrelated

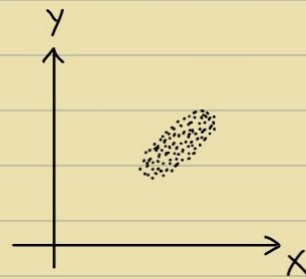
Uncorrelated $\nRightarrow$ Independence (Not always)



$$\text{cov}(X, Y) < 0$$

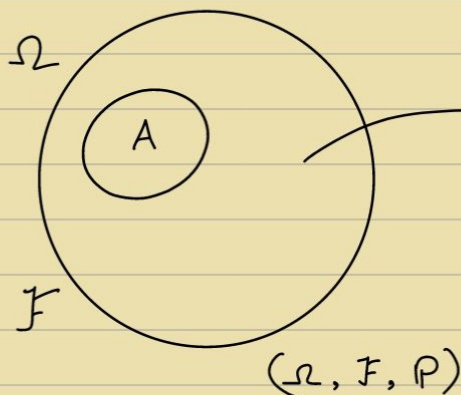


$$\text{cov}(X, Y) \approx 0$$



$$\text{cov}(X, Y) > 0$$

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$$\begin{matrix} x_1 & x_2 & \dots & x_{10} \\ p_1 & p_2 & \dots & p_{10} \end{matrix}$$

$$(\Omega', F', P_X) \quad \begin{matrix} \Omega' = \mathbb{R} \\ F = \mathcal{B}(\mathbb{R}) \end{matrix}$$

$$P_X(x) \equiv P_{X/A}(x)$$

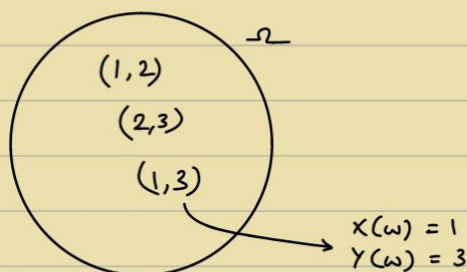
$$\frac{P(A \cap B)}{P(A)}$$

$$\sum P(A_i) \cdot P_{X/A_i}(x) = P_X(x)$$

Ex. Pick 2 integers from $\{1, 2, 3\}$ without replacement

$$\Omega = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

$$P\{\omega\} = 1/6 \quad \forall \omega \in \Omega$$



$$P_{X,Y}(x, y) = 1/6$$

$$P_X(x) = \sum_y P_{X,Y}(x, y) = 1/6 + 1/6 = 1/3$$

$$P_Y(y) = \sum_x P_{X,Y}(x, y) = 1/6 + 1/6 = 1/3$$

$$E[X] = 2$$

$$E[Y] = 2$$

$$E[XY] = \sum g(x, y) \cdot P_{X,Y}(x, y)$$

$$= 2 \cdot P_{X,Y}(1, 2) + \dots$$

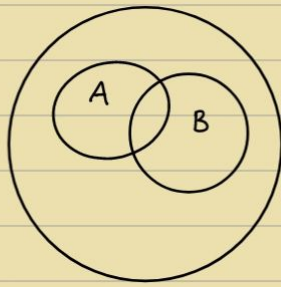
$$= 11/3$$

$$P_X(x) \cdot P_Y(y) = 1/9$$

→ R.V. X with $P_X(x)$

Event A , $A \in \mathcal{F}$

Consider event $\{\omega \in \Omega : X(\omega) = x\} \equiv \{X = x\}$



$$P(X=x|A) = \frac{P(\{X=x\} \cap A)}{P(A)} = P_{X|A}(x)$$

e.g. first no. → odd
second no. → even $\left\{ \begin{array}{l} A = \{(1,2), (3,2)\} \end{array} \right.$

$$P_{X|A}(1) = \frac{P(X=1 \cap A)}{P(A)} = \frac{1/6}{1/3} = \frac{1}{2}$$

Similarly, $P_{X|A}(3) = 1/2$

$$\left[\text{Prev. } P_X(x) = 1/3 \text{ for } x = 1, 2, 3 \right]$$

$$\begin{aligned} \sum_x P_{X|A}(x) &= \sum_x \frac{P(\{X=x\} \cap A)}{P(A)} \\ &= \frac{\sum_x P(\{X=x\} \cap A)}{P(A)} \\ &= \frac{P(A)}{P(A)} = 1 \end{aligned}$$

Ex. Roll a dice

A : roll is odd

$$P_{X|A}(x) = ?$$

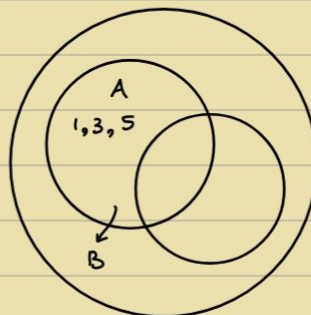
$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$P_{X|A}(x) = \frac{1/6}{1/2} = 1/3$$

$$\left\{ \frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0 \right\}$$

$$E[X|A] = 3$$

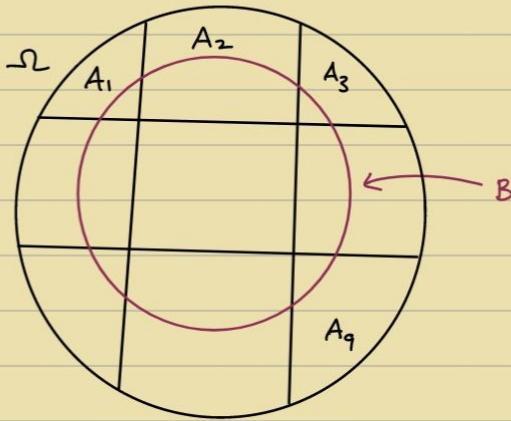
$$E[X|A] = 4 \text{ (if } A \rightarrow \text{even)}$$



$$E[X|A] = \sum_x x P_{X|A}(x)$$

Using LOTUS, $E[g(X)|A] = \sum_x g(x) \cdot P_{X|A}(x)$

- Law of Total Probability:



$$P(B) = \frac{\sum P(B \cap A_i)}{P(A_i)} \cdot P(A_i)$$

$$= \sum_i P(B|A_i) \cdot P(A_i)$$

$$P_X(x) = \sum_i P(A_i) \cdot P_{X|A_i}(x)$$

Proof: $B = \{\omega \in \Omega, X(\omega) = x\}$

$$P_X(x) = \sum_i P(A_i) \cdot \frac{P(\{X=x\} \cap A_i)}{P(A_i)}$$

$$= \sum_i P(\{X=x\} \cap A_i)$$

$$= P(\{X=x\})$$

$$B = \bigsqcup (B \cap A_i)$$

↪ Disjoint Union

$$P(B) = \sum_i \frac{P(B \cap A_i)}{P(A_i)} \cdot P(A_i)$$

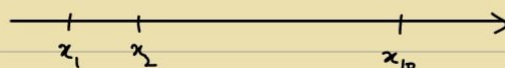
Now, $E[X] = \sum_i P(A_i) \cdot E[X|A_i]$

- e.g. $A = (X > 5) = \{6\}$
 $A \in \mathcal{F}'$

$$X \in A = \{\omega \in \Omega : X(\omega) = A\}$$

$$P\{X \in A\} = \sum_{x \in A} P_X(x)$$

$$\Omega' = \{x_1, x_2, \dots, x_{10}\}$$



$$\mathcal{F} = 2^{\Omega'}$$

$$A \in \mathcal{F}$$

Let $A = \{x_1, x_9, x_{10}\}$

$$P_{X|A}(x_1) = ? = 1/3$$

$$P_{X|A}(x_2) = ? = 0$$

$\vdots$

$$P_{X|A}(x_{10}) = ? = 1/3$$

$$P_{X|A}(x_1) = \frac{P(\{X=x_1\} \cap A)}{P(A)} = \frac{P(A \cap \{x_1\})}{P_X(A)} = \frac{P_X(x_1)}{P_X(A)}$$

$$P_X(A) = P(A^{-1})$$

$$= \{\omega \in \Omega, X(\omega) \in A\}$$

$$\text{i.e. if } x \notin A \Rightarrow P_{X|A}(x) = 0$$

$$\text{if } x \in A \Rightarrow P_{X|A}(x) = \frac{P_X(x)}{P(X \in A)}$$

Ex. Rolling a dice, $X > 2$

$$\text{i.e. } \{3, 4, 5, 6\} = A$$

$$P_{X|A}(x_1) = 0$$

$$P_{X|A}(x_2) = 0$$

$$P_{X|A}(x_3) = 1/4 = \frac{1/6}{4/6}$$

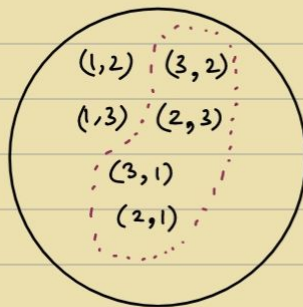
$$P_{X|A}(x_4) = 1/4$$

$$P_{X|A}(x_5) = 1/4$$

$$P_{X|A}(x_6) = 1/4$$

e.g. $X \in A$ where $A = \{2, 3\}$

$$P_{X|A}(x) = ?$$



$$P_A = 4/6$$

$$P_X(x) = 1/3$$

• Geometric R.V.:

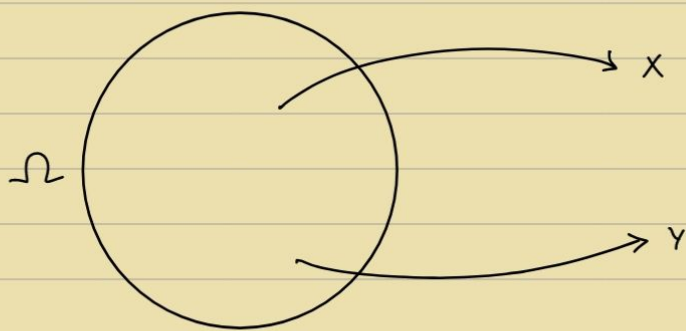
$$P_N(k) = (1-p)^{k-1} p \leftarrow \text{PMF}$$

If $A := N > n$ & $B := N = k$

$$P_{N|A}(k) = ?$$

$$\rightarrow P_{N|A}(k) = 0 \text{ if } k < n+1$$

$$P_{N|A}(k) = \frac{P(\{N > n\} \cap \{N = k\})}{P(N > n)} = \frac{P\{N = k\}}{(1-p)^n} \cdot p$$



$$P_{XY}(x, y) \quad @ Y=y$$

$$f_{XY}(x, y) \quad P_{X|A}(x)$$

$$P_{X|Y}(x|y) \rightarrow f_{X|Y}(x, y)$$

$$P(A|B) = P(X=x | Y=y) / P(Y=y)$$

$$P_{X|Y}(x|y) = \frac{P_{XY}(x, y)}{P_Y(y)}$$

$$\Rightarrow P_{XY}(x, y) = P_{X|Y}(x|y) \cdot P_Y(y)$$

$$[P(A \cap B) = P(A|B) \cdot P(B)]$$

• Independence :

$$P_{XY}(x, y) = P_{X|Y}(x|y) \cdot P_Y(y)$$

$$\Rightarrow P_X(x) \cdot P_Y(y) = P_{X|Y}(x|y) \cdot P_Y(y)$$

$$\Rightarrow P_X(x) = P_{X|Y}(x|y)$$

$$\Rightarrow P_{X|Y}(x|y) = P_X(x)$$

$$\therefore E[XY] = E[X] \cdot E[Y]$$

$$\& \text{Cov}(X, Y) = 0$$

$$\bullet P_{XY}(x, y) = P_{X|Y}(x|y) \cdot P_Y(y)$$

$$\Rightarrow P_X(x) = \sum_y P_{X|Y}(x|y) \cdot P_Y(y)$$

Law of Total Probability

Summing over x instead of y :

$$P_Y(y) = \sum_x P_{X|Y}(x|y) \cdot P_X(x)$$

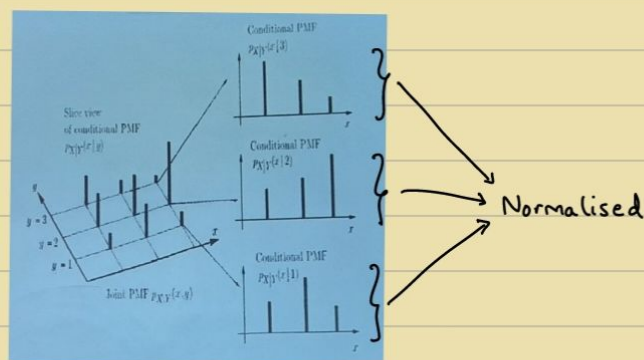
$$E[X|Y=y] = \sum_x x P_{X|Y}(x|y)$$

$$E[X] = \sum_x x P_X(x) \quad (\text{PS-34})$$

$$= \sum_x x \cdot \sum_y P_{X|Y}(x|y) \cdot P_Y(y)$$

$$= \sum_y \sum_x x P_{X|Y}(x|y) \cdot P_Y(y)$$

$$= \sum_y P_Y(y) \cdot E[X|Y=y]$$



Summary :

$$\textcircled{1} \quad P_{X|A}(x) = \frac{P_X(x)}{P(A)} \quad (\text{if } x \in A) = \frac{f_X(x)}{P(A)} \quad (\text{if } x \in A)$$

$$\textcircled{2} \quad E[X|A] = \sum_x x P_{X|A}(x) = \int_{-\infty}^{\infty} x \cdot f_{X|A}(x)$$

$$\textcircled{3} \quad P_X(x) = \sum_{i=1}^n P(A_i) P_{X|A_i}(x) = \sum_{i=1}^n P(A_i) \cdot f_{X|A_i}(x)$$

$$\textcircled{4} \quad E[X] = \sum_{i=1}^n P(A_i) \cdot E[X|A_i]$$

$$\textcircled{5} \quad P_{X,Y}(x,y) = P_{X|Y}(x|y) \cdot P_Y(y) \\ f_{X,Y}(x,y) = f_{X|Y}(x|y) \cdot f_Y(y)$$

$$\textcircled{6} \quad P_X(x) = \sum_y P_{X|Y}(x|y) \cdot P_Y(y) \\ f_X(x) = \int_y f_{X|Y}(x|y) \cdot f_Y(y) dy$$

$$\textcircled{7} \quad E[X|Y=y] = \sum_x x \cdot P_{X|Y}(x|y) \\ E[X|Y=y] = \int_x f_{X|Y}(x|y) dx$$

$$\textcircled{8} \quad E[X] = \sum_y P_Y(y) \cdot E[X|Y=y] \\ E[X] = \int_y f_Y(y) \cdot E[X|Y=y] \cdot dy$$

→ Conditional Expectation : $E[X|Y]$

$$E[X|Y=y] = \sum_x x \cdot P_{X|Y}(x|y)$$

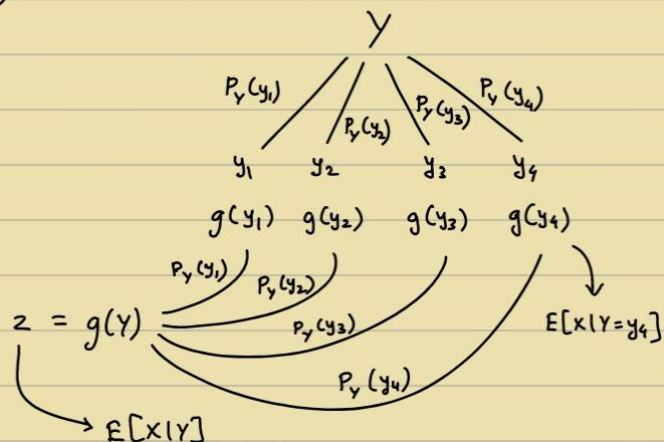
$$g(y) = E[X|Y=y] = \sum_x x \cdot P_{X|Y}(x|y)$$

$$Y = y_1, y_2, y_3, \dots, y_{10}$$

$$E[Z] = E[g(Y)]$$

$$= \sum_y P_Y(y) \cdot g(y)$$

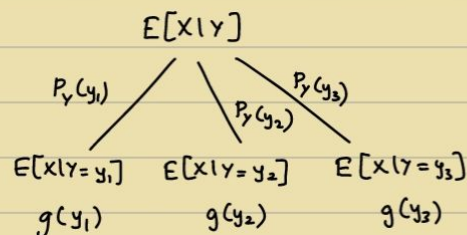
$$E[E[X|Y]] = \sum_y P_Y(y) \cdot E[X|Y=y] \\ = E[X]$$



• Ex. $Y = \begin{cases} \lambda_1 & \text{with prob } p \\ \lambda_2 & \text{with prob } 1-p \end{cases}$

$X \sim \text{Exponential}(Y)$

$$\left[\begin{array}{l} \because f_X(x) = \lambda e^{-\lambda x} \\ X \sim \text{Exp}(\lambda) \\ E[X] = 1/\lambda \end{array} \right]$$



$$E[E[X|Y]] = E[X]$$

$E[X] = ?$

$$E[X|Y = \lambda_1] = 1/\lambda_1$$

$$E[X|Y = \lambda_2] = 1/\lambda_2$$

$$\begin{aligned} E[X] &= \sum_{i=1}^2 E[X|Y=y_i] \cdot P_Y(y_i) \\ &= 1/\lambda_1 \cdot p + 1/\lambda_2 \cdot (1-p) \\ &= p(1/\lambda_1 - 1/\lambda_2) + 1/\lambda_2 \end{aligned}$$

$$f_{X|Y}(x|\lambda_1) = \lambda_1 e^{-\lambda_1 x}$$

$$f_{X|Y}(x|\lambda_2) = \lambda_2 e^{-\lambda_2 x}$$

$$\therefore E[X] = (1/\lambda_1 - 1/\lambda_2) p + 1/\lambda_2$$

$$E[X] = E[E[X|Y]]$$

↘ Misuse of Notation, but intuitively.

Ex. $Y = X_1 + X_2 + \dots + X_N$, $N > 0$

X_i 's are independent & identically distributed with mean $E[X]$

i.i.d $\rightarrow$ PMF of all X_i are same

$E[Y] = ?$

$$\begin{aligned} E[Y|N=n] &= E[X_1 + X_2 + \dots + X_n] \\ &= E[X_1] + E[X_2] + \dots + E[X_n] \\ &= n E[X] \end{aligned}$$

$$E[Y|N] = \sum_{i=1}^N E[X_i|N] = N \cdot E[X]$$

$$\begin{aligned} E[Y] &= \sum_n E[Y|N=n] \cdot P_N(n) & \text{(OR)} & E[E[Y|N]] = E[N \cdot E[X]] \\ &= E[X] \cdot E[N] & \Rightarrow & E[Y] = E[N] \cdot E[X] \end{aligned}$$

$\text{Var}[Y] = ? \leftarrow \underline{\text{H.W}}$

$$\text{Var}[Y] = E[Y^2] - E[Y]^2$$

$$\text{As } Y = \sum X_i$$

$$\& \because E[Y] = E[X] \cdot E[N]$$

$$E[Y^2] = E\left[\left(\sum X_i\right)^2\right]$$

$$= E\left[\sum (X_i)^2 + 2 \cdot \sum_i \sum_{i+1} X_i X_j\right]$$

$$= E\left[\sum (X_i)^2\right] + 2 \cdot E\left[\sum_i \sum_{i+1} X_i X_j\right]$$

$$\therefore E[Y^2] = \sum E[(X_i)^2] + 2 \sum_i \sum_{i+1} E[X_i X_j]$$

$$\text{Var}[Y] = E[Y^2] - E[Y]^2$$

$$= \sum E[(X_i)^2] + 2 \sum_i \sum_{i+1} E[X_i] \cdot E[X_j] - (E[Y])^2$$

$$= \sum E[(X_i)^2] + 2 \sum_i \sum_{i+1} E[X_i] \cdot E[X_j] - (E[\sum X_i])^2$$

$$= \sum E[(X_i)^2] + 2 \sum_i \sum_{i+1} E[X_i] \cdot E[X_j] - (\sum E[X_i])^2$$

$$= \sum E[(X_i)^2] + 2 \sum_i \sum_{i+1} E[X_i] \cdot E[X_j] - \sum (E[X_i])^2 - 2 \sum_i \sum_{i+1} E[X_i] E[X_j]$$

$$= \sum E[(X_i)^2] - \sum (E[X_i])^2$$

$$= \sum \text{Var}[X_i]$$

$$= \text{Var}[X_1] + \text{Var}[X_2] + \dots + \text{Var}[X_n]$$

$$\left. \begin{array}{l} \therefore \text{For Independent R.V.s:} \\ \text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] \end{array} \right\}$$

$$\text{Var}[Y|N=n] = \text{Var}[X_1 + X_2 + \dots + X_n]$$

$$= \text{Var}[X_1] + \text{Var}[X_2] + \dots + \text{Var}[X_n]$$

$$= n \cdot \text{Var}[X]$$

$$\text{Var}[Y|N] = \sum_{i=1}^N \text{Var}(X_i|N) + 2 \cdot \sum_{i < j} \text{cov}(X_i, X_j|N)$$

$$\text{Var}[Y] = \sum_n \text{Var}[Y|N=n] \cdot P_N(n)$$

$$\begin{aligned} & \rightarrow E[\text{Var}[Y|N]] + \text{Var}[E[Y|N]] \\ & E[\text{Var}[Y|N]] + \text{Var}\left[\sum_{i=1}^N E[X_i|N]\right] \\ & = E[N] \cdot \text{Var}[X] + \text{Var}[N] \cdot (E[X])^2 \end{aligned}$$

$$E[\text{Var}[Y|N]] = E[N \cdot \text{Var}[X]]$$

$$= E[N] \cdot \text{Var}[X]$$

$$\text{Var}[E[Y|N]] = \text{Var}[N \cdot E[X]]$$

$$= \text{Var}[N] \cdot (E[X])^2$$

2022

①

1. For a continuous non-negative random variable X prove that $E[X^2] = 2 \int_0^\infty x \bar{F}_X(x) dx$ where $\bar{F}_X(x) = 1 - F_X(x)$.

$$E[X^2] = 2 \int_0^\infty x \bar{F}_X(x) dx$$

Where, $\bar{F}_X(x) = 1 - F_X(x)$

Proof:

$$E[X] = \int_0^\infty x \cdot f_X(x) dx$$

$$F_X(x) = \int_{-\infty}^x f_X(x) dx$$

$$\Rightarrow 1 - F_X(x) = \bar{F}_X(x) = \int_x^\infty f_X(x) dx$$

$$\begin{aligned} 2 \int_0^\infty x \bar{F}_X(x) dx &= 2 \int_0^\infty x \cdot \bar{F}_X(x) dx \\ &= 2 \int_0^\infty x \cdot \int_x^\infty f_X(u) \cdot du \cdot dx \\ &= \int_0^\infty \int_0^u 2x \cdot f(u) \cdot du \cdot dx \\ &= \int_0^\infty u^2 f(u) du = E[X^2] \end{aligned}$$

②

2. Let X be a continuous random variable with distribution $F_X(\cdot)$ and density $f_X(x)$. Find the density and distribution for $Z = \sqrt{X}$.

$$Z = \sqrt{X}$$

$$F_X(x) = P(X \leq x)$$

$$F_Z(z) = P(Z \leq z) = P(\sqrt{X} \leq z) = P(X \leq z^2) = F_X(z^2)$$

$$f_Z(z) = 2z \cdot f_X(z^2) \quad \left(\text{PDF} \rightarrow \frac{d}{d(RV)} \text{CDF} \right)$$

③

3. Consider two exponential random variables X and Y with parameters λ_1 and λ_2 respectively. Consider $Z = \min(X, Y)$ (min stands for minimum). Find the probability density and cumulative distribution of Z .

$$Z = \min(X, Y)$$

$$f_X(x) = \begin{cases} \lambda_1 e^{-\lambda_1 x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} \lambda_2 e^{-\lambda_2 y}, & y > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$F_Z(z) = P(Z \leq z)$$

$$= 1 - P(Z > z)$$

$$= 1 - P(X > z \text{ \& } Y > z)$$

$$= 1 - P(X > z) \cdot P(Y > z)$$

$$= 1 - e^{-\lambda_1 z} \cdot e^{-\lambda_2 z}$$

$$F_Z(z) = 1 - e^{-(\lambda_1 + \lambda_2)z}$$

→ Exponential in $(\lambda_1 + \lambda_2)$

$$P(X \leq x) = 1 - e^{-\lambda_1 x}$$

$$P(X > x) = 1 - (1 - e^{-\lambda_1 x}) = e^{-\lambda_1 x}$$

$$f_Z(z) = (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2)z}$$

④

4. Let X and Y denote Gaussian random variables with mean μ_1 and μ_2 and standard deviation σ_1 and σ_2 respectively. Consider $Z = X + Y$. Using Moment generating functions, show that Z is also a Gaussian random variable, with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$.

$$X \sim (\mu_1, \sigma_1), Y \sim (\mu_2, \sigma_2)$$

$$Z = X + Y$$

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$M_X(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx - \frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

$$= \frac{x^2 - 2\mu x + \mu^2}{2\sigma^2} + tx$$

$$= \frac{x^2 - 2(\mu - t\sigma^2)x + \mu^2}{2\sigma^2}$$

$$Z = X + Y$$

$$M_Z(t) = E[e^{tz}]$$

$$= E[e^{t(x+y)}] = E[e^{tx} \cdot e^{ty}]$$

$$= E[e^{tx}] \cdot E[e^{ty}] = M_X(t) \cdot M_Y(t)$$

$$= e^{\mu_1 t + \frac{1}{2}\sigma_1^2 t^2} \cdot e^{\mu_2 t + \frac{1}{2}\sigma_2^2 t^2}$$

$$= e^{(\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2}$$

$$\therefore \text{Mean} \rightarrow \mu_1 + \mu_2$$

$$\text{Variance} \rightarrow \sigma_1^2 + \sigma_2^2$$

⑤

5. Let U_1 and U_2 be two independent Uniform random variables with support $[0, 1]$. Then find the cdf or pdf of $U_1 + U_2$.

$$f_{U_1}(u) = f_{U_2}(u) = \begin{cases} 1 & \text{if } u \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$Z = U_1 + U_2 : [0, 2]$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{U_1}(u_1) \cdot f_{U_2}(z - u_1) du_1$$

(I) If $0 \leq z \leq 1$

$$\Rightarrow 0 \leq u_1 \leq z$$

$$\begin{aligned} f_Z(z) &= \int_0^z f_{U_1}(u_1) \cdot f_{U_2}(z - u_1) du_1 \\ &= \int_0^z 1 \cdot du_1 = z \end{aligned}$$

(II) If $1 \leq z \leq 2$

$$\Rightarrow z-1 \leq u_1 \leq 1$$

$$\begin{aligned} f_Z(z) &= \int_{z-1}^1 f_{U_1}(u_1) \cdot f_{U_2}(z - u_1) du_1 \\ &= \int_{z-1}^1 1 \cdot du_1 = 2 - z \end{aligned}$$

$$\therefore f_Z(z) = \begin{cases} z, & \text{if } 0 \leq z \leq 1 \\ 2 - z, & \text{if } 1 \leq z \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

⑥

6. If X and Y are independent random variables, prove that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. (Recall that $\text{Var}(X) = E[(X - E[X])^2]$)

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

$$\begin{aligned} \text{Var}(X + Y) &= E[(X + Y)^2] - (E[X + Y])^2 \\ &= E[X^2 + 2XY + Y^2] - (E[X] + E[Y])^2 \\ &= E[X^2] + \cancel{E[2XY]} + E[Y^2] - (E[X])^2 - \cancel{2E[X] \cdot E[Y]} - (E[Y])^2 \\ &= E[X^2] - (E[X])^2 + E[Y^2] - (E[Y])^2 \\ &= \text{Var}(X) + \text{Var}(Y) \end{aligned}$$

⑦

1. $X_1, X_2, \dots, X_n$ are independent and identically distributed *Bernoulli*(p) random variables (i.e., they take the value 1 with probability p and 0 otherwise). Consider $S_n = \sum_{i=1}^n X_i$. Find the PMF, MGF, mean and variance of S_n .

$$\begin{aligned} M_{S_n}(t) &= E[e^{tS_n}] \\ &= E[e^{t(X_1 + X_2 + \dots + X_n)}] \end{aligned}$$

$$\begin{aligned}
&= E[e^{tx_1} \cdot e^{tx_2} \cdot \dots \cdot e^{tx_n}] \\
&= E[e^{tx_1}] \cdot E[e^{tx_2}] \cdot \dots \cdot E[e^{tx_n}] \\
&= (E[e^{tx_1}])^n = (e^0(1-p) + e^tp)^n \\
&= (1-p + e^tp)^n \\
&\therefore \text{Binomial}
\end{aligned}$$

$$\begin{aligned}
M_{S_n}(t) &= E[e^{tS_n}] \\
&= \sum_{k=0}^n e^{tk} \cdot f_X(x) = \sum_k e^{tk} \cdot \binom{n}{k} p^k (1-p)^{n-k} \\
&= (1-p + e^tp)^n
\end{aligned}$$

$$\begin{aligned}
E[S_n] &= E[X_1 + X_2 + X_3 + \dots + X_n] \\
&= E[X_1] + E[X_2] + \dots + E[X_n] \\
&= n \cdot E[X_1] = n \cdot p = np
\end{aligned}$$

$$\begin{aligned}
E[S_n] &= M'_{S_n}(t) \big|_{t=0} \\
E[S_n^2] &= M''_{S_n}(t) \big|_{t=0}
\end{aligned}$$

$$\begin{aligned}
\text{Var}(S_n) &= E[S_n^2] - (E[S_n])^2 \\
&= np + n^2 p^2 - np^2 - n^2 p^2 = np - np^2 = np(1-p)
\end{aligned}$$

⑧

2a (5 marks) The joint probability mass function of the discrete random variables X and Y are given by $p_{X,Y}(x,y) = \frac{1}{2^{x+y}}$, $x = 1, 2, \dots$ and $y = 1, 2, \dots$

- Find the expression for the marginal pmf $p_X(x)$ and $p_Y(y)$ and the conditional pmf $p_{X|Y}(x|y)$.
- Find $E[XY]$ and determine if the RV X and Y are independent.

$$(a) \quad p_{X,Y}(x,y) = \frac{1}{2^{x+y}}, \quad x=1, 2, \dots \text{ \& \& } y=1, 2, \dots$$

(i) Marginal PMF:

$$\begin{aligned}
p_X(x) &= \sum_y p_{X,Y}(x,y) \\
&= \sum_{y=1}^{\infty} \frac{1}{2^{x+y}} = \sum_y \frac{1}{2^x \cdot 2^y} = \frac{1}{2^x} \cdot \sum_y \frac{1}{2^y} = \frac{1}{2^x}
\end{aligned}$$

$$p_Y(y) = \sum_x p_{X,Y}(x,y) = \frac{1}{2^y}$$

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} = \frac{\frac{1}{2^{x+y}}}{\frac{1}{2^y}} = \frac{1}{2^x}$$

$$\begin{aligned}
(ii) \quad E[XY] &= E[X] \cdot E[Y] \quad \because X \text{ and } Y \text{ are independent} \\
&= \sum_x x \cdot \frac{1}{2^x} \cdot \sum_y y \cdot \frac{1}{2^y} \\
&= 2 \cdot 2 \\
&= 4
\end{aligned}$$

2b (5 marks) The joint pdf of random variables X and Y is given by $f_{X,Y}(x,y) = \lambda\mu e^{-\lambda x - \mu y}$, $x \geq 0, y \geq 0, \lambda > 0, \mu > 0$.

(a) Find the expression for the marginal pdf's $f_X(x)$ and $f_Y(y)$ and the joint CDF $F_{X,Y}(x,y)$

(b) Are the RV X and Y independent? Give reasons.

$$(b) f_{x,y}(x,y) = \lambda\mu e^{-\lambda x - \mu y}, \quad x \geq 0, y \geq 0, \lambda > 0, \mu > 0$$

$$\begin{aligned} f_X(x) &= \int_y f_{x,y}(x,y) dy \\ &= \int_{y=0}^{y=\infty} \lambda\mu e^{-\lambda x - \mu y} dy \\ &= \lambda\mu \cdot \int_0^{\infty} e^{-\lambda x} \cdot e^{-\mu y} dy \end{aligned}$$

$$= \lambda\mu e^{-\lambda x} \cdot \frac{-1}{\mu} (e^{-\mu y})_0^{\infty} = -\lambda e^{-\lambda x} (0 - 1) = \lambda e^{-\lambda x}$$

$$f_Y(y) = \int_x f_{x,y}(x,y) dx = \mu e^{-\mu y}$$

$$F_{x,y}(x,y) = \int_0^{\infty} \int_0^{\infty} f_{x,y}(x,y) dx dy$$

$$= (1 - e^{-\lambda x})(1 - e^{-\mu y})$$

$$= F_X(x) \cdot F_Y(y)$$

$\therefore X$ and Y are independent.

2023 :

①

1. Consider a random variable X with the following pdf. Find mean and variance for X .

$$f_X(x) = \begin{cases} 0.5\lambda e^{-\lambda x}, & x \geq 0 \\ 0.5\lambda e^{\lambda x}, & x < 0 \end{cases}$$

$$E[X] = \int_{-\infty}^0 0.5\lambda e^{\lambda x} \cdot x \, dx + \int_0^{\infty} 0.5\lambda e^{-\lambda x} \cdot x \, dx$$

$$= \frac{\lambda}{2} \left(\int_{-\infty}^0 x e^{\lambda x} \, dx + \int_0^{\infty} x e^{-\lambda x} \, dx \right)$$

$$= \frac{\lambda}{2} \left(\left[\frac{1}{\lambda} x e^{\lambda x} - \frac{1}{\lambda^2} e^{\lambda x} \right]_{-\infty}^0 + \left[-\frac{1}{\lambda} x e^{-\lambda x} - \frac{1}{\lambda^2} e^{-\lambda x} \right]_0^{\infty} \right)$$

$$= \frac{\lambda}{2} \left(\frac{-1}{\lambda^2} - \left(+\frac{1}{\lambda^2} \right) \right) = 0$$

$$\therefore E[X] = 0$$

$$E[X^2] = \int_{-\infty}^0 0.5\lambda e^{\lambda x} \cdot x^2 \, dx + \int_0^{\infty} 0.5\lambda e^{-\lambda x} \cdot x^2 \, dx$$

$$= \frac{\lambda}{2} \left(\int_{-\infty}^0 x^2 e^{\lambda x} \, dx + \int_0^{\infty} x^2 e^{-\lambda x} \, dx \right)$$

$$= \frac{2}{\lambda^2}$$

$$\text{Var}(X) = \frac{2}{\lambda^2} - 0 \quad \therefore \text{Var}(X) = \frac{2}{\lambda^2}$$

②

2. Let X be a Uniform $U[0, 1]$ random variable. Let $Y = e^{2X}$. Find pdf and cdf of Y .

$$F_X(x) = P(X \leq x)$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(e^{2X} \leq y)$$

$$= P(2X \leq \ln y)$$

$$= P\left(X \leq \frac{\ln y}{2}\right)$$

$$= F_X\left(\frac{\ln y}{2}\right)$$

$$f_X(x) = \begin{cases} 1-0 & \text{if } x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$F_X(x) = \frac{x-0}{1-0} = x, \quad x \in [0, 1]$$

$$F_Y(y) = F_X\left(\frac{\ln y}{2}\right) = \frac{\ln y}{2} \quad \text{for } 1 \leq y \leq e^2$$

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} \frac{\ln y}{2} = \frac{1}{2y} \quad \text{for } 1 \leq y \leq e^2$$

③

1. Let $Y = aX^2 + b$ where X is a continuous random variable. Derive the expression for the CDF and pdf of Y in terms of the pdf of X .

$$F_Y(y) = P(Y \leq y)$$

$$= P(aX^2 + b \leq y) = P(aX^2 \leq y - b)$$

$$\begin{aligned}
&= P\left(X^2 \leq \frac{y-b}{a}\right) \\
&= P\left(|X| \leq \sqrt{\frac{y-b}{a}}\right) \\
F_Y(y) &= F_X\left(\sqrt{\frac{y-b}{a}}\right) - F_X\left(-\sqrt{\frac{y-b}{a}}\right) \quad [\text{for } a > 0] \\
&= \int_{-\sqrt{\frac{y-b}{a}}}^{\sqrt{\frac{y-b}{a}}} f_X(x) dx \quad \& \quad 1 - \int_{-\sqrt{\frac{y-b}{a}}}^{\sqrt{\frac{y-b}{a}}} f_X(x) dx \quad (\text{if } a < 0)
\end{aligned}$$

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

⑦

2. Let X be a uniform random variable with support $[a, b]$, . Let Y be Poisson random variable with parameter λ . Derive the expression for the mean and variance of X and Y .

$$X \rightsquigarrow U[a, b]$$

$$Y \rightsquigarrow P(\lambda)$$

$$(a) \quad f_X(x) = \frac{1}{b-a} \quad \text{if } x \in [a, b], \quad 0 \text{ otherwise}$$

$$\begin{aligned}
E[X] &= \int_a^b x \cdot \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \left(\frac{x^2}{2} \right)_a^b \\
&= \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} \\
&= \frac{a+b}{2}
\end{aligned}$$

$$\begin{aligned}
E[X^2] &= \int_a^b x^2 \cdot \frac{1}{b-a} dx \\
&= \frac{1}{b-a} \left(\frac{x^3}{3} \right)_a^b \\
&= \frac{a^3 + ab^2 + b^3}{3}
\end{aligned}$$

$$\Rightarrow \text{Var}[X] = \frac{(b-a)^2}{12}$$

$$(b) \quad f_X(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad \text{for } x = 1, 2, \dots$$

$$E[X] = \sum_{i=0}^{\infty} x \cdot f_X(x) = \lambda$$

$$E[X^2] = \sum_{i=0}^{\infty} x^2 \cdot f_X(x) = \lambda^2 + \lambda \Rightarrow \text{Var}[X] = \lambda$$

2024

①

Q1: Let X be a continuous random variable with distribution $F_X(\cdot)$ and density $f_X(x)$. Find the probability density and cumulative distribution for $Y = X^2 + 4$. [Anush]

$$\begin{aligned}
 F_Y(y) &= P(Y \leq y) = P(X^2 + 4 \leq y) \\
 &= P(X^2 \leq y - 4) \\
 &= P(|X| \leq \sqrt{y-4}) \quad \text{for } y > 4 \\
 &= P(-\sqrt{y-4} \leq X \leq \sqrt{y-4}) \\
 &= F_X(\sqrt{y-4}) - F_X(-\sqrt{y-4}) \\
 &= \int_{-\sqrt{y-4}}^{\sqrt{y-4}} f_X(x) dx
 \end{aligned}$$

$$F_Y(y) = \int_{-\sqrt{y-4}}^{\sqrt{y-4}} f_X(x) dx \quad \text{for } y > 4 \quad \& \quad 0 \text{ otherwise}$$

$$\begin{aligned}
 f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} \int_{-\sqrt{y-4}}^{\sqrt{y-4}} f_X(x) dx \\
 &= \frac{1}{2\sqrt{y-4}} \cdot f_X(\sqrt{y-4}) - \frac{-1}{2\sqrt{y-4}} f_X(-\sqrt{y-4})
 \end{aligned}$$

$$\therefore f_Y(y) = \frac{1}{2\sqrt{y-4}} (f_X(\sqrt{y-4}) + f_X(-\sqrt{y-4})) \quad \text{for } y > 4, \quad 0 \text{ otherwise}$$

②

Q2: Suppose X and Y are independent exponential random variables with parameter a and b respectively. Then find $P(X < Y)$. [Ronak]

$$\begin{aligned}
 P_{X,Y}(x, y) &= \underbrace{P_X(x)}_{\rightarrow ae^{-ax}, a > 0} \cdot \underbrace{P_Y(y)}_{\rightarrow be^{-by}, b > 0}
 \end{aligned}$$

$$P(X < Y)$$

$$= P(X < \forall y_i)$$

$$= \int_y P(X < y \cap Y = y)$$

$$= \int_{y=0}^{\infty} P(X < y | Y = y) f_Y(y) dy$$

$$= \int_{y=0}^{\infty} P(X < y) f_Y(y) dy$$

$$[P(X < y) = F_X(y) \quad \& \quad \text{Independence}]$$

$$= \int_0^{\infty} (1 - e^{-ay})(be^{-by}) dy$$

$$\begin{aligned}
&= \int_0^{\infty} (be^{-by} - be^{-(a+b)y}) dy \\
&= b \cdot \left[\int_0^{\infty} e^{-by} dy - \int_0^{\infty} e^{-(a+b)y} dy \right] \\
&= b \left(\frac{1}{b} - \frac{1}{a+b} \right) \\
&= \frac{a}{a+b}
\end{aligned}$$

- ③ Q3: We are given random variables $X_1, X_2, \dots, X_n$ with finite variances, and they are not necessarily independent. We need to find the variance of $S_n = \sum_{i=1}^n X_i$ and then check how the expression changes when the X_i 's are independent. [Gopal]

$$\begin{aligned}
S_n &= X_1 + X_2 + \dots + X_n \\
\text{Var}[S_n] &= \text{Var}[X_1 + X_2 + \dots + X_n] \\
&\swarrow \\
&= E[(X_1 + X_2 + \dots + X_n)^2] - (E[X_1 + X_2 + \dots + X_n])^2 \\
E[S_n^2] - (E[S_n])^2 &= E[(X_1 + X_2 + \dots + X_n)^2] - (E[X_1] + E[X_2] + \dots + E[X_n])^2 \\
&= E\left[\sum_i X_i^2 + 2 \sum_i \sum_{i+1}^n X_i X_j\right] - \left(\sum_i E[X_i]\right)^2 \\
&= E\left[\sum_i X_i^2\right] + 2 \sum_i \sum_{i+1}^n E[X_i X_j] - \left(\sum_i E[X_i]\right)^2 \\
&= \sum_i E[X_i^2] + 2 \cdot \sum_i \sum_{i+1}^n E[X_i X_j] - \sum_i (E[X_i])^2 - 2 \sum_i \sum_{i+1}^n E[X_i] \cdot E[X_j] \\
&= \sum_i E[X_i^2] - (E[X_i])^2 + \sum_i \sum_{i+1}^n E[X_i X_j] - E[X_i] \cdot E[X_j] \\
\text{Var}[S_n] &= \sum_i \text{Var}[X_i] + 2 \sum_i \sum_{i+1}^n \text{cov}(X_i, X_j)
\end{aligned}$$

$$\text{If } X_i \text{'s are independent} \Rightarrow \text{Var}[S_n] = \sum_{i=1}^n \text{Var}[X_i]$$

- ④ Q4: Let X be a random variable having Binomial distribution with parameters N and p where N is itself a random variable having Poisson distribution with mean λ . Find the probability mass function of the random variable X . Also find $E[X]$.

$$X \rightsquigarrow \text{Binomial}(N, p)$$

$$N \rightsquigarrow \text{Poisson}(\lambda)$$

$$P_X(x) = ? \quad \& \quad E[X] = ?$$

$$\begin{aligned}
P_N(n) &= \frac{e^{-\lambda} \lambda^n}{n!}, \quad \lambda > 0 \quad \text{for } n = 0, 1, 2, \dots \\
E[N] &= \lambda
\end{aligned}$$

$$E[X] = \sum x \cdot P_X(x)$$

$$P(X=x) = P(X=x \forall n)$$

$$= \sum_{n=x}^{\infty} P(X=x \cap N=n)$$

$$= \sum_{n=x}^{\infty} P(X=x | N=n) \cdot P(N=n)$$

$$= \sum_{n=x}^{\infty} P(X=x) \cdot P(N=n)$$

$$= \sum_{n=x}^{\infty} \binom{n}{x} p^x (1-p)^{n-x} \cdot \frac{e^{-\lambda} \lambda^n}{n!}$$

$$= \sum_{n=x}^{\infty} \frac{n!}{(n-x)! x!} \cdot p^x \cdot (1-p)^{n-x} \cdot \frac{e^{-\lambda} \lambda^n}{n!}$$

$$= \frac{p^x}{x!} \cdot e^{-\lambda} \sum_{n=x}^{\infty} \frac{(1-p)^{n-x}}{(n-x)!} \cdot \lambda^n$$

$$= \frac{p^x}{x!} \cdot e^{-\lambda} \cdot \frac{\lambda^x}{e^{-\lambda(1-p)}} \cdot \sum_{n=x}^{\infty} \frac{(\lambda(1-p))^{n-x}}{(n-x)!} \cdot e^{-\lambda(1-p)}$$

$$\therefore P(X=x) = e^{-\lambda p} \cdot \frac{(\lambda p)^x}{x!} \quad \underbrace{\sum_{n=x}^{\infty} \frac{(\lambda(1-p))^{n-x}}{(n-x)!} \cdot e^{-\lambda(1-p)}}_{\sum_{n=x} \text{PMF} = 1}$$

$$\therefore E[X] = \lambda p$$

⑤

Q5: Let X be a standard normal variable (Gaussian with zero mean and unit variance). Let $Z = \sigma X + \mu$. Obtain the pdf and cdf of Z . [Kavin]

$$X \sim N(0, 1)$$

$$Z = \sigma X + \mu$$

$$F_Z(z) = P(Z \leq z)$$

$$= P(\sigma X + \mu \leq z)$$

$$= P\left(X \leq \frac{z-\mu}{\sigma}\right)$$

$$= F_X\left(\frac{z-\mu}{\sigma}\right)$$

$$\therefore F_Z(z) = \Phi\left(\frac{z-\mu}{\sigma}\right)$$

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz} \Phi\left(\frac{z-\mu}{\sigma}\right)$$

$$= \frac{1}{\sigma} f_X\left(\frac{z-\mu}{\sigma}\right)$$

$$\therefore f_Z(z) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

$$f_X(z) = \frac{d}{dz} F_X(z) = \frac{d}{dz} \Phi(z)$$

⑥

Q1: Suppose U_1 and U_2 are independent uniform random variables on the segments $[-1, 1]$ and $[0, 1]$ respectively. Let $Z = U_1 + U_2$. Derive an expression for the pdf and cdf of Z .

$$\left. \begin{array}{l} U_1 \rightarrow [-1, 1] \\ U_2 \rightarrow [0, 1] \end{array} \right\} \text{Independent}$$

$$Z = U_1 + U_2$$

$$X = U_1 \quad \& \quad Y = U_2$$

$$f_z(z) = \int_{-\infty}^{\infty} f_x(z) \cdot f_y(z-x) \cdot dx = \int_{-\infty}^{\infty} f_y(y) \cdot f_x(z-y) \cdot dy$$

$$f_z(z) = \int_{-\infty}^{\infty} f_y(y) \cdot f_x(z-y) \cdot dy$$

$$= \int_0^1 f_y(y) \cdot f_x(z-y) \cdot dy$$

$$= \int_0^1 1 \cdot f_x(z-y) \cdot dy$$

$$0 \leq y \leq 1$$

$$z \geq z-y \geq z-1$$

$$z \geq x \geq z-1$$

$$\text{If } z > 2 \Rightarrow x \notin [-1, 1] \Rightarrow f_x(x) = 0$$

$$\text{If } z < -1 \Rightarrow x \notin [-1, 1] \Rightarrow f_x(x) = 0$$

$$f_y(y) = \begin{cases} 1, & \text{if } y \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$f_x(x) = \begin{cases} \frac{1}{2}, & \text{if } x \in [-1, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$\text{case - I: } z > 2$$

$$f_z(z) = \int_0^1 f_x(z-y) \cdot dy = 0$$

$$\text{case - II: } z < -1$$

$$f_z(z) = \int_0^1 f_x(z-y) \cdot dy = 0$$

$$\text{case - III: } -1 \leq z \leq 2$$

$$\left\{ \begin{array}{l} -1 \leq z-y \leq 1 \\ -1 \leq x \leq 1 \quad \& \quad 0 \leq y \leq 1 \Rightarrow -1 \leq x+y \leq 2 \end{array} \right\}$$

$$\textcircled{a} \text{ if } z \in [-1, 0]$$

$$-2 \leq z-y \leq 0$$

$$\text{but } z-y \geq -1 \Rightarrow y \leq z+1$$

$$f_z(z) = \int_0^{z+1} f_x(z-y) \cdot dy + \int_{z+1}^1 f_x(z-y) \cdot dy$$

$$= \int_0^{z+1} f_x(z-y) \cdot dy = \int_0^{z+1} \frac{1}{2} \cdot dy = \frac{1}{2} (z+1)$$

⑥ if $z \in [0, 1]$

$$-1 \leq z-y \leq 1$$

$$\begin{aligned} f_z(z) &= \int_0^1 f_x(z-y) dy \\ &= \int_0^1 \frac{1}{2} dy = \frac{1}{2} \end{aligned}$$

⑦ If $z \in [1, 2]$

$$0 \leq z-y \leq 2$$

But $z-y \leq 1 \Rightarrow y \geq z-1$

$$\begin{aligned} f_z(z) &= \int_0^{z-1} f_x(z-y) dy + \int_{z-1}^1 f_x(z-y) dy \\ &= \int_{z-1}^1 f_x(z-y) dy \\ &= \int_{z-1}^1 \frac{1}{2} dy = \frac{1}{2} (2-z) \end{aligned}$$

$$\therefore f_z(z) = \begin{cases} \frac{1}{2}(z+1), & z \in [-1, 0] \\ \frac{1}{2}, & z \in [0, 1] \\ \frac{1}{2}(2-z), & z \in [1, 2] \\ 0, & \text{otherwise} \end{cases}$$

$$f_z(z) = \int f_z(z)$$

⑦

Q2: Let X and Y be jointly continuous random variables with joint probability density function (PDF):

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-(2x+3y)} & \text{if } x, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

We are required to solve the following:

- (a) Find $E[X]$ and $E[Y]$.
- (b) Are X and Y independent? Justify.
- (c) Find $E[Y|X > 2]$.
- (d) Find $P(X > Y)$.

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-(2x+3y)}, & \text{if } x, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

(a)

$$\begin{aligned} f_X(x) &= \int_0^{\infty} f_{X,Y}(x,y) dy \\ &= \int_0^{\infty} 6e^{-2x} \cdot e^{-3y} dy \\ &= 6e^{-2x} \int_0^{\infty} e^{-3y} dy \\ &= 6e^{-2x} \cdot \frac{-1}{3} \cdot (e^{-3y})_0^{\infty} \\ &= -2e^{-2x} \cdot (0-1) \end{aligned}$$

$$\therefore f_X(x) = 2e^{-2x}$$

$$\begin{aligned} f_Y(y) &= \int_0^{\infty} f_{X,Y}(x,y) dx \\ &= \int_0^{\infty} 6e^{-2x} \cdot e^{-3y} dx \\ &= 6e^{-3y} \int_0^{\infty} e^{-2x} dx \end{aligned}$$

$$\begin{aligned} &= 6e^{-3y} \cdot \frac{-1}{2} \cdot (e^{-2x})_0^{\infty} \\ &= -3e^{-3y} \cdot (0-1) \end{aligned}$$

$$\therefore f_Y(y) = 3e^{-3y}$$

$$(b) f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$$

$\therefore$ Independent

$$\begin{aligned}
 (c) \quad E[Y|X > 2] \\
 &= E[Y] \\
 &= 1/3
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad P(X > Y) \\
 &= P(X > \forall y) \\
 &= P(X > y \cap Y = y) \\
 &= \int_y P(X > y \cap Y = y) dy \\
 &= \int_y P(X > y | Y = y) \cdot P(Y = y) dy \\
 &= \int_y P(X > y) \cdot P(Y = y) dy \\
 &= \int_0^{\infty} e^{-2y} \cdot f_Y(y) dy \\
 &= \int_0^{\infty} e^{-2y} \cdot 3e^{-3y} dy = 3/5
 \end{aligned}$$

$$\left(\begin{aligned} P(X > y) &= 1 - P(X \leq y) \\ &= 1 - (1 - e^{-2y}) \\ &= e^{-2y} \end{aligned} \right)$$

- ① Q1: Let $Z = X_1 + X_2 + \dots + X_N$, where X_i are i.i.d. random variables and N is a positive discrete random variable. Prove that:

$$M_Z(t) = M_N(\log M_X(t)).$$

$$Z = X_1 + X_2 + \dots + X_N$$

$$X_i : \text{i.i.d. R.V}$$

$$N : \text{+ve discrete R.V}$$

To prove :

$$M_Z(t) = M_N(\log M_X(t))$$

$$\begin{aligned} \text{LHS : } M_Z(t) &= E[e^{tZ}] \\ &= E_N[E[e^{tZ} | N]] \\ &= E_N[E[e^{t \sum X_i} | N]] \\ &= E_N[E[\prod e^{tX_i} | N]] \\ &= E_N[\prod E[e^{tX_i} | N]] \\ &= E_N[(E[e^{tX_i}])^N] \\ \therefore M_Z(t) &= E_N[(M_X(t))^N] \end{aligned}$$

$$M_N(t) = E[e^{tN}]$$

$$\text{If } u = \ln(M_X(t))$$

$$\Rightarrow M_N(u) = E[(M_X(t))^N]$$

$$M_Z(t) = M_N(u) = M_N(\ln(M_X(t)))$$

$$M_Z(t) = M_N(\ln(M_X(t)))$$

$$\begin{aligned} E[Z | N=n] &= E[X_1 + X_2 + \dots + X_n] \\ &= E[X_1] + E[X_2] + \dots + E[X_n] \\ &= n E[X] \end{aligned}$$

$$E[Z | N] = N \cdot E[X]$$

$$\begin{aligned} E[Z] &= \sum_n E[Z | N=n] \cdot P_N(n) \\ &= E[X] \cdot E[N] \end{aligned}$$

(OR)

$$\begin{aligned} E[Z] &= E[E[Z | N]] = E[N \cdot E[X]] \\ &= E[N] \cdot E[X] \end{aligned}$$

- ② Q2: Let $Y = e^X$, where $X \sim \mathcal{N}(\mu, \sigma^2)$. Obtain the pdf of Y .

$$F_X(x) = P(X \leq x)$$

$$F_Y(y) = P(Y \leq y)$$

$$= P(e^X \leq y)$$

$$= P(X \leq \ln(y))$$

$$= F_X(\ln(y)) \quad , \quad y > 0$$

$$\frac{d}{dy} F_Y(y) = \frac{d}{dy} (F_X(\ln(y)))$$

$$f_Y(y) = \frac{1}{y} \cdot f_X(\ln(y))$$

$$\therefore f_Y(y) = \frac{1}{y} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(\ln(y) - \mu)^2}{2\sigma^2}} \quad , \quad y > 0$$

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

- ⑤ Q5: Let X and Y be independent random variables with common distribution function F .

(a) PDF of $Z_1 = \max(X, Y)$

(b) PDF of $Z_2 = \min(X, Y)$

Sol:

$$\begin{aligned}
 (a) \quad F_{Z_1}(z_1) &= P(Z \leq z_1) \\
 &= P(\max(X, Y) \leq z_1) \\
 &= P(X \leq z_1 \text{ and } Y \leq z_1) \\
 &= P(X \leq z_1) \cdot P(Y \leq z_1) \\
 &= F_X(z_1) \cdot F_Y(z_1)
 \end{aligned}$$

$$F_{Z_1}(z_1) = F(z_1)^2$$

$$\begin{aligned}
 \frac{d}{dz_1} F_{Z_1}(z_1) &= \frac{d}{dz_1} F(z_1)^2 \\
 f_{Z_1}(z_1) &= 2 \cdot F(z_1) \cdot f(z_1)
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad F_{Z_2}(z_2) &= P(Z \leq z_2) \\
 &= P(\min(X, Y) \leq z_2) \\
 &= 1 - P(\min(X, Y) > z_2) \\
 &= 1 - P(X > z_2 \cap Y > z_2) \\
 &= 1 - P(X > z_2) \cdot P(Y > z_2) \\
 &= 1 - (1 - P(X \leq z_2)) \cdot (1 - P(Y \leq z_2)) \\
 &= 1 - (1 - F_X(z_2))(1 - F_Y(z_2)) \\
 &= 1 - (1 - F_X(z_2) - F_Y(z_2) + F_X(z_2) \cdot F_Y(z_2)) \\
 &= F_X(z_2) + F_Y(z_2) - F_X(z_2) \cdot F_Y(z_2) \\
 &= 2F(z_2) - F(z_2)^2
 \end{aligned}$$

$$\begin{aligned}
 f_{Z_2}(z_2) &= \frac{d}{dz_2} F_{Z_2}(z_2) = \frac{d}{dz_2} (2F(z_2) - F(z_2)^2) \\
 &= 2f_{Z_2}(z_2) - 2F(z_2) \cdot f_{Z_2}(z_2)
 \end{aligned}$$

⑩

$$f_X(x) = \begin{cases} \lambda e^{-\lambda(x-\mu)}, & \text{if } x \geq \mu \\ 0, & \text{otherwise} \end{cases}$$

(a) $E[e^{tx}] = ?$

$$= \int_{\mu}^{\infty} e^{tx} \cdot \lambda e^{-\lambda(x-\mu)} dx$$

$$= \lambda \int_{\mu}^{\infty} e^{tx} e^{-\lambda x} \cdot e^{\lambda \mu} dx$$

$$= \lambda \cdot e^{\lambda \mu} \int_{\mu}^{\infty} e^{(t-\lambda)x} dx$$

$$= \lambda \cdot e^{\lambda \mu} \cdot \frac{1}{t-\lambda} \cdot \left(e^{(t-\lambda)x} \right)_{\mu}^{\infty}$$

$$= \lambda \cdot e^{\lambda \mu} \cdot \frac{1}{\lambda-t} (0 - (-1)) , \text{ for } t < \lambda$$

$$= \frac{\lambda e^{\lambda \mu}}{\lambda-t} , \text{ for } t < \lambda \quad \& \quad 0 \text{ if } \lambda < t$$

$$(b) E[X] = M'_x(0)$$

$$M'_x(t) = \frac{d}{dt} \lambda e^{\mu t} (\lambda - t)^{-1} = \frac{\mu \cdot \lambda \cdot e^{\mu t}}{\lambda - t} + \frac{\lambda e^{\mu t}}{(\lambda - t)^2}$$

$$E[X] = M'_x(0) = \frac{\lambda \mu}{\lambda} + \frac{\lambda}{\lambda^2} = \mu + \frac{1}{\lambda}$$

$$E[X^2] = M''_x(0)$$

$$M''_x(t) = \frac{\mu^2 \lambda e^{\mu t}}{\lambda - t} + 2 \cdot \frac{\mu \lambda \cdot e^{\mu t}}{(\lambda - t)^2} + \frac{2 \lambda e^{\mu t}}{(\lambda - t)^3}$$

$$\Rightarrow M''_x(0) = \frac{\mu^2 \lambda}{\lambda} + 2 \cdot \frac{\mu \lambda}{\lambda^2} + \frac{2 \lambda}{\lambda^2}$$

$$= \left(\mu + \frac{1}{\lambda} \right)^2 + \frac{1}{\lambda^2}$$

$$\begin{aligned} \text{Var}[X] &= E[X^2] - (E[X])^2 = \left(\mu + \frac{1}{\lambda} \right)^2 + \frac{1}{\lambda^2} - \left(\mu + \frac{1}{\lambda} \right)^2 \\ &= \frac{1}{\lambda^2} \end{aligned}$$

2023 end

② $N(\mu, \sigma^2)$

$$M_X(t) = E[e^{tx}]$$

$$= \int e^{tx} \cdot p_X(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \cdot \int_{-\infty}^{\infty} e^{tx} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{2\sigma^2 tx - (x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t\sigma^2-\mu)^2}{2\sigma^2}} \cdot e^{\frac{\sigma^2 t^2}{2} + \mu t} dx$$

$$= \frac{e^{\frac{\sigma^2 t^2}{2} + \mu t}}{\sigma\sqrt{2\pi}} \cdot \sigma\sqrt{2\pi}$$

$$= e^{\frac{\sigma^2 t^2}{2} + \mu t}$$

$$2\sigma^2 tx - x^2 + 2\mu x - \mu^2$$

$$- \left(x - (t\sigma^2 + \mu) \right)^2 + t^2\sigma^4 + 2\mu t\sigma^2$$

$$u = \frac{x - t\sigma^2 - \mu}{\sigma}$$

$$du = dx$$

$$M'(t) = (\sigma^2 t + \mu) \cdot e^{\frac{\sigma^2 t^2}{2} + \mu t}$$

$$M'(0) = \mu$$

$$M''(t) = (\sigma^2 t + \mu)^2 e^{\frac{\sigma^2 t^2}{2} + \mu t} + e^{\frac{\sigma^2 t^2}{2} + \mu t} \cdot \sigma^2$$

$$M''(0) = \mu^2 + \sigma^2$$

②

Question 2

If A and B are exponential random variables with parameters a and b respectively, then prove $P(A < B) = E[e^{-bA}]$. Further show that this is equal to $a/(a+b)$.

(4 marks)

$$\begin{aligned}
 P(A < B) &= P(A < b \ \forall b) \\
 &= \int_x P(B > a \cap A = x) \\
 &= \int_x P(B > a | A = x) \cdot f_A(x) \, dx \quad \left(\begin{array}{l} A \rightarrow ae^{-ax} = f_A(x) \\ B \rightarrow be^{-bx} = f_B(x) \end{array} \right) \\
 &= \int_x P(B > a) \cdot f_A(x) \, dx = \int_x e^{-bx} \cdot f_A(x) \, dx \\
 &= E[e^{-bA}]
 \end{aligned}$$

$$E[e^{-bA}] \rightarrow \int_x e^{-bx} \cdot ae^{-ax} \, dx = \int_0^\infty a \cdot e^{-(a+b)x} \, dx = \frac{a}{a+b}$$

③

Question 3

Derive the expression for the Moment generating function of the following random variables

1. Standard Normal with mean 0 and variance 1 (3 marks)
2. Poisson random variable with parameter λ (3 marks)

$$M_X(t) = E[e^{tx}]$$

(i) $N(0, 1)$

$$M_X(t) = \int_x e^{tx} \cdot p_X(x) \, dx$$

$$= \int_x e^{tx} \cdot \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \, dx$$

$$= \int_x e^{tx} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \int_x e^{tx - \frac{x^2}{2}} \, dx$$

$$= \frac{e^{t^2/2}}{\sqrt{2\pi}} \int_x e^{-\frac{(x-t)^2}{2}} \, dx$$

$$= \frac{e^{t^2/2}}{\sqrt{2\pi}} \cdot \sqrt{2\pi}$$

$$u = \frac{x-t}{\sqrt{2}}$$

$$= -\frac{x^2 - 2tx + t^2}{2} + \frac{t^2}{2}$$

$$= -\frac{(x-t)^2}{2} + \frac{t^2}{2}$$

$$\therefore M_X(t) = e^{t^2/2}$$

$$(ii) P_X(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$M_X(t) = E[e^{tx}]$$

$$= \sum e^{tx} \cdot P_X(x) \, dx$$

$$= \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \cdot e^{-\lambda}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$\therefore M[e^{tx}] = e^{\lambda e^t - \lambda}$$

④

Question 4

Let X be an exponential random variable with parameter 1. Find

1. Conditional PDF and CDF given $X > 1$
2. $E[X|X > 1]$

$$f_X(x) = e^{-x} \text{ for } x > 0, 0 \text{ otherwise}$$

$$(i) P(X=x|X>1) = f_{X|X>1}(x)$$

$$X > 1 : P(A) = \int_1^{\infty} e^{-x} \, dx = \frac{1}{e}$$

$$f_{X|X>1}(x) = \frac{f_X(x)}{P_e} = \frac{e^{-x}}{e^{-1}} = e^{1-x} \text{ for } x > 1$$

$$F_{X|X>1}(x) = \frac{F_X(x) - F_X(1)}{P(A)} = 1 - e^{1-x}, \text{ for } x > 1$$

$$(ii) E[X|X>1] = \int_1^{\infty} x \cdot f_{X|X>1}(x) \, dx$$

$$= \int_1^{\infty} x \cdot e^{1-x} \, dx$$

$$= e \int_1^{\infty} x \cdot e^{-x} \, dx = e \cdot \frac{2}{e} = 2$$

⑤

Question 5

Let X and Y be independent and identically distributed discrete random variables taking values on positive integers. Their pmf is $p(x) = C2^{-x}$ for $x \geq 1$. Find

1. The value of C that makes it a valid pmf
2. $P(\min\{X, Y\} \leq x)$
3. $P(X \text{ divides } Y)$

X and $Y \leftarrow \text{i.i.d}$

$$P(x) = c \cdot 2^{-x} \text{ for } x \geq 1$$

$$\begin{aligned} \text{(a)} \quad \int_1^{\infty} c \cdot 2^{-x} dx &\Rightarrow c \cdot \int_1^{\infty} 2^{-x} dx = 1 \\ &\Rightarrow c \cdot \left(\frac{-2^{-x}}{\ln(2)} \right) \Big|_1^{\infty} = 1 \\ &\Rightarrow \frac{c}{2 \ln 2} = 1 \Rightarrow c = 2 \ln 2 \end{aligned} \quad \left. \vphantom{\int_1^{\infty}} \right\} \text{ If Continuous}$$

$$\begin{aligned} \sum_{x=1}^{\infty} c \cdot 2^{-x} = 1 &\Rightarrow c \left(\frac{1}{2} + \frac{1}{4} + \dots \right) = 1 \\ &\Rightarrow \underline{\underline{c = 1}} \end{aligned} \quad \left. \vphantom{\sum_{x=1}^{\infty}} \right\} \text{ Discrete}$$

$$\therefore P(x) = 2^{-x} \quad \forall x \geq 1$$

$$\begin{aligned} \text{(ii)} \quad P(\min\{X, Y\} \leq x) &= 1 - P(\min(X, Y) > x) \\ &= 1 - P(X > x \cap Y > x) \\ &= 1 - P(X > x) \cdot P(Y > x) \\ &= 1 - (1 - P(X \leq x)) \cdot (1 - P(Y \leq x)) \\ &= 1 - (1 - F_X(x))(1 - F_Y(x)) \\ &= 1 - (1 - F_X(x) - F_Y(x) + F_X(x) \cdot F_Y(x)) \\ &= F_X(x) + F_Y(x) - F_X(x) \cdot F_Y(x) \\ &= 2F(x) - F(x)^2 \end{aligned}$$

$$\begin{aligned} F_X(x) = P(X \leq x) &= \sum_{x=-\infty}^x 2^{-x} = 1 - 2^{-x} \\ \therefore F(x) &= 1 - 2^{-x} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P(X \text{ divides } Y) &= \sum \sum P(X=n \cap Y=kn) \\ &= \sum 2^{-n} \sum 2^{-kn} \\ &= \sum 2^{-n} \cdot \frac{2^{-n}}{1 - 2^{-n}} \\ &= \sum \frac{2^{-n}}{2^n - 1} \\ &= \sum \frac{1}{2^n - 1} - \frac{1}{2^n} \\ &= \sum_{n=2}^{\infty} \frac{1}{2^n - 1} \end{aligned}$$

⑥

Question 1

Let X and Y be independent Poisson random variables with parameters λ_1 and λ_2 . Let $Z = X + Y$. Then find the pmf of Z .

$$\left. \begin{aligned} P_X(x) &= \frac{e^{-\lambda_1} \lambda_1^x}{x!} \\ P_Y(y) &= \frac{e^{-\lambda_2} \lambda_2^y}{y!} \end{aligned} \right\} Z = X + Y$$

Method - I :

$$M_Z(t) = E[e^{tz}]$$

$$= E[e^{tx+ty}]$$

$$= E[e^{tx} \cdot e^{ty}]$$

$$= E[e^{tx}] \cdot E[e^{ty}]$$

$$= e^{\lambda_1(t-1)} \cdot e^{\lambda_2(t-1)}$$

$$= e^{(\lambda_1+\lambda_2)(t-1)}$$

$$\begin{aligned} E[e^{tz}] &= \sum_x e^{tx} \cdot \frac{e^{-\lambda_1} \lambda_1^x}{x!} \\ &= e^{-\lambda_1} \cdot \sum_x \frac{(e^t \lambda_1)^x}{x!} \\ &= e^{-\lambda_1} \cdot e^{t\lambda_1} \\ &= e^{\lambda_1(t-1)} \end{aligned}$$

$$Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

$$\therefore P_Z(z) = \frac{e^{-(\lambda_1+\lambda_2)} (\lambda_1+\lambda_2)^z}{z!} \quad (\lambda_1+\lambda_2 > 0)$$

Method - II :

$$P(Z = X + Y = k) = \sum_{i=0}^k P(X+Y=k \cap X=i)$$

$$= \sum_{i=0}^k P(X=i \cap Y=k-i)$$

$$= \sum_i P(X=i) \cdot P(Y=k-i)$$

$$= \sum_i \frac{e^{-\lambda_1} \lambda_1^i}{i!} \cdot \frac{e^{-\lambda_2} \lambda_2^{(k-i)}}{(k-i)!}$$

$$= \frac{(\lambda_1 + \lambda_2)^k}{k!} \cdot e^{-(\lambda_1 + \lambda_2)}$$

$$f_X(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

$$\therefore p_X(z) = \sum_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x)$$

⑨ Question 4

Let X have a Poisson distribution with parameter Λ , where Λ is an exponential random variable with parameter μ . Show that X has a geometric distribution.

$$P_X(x) = \frac{e^{-\Lambda} \cdot \Lambda^x}{x!}, \quad \Lambda > 0$$

$$f_\Lambda(\lambda) = \mu e^{-\mu\lambda}, \quad \mu > 0$$

$$\text{Now, } P(X=x) = P(X=x \cap \Lambda = \forall \lambda)$$

$$= \int_{\lambda} P(X=x \cap \Lambda=\lambda) d\lambda$$

$$= \int_{\lambda=0}^{\infty} P(X=x | \Lambda=\lambda) \cdot f_\Lambda(\lambda) d\lambda$$

$$= \int_{\lambda=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} \cdot f_\Lambda(\lambda) d\lambda, \quad x \geq 0$$

$$= \int_{\lambda=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} \cdot \mu e^{-\mu\lambda} d\lambda$$

$$= \frac{\mu}{x!} \cdot \int_0^{\infty} e^{-\lambda} \cdot \lambda^x \cdot e^{-\mu\lambda} d\lambda$$

$$= \frac{\mu}{x!} \int_0^{\infty} \lambda^x \cdot e^{-\lambda(\mu+1)} d\lambda$$

$$= \frac{\mu}{x!} \cdot \frac{x!}{(1+\mu)^{x+1}}$$

$$= \frac{1}{(1+\mu)^x} \cdot \frac{\mu}{\mu+1}$$

$$\therefore P(X=x) = \left(\frac{1}{1+\mu}\right)^x \cdot \left(1 - \frac{1}{1+\mu}\right)$$

$$\therefore X \sim \text{Geo}\left(\frac{1}{1+\mu}\right)$$

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$$\rightarrow Z = X + Y$$

 $X, Y, Z \rightarrow \text{discrete RV}$

 if X and Y are dependent:

$$P_Z(z) = \sum_{\substack{x,y \\ (x+y=z)}} P_{XY}(x,y)$$

 If X and Y are independent,

$$\begin{aligned} P_Z(z) &= \sum_{\substack{x,y \\ (x+y=z)}} P_X(x) \cdot P_Y(y) \\ &= \sum_x P_X(x) \cdot P_Y(z-x) \end{aligned}$$

 When X and Y are dependent,

$$\begin{aligned} P_Z(z) &= \sum_{\substack{x,y \\ (x+y=z)}} P_{XY}(x,y) \\ &= \sum_{\substack{x,y \\ (x+y=z)}} P_X(x) \cdot P_{Y|X}(y|x) \\ &= \sum_x P_X(x) \cdot P_{Y|X}(z-x|x) \end{aligned}$$

 $X, Y, Z \rightarrow \text{continuous RV}$

 if X and Y are dependent:

$$f_Z(z) = \int_{\substack{x,y \\ (x+y=z)}} f_{XY}(x,y) dx dy$$

 If X and Y are independent,

$$\begin{aligned} f_Z(z) &= \int_{\substack{x,y \\ (x+y=z)}} f_X(x) \cdot f_Y(y) dx dy \\ &= \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x) dx \end{aligned}$$

 When X and Y are dependent,

$$\begin{aligned} f_Z(z) &= \int_{\substack{x,y \\ (x+y=z)}} f_{XY}(x,y) dx dy \\ &= \int_{\substack{x,y \\ (x+y=z)}} f_X(x) \cdot f_{Y|X}(y|x) dx \cdot dy \\ &= \int_{-\infty}^{\infty} f_X(x) \cdot f_{Y|X}(z-x|x) dx \end{aligned}$$

Refer: Convolution of box signals with itself

 ✓ H.W: X and $Y \rightarrow \text{independent \& Uniform}[0,1]$

 Find PDF and CDF of $Z = X + Y$

$$\text{Sol: } f_X(x) = \begin{cases} 1, & x \in [0,1] \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} 1, & y \in [0,1] \\ 0, & \text{otherwise} \end{cases}$$

$$\left. \begin{aligned} f_{XY}(x,y) &= f_X(x) \cdot f_Y(y) \\ &= 1, \text{ if } x \in [0,1] \text{ and } y \in [0,1] \\ &= 0, \text{ otherwise} \end{aligned} \right\}$$

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x) dx \\ &= \int_0^1 1 \cdot f_Y(z-x) dx \end{aligned} \quad z \in [0,2]$$

 If $z \in [0,1]$:

 $0 \leq z-x \leq 1$ (from diagram)

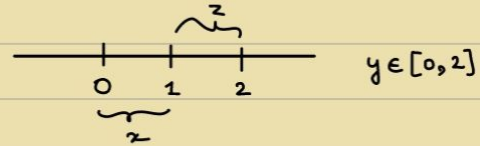
 But $0 \leq z-x \leq 1$ (from $y = z-x$)


$$\Rightarrow \int_0^1 1 \, dx = (x)_0^1 = 1$$

If $z \in [1, 2]$:

$$\therefore 0 \leq z-x \leq 2$$

$$\text{But } z-x \leq 1 \Rightarrow x \geq z-1$$



$$\Rightarrow \int_0^{z-1} f_y(z-x) \, dx + \int_{z-1}^1 f_y(z-x) \, dx$$

$$\Rightarrow \int_{z-1}^1 f_y(z-x) \, dx$$

$$\Rightarrow (x)_{z-1}^1 \Rightarrow 1 - (z-1) = 2-z$$

$$f_z(z) = \begin{cases} 1, & \text{if } z \in [0, 1] \\ 2-z, & \text{if } z \in [1, 2] \\ 0, & \text{otherwise} \end{cases}$$

$$F_z(z) = \int_{-\infty}^z f_z(z) \, dz$$

$$\text{If } z < 0 : F_z(z) = 0$$

$$\text{If } z \in [0, 1] :$$

$$\begin{aligned} F_z(z) &= \int_{-\infty}^0 f_z(z) \, dz + \int_0^z f_z(z) \, dz \\ &= \int_0^z 1 \, dz = z \end{aligned}$$

$$\text{If } z \in [1, 2] :$$

$$\begin{aligned} F_z(z) &= \int_{-\infty}^0 f_z(z) \, dz + \int_0^1 f_z(z) \, dz + \int_1^z f_z(z) \, dz \\ &= 1 + (2z - z^2/2)_1^z \\ &= 1 + 2z - z^2/2 - 2 + 1/2 \\ &= -z^2/2 + 2z - 1/2 \end{aligned}$$

$$\text{If } z > 2 :$$

$$\begin{aligned} F_z(z) &= \int_{-\infty}^0 f_z(z) \, dz + \int_0^1 f_z(z) \, dz + \int_1^2 f_z(z) \, dz + \int_2^z f_z(z) \, dz \\ &= 1 + (2z - z^2/2)_1^2 \end{aligned}$$

$$= 1 + 4 - 2 - 2 + \frac{1}{2}$$

$$= \frac{3}{2}$$

$$\therefore F_2(z) = \begin{cases} 0, & \text{if } z < 0 \\ z, & \text{if } z \in [0, 1] \\ -\frac{z^2}{2} + 2z - \frac{1}{2}, & \text{if } z \in [1, 2] \\ \frac{3}{2}, & \text{if } z > 2 \end{cases}$$

✓ H.W: X and $Y \rightarrow$ outcomes of independent rolls of dice
 $Z = X + Y$, find PDF and CDF of Z .

Sol: $P(X=x) = \frac{1}{6} \quad \forall x \in [1, 6], i \in \mathbb{N}$
 $P(Y=y) = \frac{1}{6} \quad \forall y \in [1, 6], i \in \mathbb{N}$

$$P_2(z) = \sum_{\substack{x, y \\ x+y=z}} P_x(x) \cdot P_y(y)$$

$$= \sum_x P_x(x) \cdot P_y(z-x)$$

$$= P_x(1) \cdot P_y(z-1) + P_x(2) \cdot P_y(z-2) + \dots + P_x(6) \cdot P_y(z-6)$$

$$= \frac{1}{6} (P_y(z-1) + P_y(z-2) + \dots + P_y(z-6))$$

$$\left. \begin{array}{l} 1 \leq x \leq 6 \\ 1 \leq y \leq 6 \end{array} \right\} \Rightarrow 2 \leq z \leq 12$$

If $z = 2 : P_2(z) = \frac{1}{6} (P_y(1)) = \frac{1}{36}$

If $z = 3 : P_2(z) = \frac{1}{6} (P_y(2) + P_y(1)) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6}) = \frac{1}{18}$

If $z = 4 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{1}{12}$

If $z = 5 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{1}{9}$

If $z = 6 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{5}{36}$

If $z = 7 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{1}{6}$

If $z = 8 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{5}{36}$

If $z = 9 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{1}{9}$

If $z = 10 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6} + \frac{1}{6}) = \frac{1}{12}$

If $z = 11 : P_2(z) = \frac{1}{6} (\frac{1}{6} + \frac{1}{6}) = \frac{1}{18}$

If $z = 12 : P_2(z) = \frac{1}{6} (\frac{1}{6}) = \frac{1}{36}$

$$P_2(z) = \begin{cases} 1/6, & \text{if } z = 7 \\ 5/36, & \text{if } z = 6 \text{ or } 8 \\ 1/9, & \text{if } z = 5 \text{ or } 9 \\ 1/12, & \text{if } z = 4 \text{ or } 10 \\ 1/18, & \text{if } z = 3 \text{ or } 11 \\ 1/36, & \text{if } z = 2 \text{ or } z = 12 \\ 0, & \text{otherwise} \end{cases}$$

$$F_2(z) = \sum_{i=2}^z f_z(z)$$

$$\text{If } z = 2 \Rightarrow F_2(z) = 1/36$$

$$\text{If } z = 3 \Rightarrow F_2(z) = 1/36 + 1/18 = 1/12$$

$$\text{If } z = 4 \Rightarrow F_2(z) = 1/12 + 1/12 = 1/6$$

$$\text{If } z = 5 \Rightarrow F_2(z) = 1/6 + 1/9 = 5/18$$

$$\text{If } z = 6 \Rightarrow F_2(z) = 5/18 + 5/36 = 5/12$$

$$\text{If } z = 7 \Rightarrow F_2(z) = 5/36 + 1/6 = 7/12$$

$$\text{If } z = 8 \Rightarrow F_2(z) = 7/12 + 5/36 = 13/18$$

$$\text{If } z = 9 \Rightarrow F_2(z) = 13/18 + 1/9 = 5/6$$

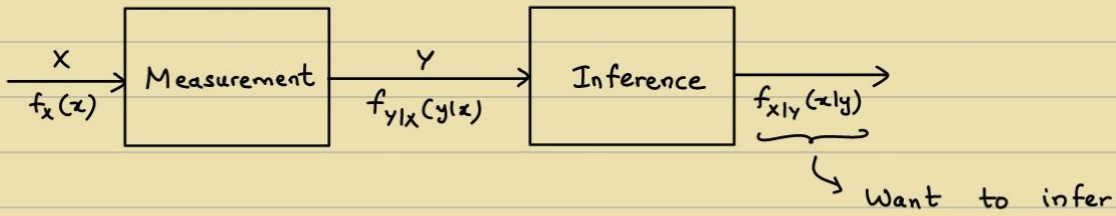
$$\text{If } z = 10 \Rightarrow F_2(z) = 5/6 + 1/12 = 11/12$$

$$\text{If } z = 11 \Rightarrow F_2(z) = 11/12 + 1/18 = 35/36$$

$$\text{If } z = 12 \Rightarrow F_2(z) = 35/36 + 1/36 = 1$$

$$F_2(z) = \begin{cases} 1/36, & \text{If } z = 2 \\ 1/12, & \text{If } z = 3 \\ 1/6, & \text{If } z = 4 \\ 5/18, & \text{If } z = 5 \\ 5/12, & \text{If } z = 6 \\ 7/12, & \text{If } z = 7 \\ 13/18, & \text{If } z = 8 \\ 5/6, & \text{If } z = 9 \\ 11/12, & \text{If } z = 10 \\ 35/36, & \text{If } z = 11 \\ 1, & \text{If } z = 12 \\ 0, & \text{Otherwise} \end{cases}$$

→ Inference Problem :



$X : f_X(x) \rightarrow$ Not observable
but of interest } $\rightarrow$ Can be calculation

$Y = y \rightarrow$ Observable

$f_{Y|X}(y|x) \rightarrow$ Given

$f_{X|Y}(x|y) \rightarrow$ Need to find

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_{Y|X}(y|x) \cdot f_X(x)}{\underbrace{f_Y(y)}} \rightarrow \int_x f_{Y|X}(y|t) \cdot f_X(t) dt$$

If Discrete,

$$P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)} = \frac{P_{Y|X}(y|x) \cdot P_X(x)}{P_Y(y)} = \frac{P_{Y|X}(y|x) \cdot P_X(x)}{\sum_i P_{Y|X}(y|i) \cdot P_X(i)}$$

Ex.

$Y \sim \text{Exp}(\lambda)$

$\lambda \rightarrow U[1, 1.5]$

$$f_\lambda(\lambda) = \begin{cases} 2, & \lambda \in [1, 1.5] \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0, \quad \lambda \geq 0$$

$$f_Y(y|\lambda = \lambda) = \lambda e^{-\lambda y}$$

$$\Rightarrow f_Y(y) = \int_{\forall \lambda} f(y|\lambda = \lambda) \cdot f_\lambda(\lambda) \cdot d\lambda$$

$$= \int_{\lambda} f_{Y|\lambda}(y|\lambda) \cdot f_\lambda(\lambda) \cdot d\lambda$$

$$= \int_1^{1.5} \lambda e^{-\lambda y} \cdot (2) d\lambda$$

$$= 2 \left(\frac{-\lambda}{y} e^{-\lambda y} - \frac{1}{y^2} e^{-\lambda y} \right) \Big|_1^{1.5}$$

$$\leftarrow f_{\lambda|Y}(\lambda|y) = \frac{f_{Y|\lambda}(y|\lambda) \cdot f_\lambda(\lambda)}{f_Y(y)}$$

$$\begin{pmatrix} D & I \\ + \lambda & e^{-y\lambda} \\ - 1 & -\frac{1}{y} e^{-\lambda y} \\ + 0 & \frac{1}{y^2} e^{-\lambda y} \end{pmatrix}$$

$$\begin{aligned}
 &= 2 \left(\frac{-\lambda}{y} e^{-\lambda y} - \frac{1}{y^2} e^{-\lambda y} \right)^{1.5} \\
 &= 2 \left[\left(\frac{-1.5}{y} e^{-1.5y} - \frac{1}{y^2} e^{-1.5y} \right) - \left(\frac{-1}{y} e^{-y} - e^{-y} \right) \right] \\
 \therefore f_Y(y) &= 2 \left(\frac{-3}{2y} e^{-3y/2} - \frac{1}{y^2} e^{-3y/2} + \frac{e^{-y}}{y} + e^{-y} \right)
 \end{aligned}$$

Note :

$$P(N=n | Y=y) = \frac{f_{Y|N}(y|n) \cdot P_N(n)}{f_Y(y)} = \frac{f_{Y|N}(y|n) \cdot P_N(n)}{\sum_t f_{Y|N}(y|t) \cdot P_N(t)}$$

$\searrow$
 $\sum_t f_{Y|N}(y|t) \cdot P_N(t)$

$$f(N=n | Y=y) = \frac{P(N=n | Y=y) \cdot f_Y(y)}{P_N(n)} = \frac{P(N=n | Y=y) \cdot f_Y(y)}{\int_{-\infty}^{\infty} P(N=n | Y=t) f_Y(t) dt}$$

$\searrow$
 $\int_t P(N=n | Y=t) \cdot f_Y(t) dt$

✓ H.W
(Q)

$X = 1$ with probability p
 $X = -1$ with probability $1-p$
 $N \sim N(0, 1)$

$Y = X + N$, Observe $Y = y$

Prove that

$$P(X=1 | Y=y) = \frac{pe^y}{pe^y + (1-p)e^{-y}}$$

As $y \rightarrow -\infty$, Prob. $\rightarrow 0$
 $y \rightarrow \infty$, Prob. $\rightarrow 1$

Sol :

$$P(X=1 | Y=y) = \frac{f_{Y|X}(y|1) \cdot P_X(1)}{f_Y(y)} = \frac{(p) \cdot f_{Y|X}(y|1)}{(p) \cdot f_{Y|X}(y|1) + (1-p) \cdot f_{Y|X}(y|-1)}$$

$\searrow$
 $f_{Y|X}(y|1) \cdot P_X(1) + f_{Y|X}(y|-1) \cdot P_X(-1)$

$$Y = N + X$$

$$\left[f_Y(y) = \int_{-\infty}^{\infty} P_X(x) \cdot f_N(y-x) dx \right] \leftarrow \text{Intuition}$$

$$\therefore f_{Y|X}(y|X=x) = P_X(x) \cdot f_N(y-x) = p^{(1+x)/2} \cdot (1-p)^{(1-x)/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-x)^2}{2}}$$

$$\therefore P(X=1 | Y=y) = \frac{(p) \cdot (p) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-1)^2}{2}}}{(p) \cdot (p) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-1)^2}{2}} + (1-p)(1-p) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(y+1)^2}{2}}}$$

$$= \frac{p \cdot e^{-\frac{(y-1)^2}{2}}}{p \cdot e^{-\frac{(y-1)^2}{2}} + (1-p) \cdot e^{-\frac{(y+1)^2}{2}}} = \frac{p}{p + (1-p) \cdot e^{-\frac{(y+1)^2}{2} - \frac{(y-1)^2}{2}}}$$

→ Variance of sum of R.V:

$$S_n = X_1 + X_2 + X_3 + \dots + X_n$$

$$\text{Var}[S_n] = ?$$

$$\begin{aligned}\text{Var}[S_n] &= E[(X_1 + X_2 + \dots + X_n)^2] - (E[X_1 + X_2 + \dots + X_n])^2 \\&= E[(X_1 + X_2 + \dots + X_n)^2] - (E[X_1] + E[X_2] + \dots + E[X_n])^2 \\&= E\left[\sum_{i=1}^n X_i^2 + 2 \sum_{i=1}^n \sum_{j=i+1}^n X_i X_j\right] - \left(\sum_{i=1}^n E[X_i]\right)^2 \\&= E\left[\sum_{i=1}^n X_i^2\right] + 2 \sum_{i=1}^n \sum_{j=i+1}^n E[X_i X_j] - \left(\sum_{i=1}^n E[X_i]\right)^2 \\&= \sum_{i=1}^n E[X_i^2] + 2 \cdot \sum_{i=1}^n \sum_{j=i+1}^n E[X_i X_j] - \sum_{i=1}^n (E[X_i])^2 - 2 \sum_{i=1}^n \sum_{j=i+1}^n E[X_i] \cdot E[X_j] \\&= \sum_{i=1}^n E[X_i^2] - (E[X_i])^2 + \sum_{i=1}^n \sum_{j=i+1}^n E[X_i X_j] - E[X_i] \cdot E[X_j]\end{aligned}$$

$$\text{Var}[S_n] = \sum_{i=1}^n \text{Var}[X_i] + 2 \sum_{i=1}^n \sum_{j=i+1}^n \text{cov}(X_i, X_j)$$

$$\text{If } X_i \text{'s are independent} \Rightarrow \text{Var}[S_n] = \sum_{i=1}^n \text{Var}[X_i]$$

$$\bullet \text{cov}(X, Y) = E[XY] - E[X] \cdot E[Y]$$

Properties : (Prove: H.W)

$$(1) \text{cov}(X, X) = \text{Var}[X] = E[(X - E[X])^2]$$

$$(2) \text{If } X \text{ and } Y \text{ are independent, } \text{cov}(X, Y) = 0$$

$$(3) \text{cov}(X, Y) = \text{cov}(Y, X)$$

$$(4) \text{cov}(aX, Y) = a \cdot \text{cov}(X, Y)$$

$$(5) \text{cov}(X + a, Y) = a \cdot \text{cov}(X, Y)$$

$$(6) \text{cov}(X + Z, Y) = \text{cov}(X, Y) + \text{cov}(Z, Y)$$

$$(7) \text{cov}\left(\sum_{i=1}^m a_i X_i, \sum_{j=1}^n b_j Y_j\right) = \sum_{i=1}^m \sum_{j=1}^n a_i b_j \text{cov}(X_i, Y_j)$$

$$\text{Ex. } Z = \sum a_i X_i$$

$$\text{Var}[Z] = \text{cov}(Z, Z)$$

$$= \text{cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^n a_j X_j\right)$$

$$= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{cov}(X_i, X_j)$$

$$= \sum_{i=1}^n a_i^2 \cdot \text{Var}[X_i] + \sum_{\substack{i, j \\ (i \neq j)}} a_i a_j \cdot \text{cov}(X_i, X_j)$$

Q) Let $\{X_i, i=1, 2, \dots, n\}$ be i.i.d

$$S_n = \frac{\sum_{i=1}^n X_i}{n}$$

$$\text{Var}[Z] = \sum_{i=1}^n \text{Var}[X_i] = n \cdot \text{Var}[X]$$

$$\begin{aligned}\text{Var}[S_n] &= \text{Var}\left[\frac{Z}{n}\right] = \frac{1}{n^2} \text{Var}[Z] \\ &= \frac{1}{n^2} \cdot n \cdot \text{Var}[X] \\ &= \frac{1}{n} \cdot \text{Var}[X]\end{aligned}$$

$$E[S_n] = E[X]$$

$$\text{Var}[S_n] = \frac{1}{n} \cdot \text{Var}[X]$$

— Properties :

$$(1) \text{cov}(X, Y) = E[XY] - E[X] \cdot E[Y]$$

$$\begin{aligned}\text{cov}(X, X) &= E[X^2] - (E[X])^2 \\ &= \text{Var}[X]\end{aligned}$$

$$(2) \text{Independence} \Rightarrow E[XY] = E[X] \cdot E[Y]$$

$$\begin{aligned}\text{cov}(X, Y) &= E[XY] - E[X] E[Y] \\ &= E[X] E[Y] - E[X] \cdot E[Y] = 0\end{aligned}$$

$$\begin{aligned}(3) \text{cov}(X, Y) &= E[XY] - E[X] \cdot E[Y] \\ &= E[YX] - E[Y] \cdot E[X] \\ &= \text{cov}(Y, X)\end{aligned}$$

$$\begin{aligned}(4) \text{cov}(aX, Y) &= E[(aX)Y] - E[aX] \cdot E[Y] \\ &= E[aXY] - a \cdot E[X] \cdot E[Y] \\ &= a \{E[XY] - E[X] \cdot E[Y]\} \\ &= a \cdot \text{cov}(X, Y)\end{aligned}$$

$$\begin{aligned}(5) \text{cov}(X+a, Y) &= E[(X+a)Y] - E[X+a] \cdot E[Y] \\ &= E[XY + aY] - (E[X] + a) E[Y] \\ &= E[XY] + aE[Y] - E[X] \cdot E[Y] - aE[Y] \\ &= a \{E[XY] - E[X] \cdot E[Y]\} \\ &= a \cdot \text{cov}(X, Y)\end{aligned}$$

$$\begin{aligned}
 (6) \text{ cov}(X+Z, Y) &= E[(X+Z)Y] - E[X+Z] \cdot E[Y] \\
 &= E[XY + ZY] - (E[X] + E[Z]) \cdot E[Y] \\
 &= E[XY] + E[ZY] - E[X] \cdot E[Y] - E[Z] \cdot E[Y] \\
 &= (E[XY] - E[X] \cdot E[Y]) + (E[ZY] - E[Z] \cdot E[Y]) \\
 &= \text{cov}(X+Y) + \text{cov}(Z+Y)
 \end{aligned}$$

$$\text{Similarly, } \text{cov}(X, Y+Z) = \text{cov}(X, Y) + \text{cov}(X, Z)$$

$$\begin{aligned}
 (7) \text{ cov}\left(\sum_{i=1}^m a_i X_i, \sum_{j=1}^n b_j Y_j\right) &= \text{cov}(a_1 X_1 + a_2 X_2 + \dots + a_m X_m, b_1 Y_1 + b_2 Y_2 + \dots + b_n Y_n) \\
 &\quad [\text{from property -6}] \\
 &= \text{cov}(a_1 X_1, b_1 Y_1) + \text{cov}(a_1 X_1, b_2 Y_2) + \dots + \text{cov}(a_1 X_1, b_n Y_n) \\
 &\quad + \text{cov}(a_2 X_2, b_1 Y_1) + \text{cov}(a_2 X_2, b_2 Y_2) + \dots + \text{cov}(a_2 X_2, b_n Y_n) \\
 &\quad + \dots + \text{cov}(a_m X_m, b_1 Y_1) + \dots + \text{cov}(a_m X_m, b_n Y_n) \\
 &= \sum_{i=1}^m \sum_{j=1}^n a_i b_j \text{cov}(X_i, Y_j)
 \end{aligned}$$

1/10

→ MGF: Moment Generating Function

$$M_X: \mathbb{R} \rightarrow \infty,$$

$$M_X(t) = \underbrace{E[e^{tx}]}$$

$$\begin{aligned}
 \swarrow E[X] &= \int_{-\infty}^{\infty} x \cdot f_X(x) dx \Rightarrow E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx \\
 &= \int_{-\infty}^{\infty} x \cdot f_X(x) dx &= \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx
 \end{aligned}$$

• Ex. $X \sim \exp(\lambda)$

$$M_X(t) = ?$$

Sol:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned}
 M_X(t) &= E[e^{tx}] \\
 &= \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx \\
 &= \int_{-\infty}^0 e^{tx} \cdot f_X(x) dx + \int_0^{\infty} e^{tx} \cdot f_X(x) dx \\
 &= \int_0^{\infty} e^{tx} \cdot \lambda e^{-\lambda x} dx \\
 &= \lambda \cdot \int_0^{\infty} e^{-(\lambda-t)x} dx
 \end{aligned}$$

$$= \lambda \cdot \frac{-1}{\lambda - t} \cdot \left(e^{-(\lambda - t)x} \right)_0^\infty$$

$$= \frac{-\lambda}{\lambda - t} (0 - (1)) \quad (\text{for } \lambda > t)$$

$$\boxed{\therefore M_X(t) = \frac{\lambda}{\lambda - t}} \quad \text{Only for } \lambda > t, \infty \text{ otherwise}$$

• If $X \rightarrow$ discrete,

$$M_X(t) = \sum_{x \in \Omega'} e^{tx} \cdot P_X(x)$$

If $X \rightarrow$ continuous,

$$M_X(t) = \int_x e^{tx} \cdot f_X(x) dx$$

H.W: Find MGF $\forall$ cont. R.Vs (i.e. Uniform, Normal)

(1) Uniform R.V :

$$f_X(x) = \begin{cases} \frac{1}{\beta - \alpha}, & x \in [\alpha, \beta] \\ 0, & \text{otherwise} \end{cases}$$

$$M_X(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx$$

$$= \int_{\alpha}^{\beta} e^{tx} \cdot \frac{1}{\beta - \alpha} dx$$

$$= \frac{1}{\beta - \alpha} \cdot \int_{\alpha}^{\beta} e^{tx} dx$$

$$= \frac{1}{\beta - \alpha} \cdot \frac{1}{t} \cdot \left(e^{tx} \right)_\alpha^\beta$$

$$M_X(t) = \frac{e^{t\beta} - e^{t\alpha}}{t(\beta - \alpha)}$$

(2) Normal R.V :

$$f_X(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

$$M_X(t) = E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x - \mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} \cdot e^{-\frac{(x - \mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(tx - \frac{(x - \mu)^2}{2\sigma^2} \right)} dx$$

$$\begin{aligned}
&= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right)} dx \\
&= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{(x-(\mu+\sigma^2t))^2}{2\sigma^2}} \cdot e^{\mu t + \frac{\sigma^2 t^2}{2}} dx \\
&= \frac{e^{\mu t + \frac{\sigma^2 t^2}{2}}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2\sigma^2}} du
\end{aligned}$$

$$\begin{cases} u = \frac{x - (\mu + \sigma^2 t)}{\sigma} \\ du = \frac{1}{\sigma} dx \end{cases}$$

$$= \frac{e^{\mu t + \frac{\sigma^2 t^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} du$$

$$\boxed{\therefore M_x(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}} \leftarrow N(\mu, \sigma^2)$$

$$\begin{aligned}
&\frac{2\sigma^2 tx - (x-\mu)^2}{2\sigma^2} \\
&= \frac{2\sigma^2 tx - x^2 + 2x\mu - \mu^2}{2\sigma^2} \\
&= -\frac{x^2 - 2x(\mu + \sigma^2 t) + \mu^2}{2\sigma^2} \\
&= -\frac{x^2 - 2x(\mu + \sigma^2 t) + (\mu + \sigma^2 t)^2}{2\sigma^2} \\
&\quad + \frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2} \\
&= \frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2}
\end{aligned}$$

→ Region of Convergence:

$$D_x = \{t \mid M_x(t) < \infty\}$$

• For Bernoulli R.V:

$$x = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

$$P_x(x) = p^x (1-p)^{1-x}$$

$$M_x(t) = \sum_x e^{tx} P_x(x)$$

$$= (e^0) \cdot (1-p) + (e^t)p$$

$$= p \cdot e^t + (1-p)$$

$$\begin{aligned}
&p^{ax+b} (1-p)^{cx+d} \\
x=0 &\Rightarrow p^b (1-p)^d = (1-p) \\
&\therefore b=0; d=1
\end{aligned}$$

$$\begin{aligned}
x=1 &\Rightarrow p^a (1-p)^{c+1} = p \\
&\therefore a=1; c=-1
\end{aligned}$$

• Advantages:

$$(1) \Psi(it) = E[ie^{itX}]$$

(2) Z-transform $\leftarrow$ Refer

- Refer Beta distribution

H.W : Find MGF of Binomial(n, p) and Poisson(λ)

$$(1) P_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x \in [0, n]$$

$$M_X(t) = E[e^{tX}]$$

$$= \sum_x e^{tx} \cdot P_X(x) dx$$

$$= \sum_{x=0}^n e^{tx} \cdot \binom{n}{x} p^x (1-p)^{n-x}$$

$$\left\{ = \binom{n}{0} (1-p)^n + e^t \binom{n}{1} p^1 (1-p)^{n-1} + e^{2t} \binom{n}{2} p^2 (1-p)^{n-2} + \dots + e^{nt} \binom{n}{n} p^n \right\}$$

$$= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x}$$

$$\therefore M_X(t) = (pe^t + 1-p)^n$$

(2) For poisson ,

$$\text{Intuition : } np = \lambda \Rightarrow \text{MGF} = (\lambda/n e^t + 1 - \lambda/n)^n$$

$$P_X(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad (\lambda > 0, x \geq 0)$$

$$M_X(t) = E[e^{tX}]$$

$$= \sum_x e^{tx} \cdot P_X(x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot (\lambda e^t)^x}{x!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$\therefore M_X(t) = e^{\lambda(e^t - 1)}$$

$$\left\{ \begin{array}{l} \text{Opt for courses :} \\ \text{Real Analysis} \\ \text{Functional Analysis} \end{array} \right\}$$

• Theorem :

If $M_X(t)$ is finite $\forall |t| \leq \varepsilon$ for some $\varepsilon > 0$,

Then, $M_X(t)$ is differentiable on $(-\varepsilon, \varepsilon)$

$$M_X^{(r)}(t) = E[e^{tx} x^r] \quad \forall r$$

Proof :

$$M_X(t) = E[e^{tx}]$$

$$M_X'(t) = \frac{d}{dt} E[e^{tx}] = E\left[\frac{d}{dt} e^{tx}\right] \\ = E[X e^{tx}]$$

$$M_X''(t) = \frac{d}{dt} E[X e^{tx}] = E[X^2 e^{tx}]$$

$$\left\{ \because \frac{d}{dt} e^{tx} = x e^{tx} \right\}$$

Operates term-by-term

In case of discrete :

$$M_X(t) = E[e^{tx}] \\ = \sum_x e^{tx} p_X(x) \\ \Rightarrow \frac{d}{dt} M_X(t) = \frac{d}{dt} E[e^{tx}] \\ = \frac{d}{dt} \sum_x e^{tx} p_X(x) \\ = \sum_x \frac{d}{dt} (e^{tx} \cdot p_X(x)) \\ = \sum_x p_X(x) \cdot x \cdot e^{tx} \\ \therefore \frac{d}{dt} M_X(t) = E[X e^{tx}]$$

In case of continuous :

$$M_X(t) = E[e^{tx}] \\ = \int_x e^{tx} f_X(x) dx \\ \frac{d}{dt} M_X(t) = \frac{d}{dt} E[e^{tx}] \\ = \frac{d}{dt} \int_x e^{tx} f_X(x) dx \\ = \int_x \frac{d}{dt} (e^{tx} \cdot f_X(x)) dx \\ = \int_x f_X(x) \cdot x e^{tx} dx \\ \therefore \frac{d}{dt} M_X(t) = E[X e^{tx}]$$

- Now, set $t=0$:

$$M_X(0) = E[1] = 1$$

$$M_X'(0) = E[X]$$

$$M_X''(0) = E[X^2]$$

$$\therefore M_X^{(r)}(0) = E[X^r]$$

$$\left\{ \Rightarrow \text{Var}[X] = M_X''(0) - (M_X'(0))^2 \right.$$

e.g. In case of exponential :

$$M_X(t) = \frac{\lambda}{\lambda - t} \quad [\lambda > t]$$

$$\Rightarrow M_X'(t) = \frac{(-1)(-\lambda)}{(\lambda - t)^2} = \frac{\lambda}{(\lambda - t)^2} \quad \& \quad M_X'(0) = \frac{1}{\lambda} \leftarrow E[X]$$

$$\Rightarrow M_X''(t) = \frac{d}{dt} \left(\frac{\lambda}{(\lambda - t)^2} \right)$$

$$= \frac{-2\lambda \cdot (-1)}{(\lambda - t)^3} = \frac{2\lambda}{(\lambda - t)^3} \quad \& \quad M_X''(0) = \frac{2}{\lambda^2} \leftarrow E[X^2]$$

• Sum of independent R.V.s:

$$Z = X + Y$$

$$\text{PDF of } Z = ?$$

Sol: $f_z(z) = \int_{-\infty}^{\infty} f_x(x) \cdot f_{y|x}(z-x) dx \leftarrow \text{Convolution}$

If X and Y are independent,

$$\text{MGF of } Z = M_z(t)$$

$$= E[e^{tz}]$$

$$= E[e^{t(x+y)}]$$

$$= E[e^{tx+ty}]$$

$$= E[e^{tx} \cdot e^{ty}]$$

$$= E[e^{tx}] \cdot E[e^{ty}]$$

$$\therefore M_z(t) = M_x(t) \cdot M_y(t)$$

Proof?

i.e. if X and Y are independent,

are e^{tx} and e^{ty} independent?

General Question:

are $g(X)$ and $h(Y)$ independent?

$$E[g(X) \cdot h(Y)]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x) \cdot h(y) \cdot f_{xy}(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x) \cdot f_x(x) \cdot h(y) \cdot f_y(y) \cdot dx dy$$

($\because$ Independent)

$$= \int_{-\infty}^{\infty} g(x) \cdot f_x(x) dx \cdot \int_{-\infty}^{\infty} h(y) \cdot f_y(y) dy$$

$$= E[XY]$$

$$= E[X] \cdot E[Y] \quad (\because \text{Independence})$$

Now, When $Z = X_1 + X_2 + \dots + X_n$

(Where X_i 's are i.i.d)

$$\text{Then, } M_z(t) = \prod_i M_{X_i}(t) = (M_x(t))^n$$

$\hookrightarrow E[e^{tz}]$

Now, if $Z = X_1 + X_2 + \dots + X_N$

where $N \rightarrow$ positive Discrete R.V.

(Where X_i 's are i.i.d, PMF: X)

& PMF of $N = P_N(n)$

MGF of $Z = ?$

Sol: As $E[Z] = E_N[E[Z|N=n]]$

$$= E_N[nE[X]]$$

$$= E[N] \cdot E[X]$$

Similarly, for $M_z(t)$:

$$M_z(t) = E[e^{tz}]$$

$$= E_N[E[e^{tz}|N=n]]$$

$$= E_N[(M_x(t))^n]$$

$$= \sum_n (M_x(t))^n \cdot P_N(n)$$

$$\left\{ \begin{aligned} E[e^{tz}|N=n] &= E[e^{t(X_1+X_2+\dots+X_n)}] \\ &= E[e^{tx_1} \cdot e^{tx_2} \cdot \dots \cdot e^{tx_n}] \end{aligned} \right\}$$

H.W:

Prove that:

$$M_z(t) = M_N(\log(M_x(t)))$$

$$\begin{aligned}
\text{LHS : } M_Z(t) &= E[e^{tz}] \\
&= E_N[E[e^{tz} | N=n]] \\
&= E_N[(M_X(t))^n] \\
&= E_N\left[e^{\ln(M_X(t))^n}\right] \\
&= E_N\left[e^{\ln(M_X(t)) \cdot n}\right] \\
&= M_N(\log(M_X(t))) \quad \left\{ \text{As } M_N(s) = E_N[e^{sN}] \right\} \\
&= \text{RHS}
\end{aligned}$$

$\therefore \text{LHS} = \text{RHS}$

Hence Proved

→ Next Topics:

- Stochastic Simulation
- Convergence of R.V.

→ Generating samples using Uniform distribution:

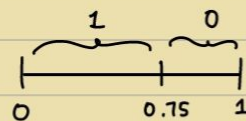
Ex. flipping a biased coin:

$$X = \begin{cases} 1 & \text{with probability } p \text{ [HEAD]} \\ 0 & \text{with probability } 1-p \text{ [TAIL]} \end{cases}$$

$$U \sim 0.25, 0.75$$

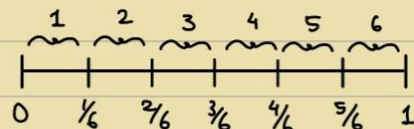
If $p = 0.75$:

$$\begin{array}{l}
0.25 \rightarrow 1 \\
0.7 \rightarrow 1 \\
0.8 \rightarrow 0
\end{array}
\left. \vphantom{\begin{array}{l} 0.25 \\ 0.7 \\ 0.8 \end{array}} \right\} \text{Mapping}$$



Ex. Rolling a dice:

$$Y = \begin{cases} 1 & \text{with probability } 1/6 \\ 2 & \text{with probability } 1/6 \\ 3 & \text{with probability } 1/6 \\ 4 & \text{with probability } 1/6 \\ 5 & \text{with probability } 1/6 \\ 6 & \text{with probability } 1/6 \end{cases}$$



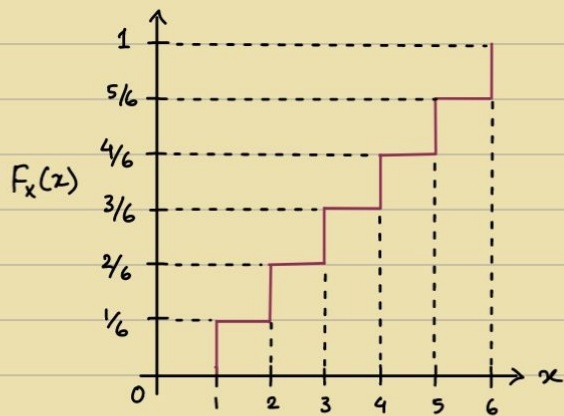
$$P\left(\frac{1}{6} \leq U \leq \frac{2}{6}\right) = \int_{\frac{1}{6}}^{\frac{2}{6}} f_X(x) dx = \frac{1}{6}$$

[Refer : Simulations book by Sheldon Ross]

(Stochastic Process by Sheldon Ross ← Next course on Probability)

- If X has support set as $\{x_1, x_2, x_3, \dots, x_n\}$

By Inverse Transform Method :



$$F_X(u) = \{u : F_X(x) > u\}$$

$$X = \begin{cases} x_0, & \text{if } u < p_0 \\ x_1, & \text{if } p_0 \leq u < p_0 + p_1 \\ x_2, & \text{if } p_0 + p_1 \leq u < p_0 + p_1 + p_2 \\ \vdots & \\ x_j, & \text{if } \sum_{i=0}^{j-1} p_i \leq u < \sum_{i=0}^j p_i \end{cases}$$

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- $X \rightsquigarrow \exp(\lambda)$

$$F_X(x) = 1 - e^{-\lambda x}$$

$$\hookrightarrow u_0, u_1, u_2, \dots, u_{100}$$

$$x_0, x_1, x_2, \dots, x_{100}$$

- Obtain samples from a continuous R.V.:

$$\left\{ \begin{array}{c} \text{H.W. : } N(0,1) \rightarrow \text{Exp}(\lambda) ? \\ \quad \quad \quad \searrow \quad \nearrow \\ \quad \quad U[0,1] \end{array} \right\}$$

Let $U[0,1]$

Consider continuous R.V. X , $F_X(x)$

Now, $\hat{X} := F_X^{-1}(z)$

e.g. $F_X(x) = 1 - e^{-\lambda x}$

$$F_X^{-1}(x) = ?$$

$$\Rightarrow x = 1 - e^{-\lambda F_X(x)}$$

$$\Rightarrow 1 - x = e^{-\lambda F_X^{-1}(x)}$$

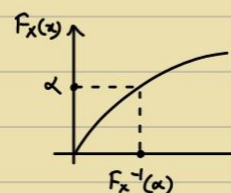
$$\Rightarrow F_X^{-1}(x) = \frac{-1}{\lambda} \ln(1-x) \Rightarrow F_X^{-1}(\alpha) = \frac{-1}{\lambda} (1-\alpha)$$

As functions of random variables are random variables,

$$F_X^{-1}(U) = \frac{-1}{\lambda} (1-U)$$

$$\downarrow$$

$$\hat{X}$$



Now, $\hat{X} = F_X^{-1}(U)$

$$P(\hat{X} = x) = P(F_X^{-1}(U) \leq x)$$

$$= P(U \leq F_X(x))$$

$$= F_X(x)$$

- Workflow:

Given $u_0 \Rightarrow F_X^{-1}(u_0) = \frac{-1}{\lambda} \ln(1-u_0)$

obtain x_0 Claim: x_0 comes from X
 $\therefore P(\hat{X} \leq x_0) = P(X \leq x_0) \quad \forall x_0 \in X$

Now $P(\hat{X} \leq x_0) \rightarrow F_X(x_0)$

$\therefore \hat{X}$ and X have same CDF $(u_0 \rightarrow x_0)$

(Refer : Inverse Transform Sampling)

- Lemma : $\hat{X} = F_X^{-1}(U)$ has distribution $F_X(\cdot)$
CDF of $Y = F_X(\hat{X})$

$$X \rightarrow F_X(x) \text{ \& } F_X(x) = g(x)$$

$$\text{e.g. } F_X(x) = 1 - e^{-\lambda x}$$

$$z = g(x) = F_X(x) = 1 - e^{-\lambda x}$$

$$P(Z \leq z) = P(F_X(\hat{X}) \leq z)$$

$$Z = F_X(\hat{X})$$

$$P(Z \leq z) = P(F_X(\hat{X}) \leq z)$$

$$= P(F_X(F_X^{-1}(U)) \leq z)$$

$$= P(U \leq z)$$

$$\therefore Z \rightsquigarrow \text{Uniform}$$

for $X \rightsquigarrow U[a, b]$:

$$F_X(x) = \frac{x-a}{b-a}$$

$$F_X(X) = \frac{X-a}{b-a} \quad \& \quad Z = F_X(X) \\ = \frac{X-a}{b-a}$$

- Hypothesis :

$$D \sim \{d_1, d_2, \dots, d_{100}\}$$

Does it come from $X \rightsquigarrow \exp(\lambda)$

$$\text{Sol: } F_X(x) = 1 - e^{-\lambda x}$$

$$Z = 1 - e^{-\lambda x}$$

$$\left(\begin{array}{l} z_1 = 1 - e^{-\lambda d_1}, \\ z_2 = 1 - e^{-\lambda d_2}, \dots \end{array} \right)$$

$$F_X(X) \rightsquigarrow U \quad \left[\text{Universality of Uniform / Probability integral Transform} \right]$$

$$\{d_1, d_2, \dots, d_{100}\}$$

Plot histogram / CDF distribution & if it looks like Uniform distribution,

$\therefore$ Hypothesis is true.

→ Evaluating Integrals via Monte Carlo approach :

$$\theta = \int_0^1 g(x) dx \text{ using samples from } U[0,1]$$

$$X \rightsquigarrow U[0,1] \Rightarrow f_X(x) = 1$$

$$\theta = \int_0^1 g(x) dx = \int_0^1 f_X(x) \cdot g(x) dx \quad \left[\because f_X(x) = 1 \right]$$

$$= E[g(U)] \quad \left\{ U \rightarrow U[0,1] \right\}$$

$$u_1 \quad u_2 \quad u_3 \quad \dots \quad u_{100}$$

$$g(u_1) \quad g(u_2) \quad g(u_3) \quad \dots \quad g(u_{100}) \leftarrow E[g(u)]$$

$$S_n = \frac{\sum_{i=1}^{100} g(u_i)}{100}$$



Law of Large numbers

$$\left\{ \begin{array}{l} \text{Intuition: } S_n = \frac{\sum x_i}{n} \quad (x_i : \text{i.i.d}) \\ E[S_n] = E[X] = \mu \\ \text{Var}[S_n] = \frac{1}{n} \text{Var}[X] \end{array} \right\}$$

As n increases $\Rightarrow S_n$ becomes more deterministic (goes to μ)
(Variance keeps dropping)

$$\theta = E[g(u)]$$

Suppose X_i 's are i.i.d & $S_n = \frac{\sum_{i=1}^n X_i}{n}$, Then $\frac{S_n}{n} \rightarrow E[X]$

$$\text{As } n \rightarrow \infty : \sum_{i=1}^n \frac{g(u_i)}{n} \rightarrow E[g(u)] = \theta$$

$$\text{H.W : } \int_a^b g(x) dx = ?$$

• Importance Sampling : [Imp. of Reinforcement Learning]

Compute $E[h(x)]$, X has PDF $f(\cdot)$

{No access to X , $f_x(\cdot)$ }

$$E[h(X)] = \frac{1}{n} \cdot \sum_{i=1}^n h(x_i)$$

$$E[h(X)] = \int_{-\infty}^{\infty} f_X(x) \cdot h(x) dx$$

$$X \rightsquigarrow \begin{array}{cccc} x_1 & x_2 & x_3 & x_{100} \\ h(x_1) & h(x_2) & h(x_3) & \dots h(x_{100}) \end{array}$$

$$\underbrace{Y \rightsquigarrow y_1 \quad y_2 \quad y_3 \quad \dots \quad y_{100}}_{\text{Given}} \quad \text{coming from } g(\cdot)$$

$$E[h(X)] = \int_{-\infty}^{\infty} \frac{f_X(x) \cdot h(x)}{g(x)} \cdot g(x) dx$$

$$= E_Y \left[\frac{f_X(x) \cdot h(x)}{g(x)} \right]$$

$$E[p(Y)] = \int_{-\infty}^{\infty} g(y) \cdot p(y) dy$$

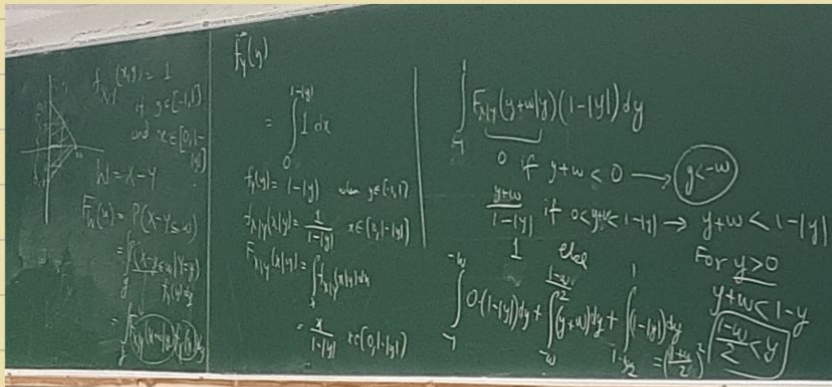
$$p(y) = \frac{f_X(y) \cdot h(y)}{g(y)}$$

Now use LLN to obtain $E[h(X)]$

$\hookrightarrow$ Law of Large Numbers

Tutorial - 6

Q1: Let X and Y be the Cartesian coordinates of a randomly chosen point (according to a uniform PDF) in the triangle with vertices at $(0, 1)$, $(0, -1)$, and $(1, 0)$. Find the CDF and the PDF of $Z = |X - Y|$.



← Method will not be given in solutions.pdf

Q2: Suppose $X \sim \text{Binomial}(n_1, p)$ and $Y \sim \text{Binomial}(n_2, p)$ are two independent random variables.

- Find the MGF of X .
- Let $Z = X + Y$. Find the MGF of Z by using the MGFs of X and Y .
- By recognizing the form of the resulting MGF, identify the distribution of Z (including its parameters). What does this result imply about the sum of independent Binomial random variables with the same success probability?

$$\begin{aligned}
 (a) \quad M_X(t) &= E[e^{tx}] \\
 &= \sum_x e^{tx} \cdot p_x(x) \, dx \\
 &= \sum_{x=0}^n e^{tx} \cdot \binom{n}{x} p^x (1-p)^{n-x} \\
 &= \left\{ \binom{n}{0} (1-p)^n + e^t \binom{n}{1} p^1 (1-p)^{n-1} + e^{2t} \binom{n}{2} p^2 (1-p)^{n-2} + \dots + e^{nt} \binom{n}{n} p^n \right\} \\
 &= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x}
 \end{aligned}$$

$$\therefore M_X(t) = (pe^t + 1 - p)^n$$

$$\begin{aligned}
 (b) \quad M_Z(t) &= M_X(t) \cdot M_Y(t) \\
 &= (pe^t + 1 - p)^{n_1} \cdot (pe^t + 1 - p)^{n_2} \\
 &= (pe^t + 1 - p)^{n_1 + n_2}
 \end{aligned}$$

$$(c) \quad Z \rightsquigarrow \text{Binomial}(n_1 + n_2, p)$$

$$\left(\because Z = X + Y \right) \left\{ \begin{aligned} M_Z(t) &= E[e^{tZ}] \\ &= E[e^{tx+ty}] \\ &= E[e^{tx} \cdot e^{ty}] \\ &= E[e^{tx}] \cdot E[e^{ty}] \\ &= M_X(t) \cdot M_Y(t) \end{aligned} \right\}$$

Q3: Let X be a continuous random variable with the PDF

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad x \in \mathbb{R}, \lambda > 0.$$

Find the MGF of X , $M_X(s)$.

$X \rightarrow \text{cont.}$

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad x \in \mathbb{R}, \lambda > 0$$

$$M_X(s) = ?$$

$$= E[e^{sX}]$$

$$= \int e^{sx} \cdot \frac{\lambda}{2} e^{-\lambda|x|} dx$$

$$= \int_{-\infty}^0 e^{sx} \cdot \frac{\lambda}{2} e^{\lambda x} dx + \int_0^{\infty} e^{sx} \cdot \frac{\lambda}{2} e^{-\lambda x} dx$$

$$= \frac{\lambda}{2} \cdot \underbrace{\int_{-\infty}^0 e^{(s+\lambda)x} dx}_{s+\lambda > 0 \Rightarrow s > -\lambda} + \frac{\lambda}{2} \underbrace{\int_0^{\infty} e^{(s-\lambda)x} dx}_{s-\lambda < 0 \Rightarrow s < \lambda}$$

$$\Leftarrow \text{for convergence} \\ \therefore |s| < \lambda$$

$$= \frac{\lambda}{2} \left\{ \left(\frac{1}{s+\lambda} \right) e^{(s+\lambda)x} \right\}_{-\infty}^0 + \left(\frac{1}{s-\lambda} \right) e^{(s-\lambda)x} \right\}_0^{\infty}$$

$$= \frac{\lambda}{2} \left[\frac{1}{s+\lambda} - \frac{1}{s-\lambda} \right]$$

$$= \frac{\lambda}{2} \cdot \frac{-2\lambda}{s^2 - \lambda^2}$$

$$= \frac{-2\lambda^2}{s^2 - \lambda^2} = M_X(s)$$

Q4: Let $M_X(s)$ be finite for $s \in [c, c]$ where $c > 0$. Prove:

(a)

$$\lim_{n \rightarrow \infty} \left[M_X\left(\frac{s}{n}\right) \right]^n = e^{sE[X]}$$

(b) Now assume $E[X] = 0$ and $\text{Var}[X] = 1$, then

$$\lim_{n \rightarrow \infty} \left[M_X\left(\frac{s}{\sqrt{n}}\right) \right]^n = e^{\frac{s^2}{2}}.$$

(c) We know that for $X \sim N(0, 1)$, we have $M_X(s) = e^{\frac{s^2}{2}}$. What can you say about the expression you derived above?

$$(a) \lim_{n \rightarrow \infty} \left[M_X\left(\frac{s}{n}\right) \right]^n \quad s \in (-c, c), \quad c > 0$$

$$M_X(s) \text{ is finite}$$

$$M_X\left(\frac{s}{n}\right) = E\left[e^{\frac{sX}{n}}\right]$$

$$\therefore \text{LHS} : \lim_{n \rightarrow \infty} \left(E\left[e^{\frac{sX}{n}}\right] \right)^n$$

$$= e^{\ln \left(\lim_{n \rightarrow \infty} \left(M_X\left(\frac{s}{n}\right) \right)^n \right)} = e^{\lim_{n \rightarrow \infty} n \cdot \ln \left(M_X\left(\frac{s}{n}\right) \right)}$$

$$= e^{\lim_{n \rightarrow \infty} \frac{\ln \left(M_X\left(\frac{s}{n}\right) \right)}{1/n}}$$

$$\ln(M_X(0)) = \ln(E[e^0])$$

$$= \ln(1) = 0$$

$\therefore \frac{0}{0}$ situation $\rightarrow$ Use L'Hospital's rule

$$\Rightarrow \text{LHS} = e^{\lim_{n \rightarrow \infty} \frac{\frac{1}{M_X(s/n)} \cdot \left(-\frac{s}{n^2}\right) \cdot M_X'(s/n)}{-1/n^2}}$$

$$= e^{\lim_{n \rightarrow \infty} s \cdot \frac{M_X'(s/n)}{M_X(s/n)}}$$

$$= e^{s \cdot \frac{E[X]}{1}} = e^{sE[X]} = \text{RHS}$$

Hence Proved !

$$(b) E[X] = 0$$

$$\text{Var}[X] = 1$$

$$\lim_{n \rightarrow \infty} \left(M_X\left(\frac{s}{\sqrt{n}}\right) \right)^n = e^{\lim_{n \rightarrow \infty} n \cdot \ln \left(M_X\left(\frac{s}{\sqrt{n}}\right) \right)}$$

$$= e^{\lim_{n \rightarrow \infty} \frac{\ln \left(M_X\left(\frac{s}{\sqrt{n}}\right) \right)}{1/n}}$$

$$= e^{\lim_{n \rightarrow \infty} \frac{\frac{1}{M_X(s/\sqrt{n})} \cdot \left(-\frac{s}{2n\sqrt{n}}\right) \cdot M_X'(s/\sqrt{n})}{-1/n^2}}$$

$$= e^{\frac{s}{2} \cdot \lim_{n \rightarrow \infty} \frac{\sqrt{n} M_X'(s/\sqrt{n})}{M_X(s/\sqrt{n})}}$$

$$M_X(0) = E[e^0] = 1$$

$$= e^{\frac{s}{2} \cdot \lim_{n \rightarrow \infty} \frac{M_X'(s/\sqrt{n})}{1/\sqrt{n}}}$$

Apply L'Hospital's rule again

$$= e^{\frac{s}{2} \cdot \lim_{n \rightarrow \infty} \frac{M_X''(s/\sqrt{n}) \cdot \left(-\frac{s}{2n\sqrt{n}}\right)}{-1/2n\sqrt{n}}}$$

$$= e^{s^2/2 \cdot M_X''(0)}$$

$$= e^{s^2/2} = \text{RHS}$$

Hence Proved !

$$(c) \lim_{n \rightarrow \infty} \left(M_X\left(\frac{s}{\sqrt{n}}\right) \right)^n = \underbrace{e^{s^2/2}}_{\text{LHS}} \cdot \underbrace{M_X(s)}_{\text{RHS}}$$

Q5: A discrete random variable X can take one of three values with the following probabilities:

$$p_X(1) = 0.4, \quad p_X(5) = 0.3, \quad p_X(10) = 0.3.$$

Describe, step-by-step, the inverse transform method to generate a random sample for X using a random number u drawn from a Uniform[0, 1] distribution. Provide the specific ranges of u that would correspond to each value of X .

$$P_X(x) = \begin{cases} 0.4, & \text{if } x = 1 \\ 0.3, & \text{if } x = 5 \text{ or } 10 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{array}{ccc} x_j & 1 & 5 & 10 \\ p_j & 0.4 & 0.3 & 0.3 \end{array}$$

$$\{ P(X = x_j) = p_j \}$$

$$X = \begin{cases} 1 & \text{if } u < 0.4 \\ 5 & \text{if } 0.4 \leq u < 0.7 \\ 10 & \text{if } 0.7 \leq u < 1 \end{cases}$$

$\downarrow$
 $F_X^{-1}(u)$

Q6: Let the CDF of a continuous random variable X be

$$F(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right), \quad -\infty < x < \infty.$$

(a) Find the inverse CDF $F^{-1}(u)$ for $u \in (0, 1)$.

(b) Using the inverse transform method, compute the sample x when $U = 0.75$.

$$F_X(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right), \quad -\infty < x < \infty$$

$$F^{-1}(u), \quad u \in (0, 1)$$

$$u = \frac{1}{2} \left(1 + \frac{y}{\sqrt{1+y^2}} \right) \Rightarrow 2u - 1 = \frac{y}{\sqrt{1+y^2}} \quad [y = F^{-1}(u)]$$

$$\Rightarrow (2u - 1)^2 = \frac{y^2}{1+y^2}$$

$$\Rightarrow (1+y^2)(4u^2 - 4u + 1) = y^2$$

$$\Rightarrow (2u-1)^2 + y^2(4u^2 - 4u + 1 - 1) = 0$$

$$\Rightarrow (2u-1)^2 + y^2(4u)(u-1) = 0$$

$$\Rightarrow y^2(4u)(u-1) = -(2u-1)^2$$

$$\therefore y^2 = \frac{-(2u-1)^2}{(4u)(u-1)} = \frac{(2u-1)^2}{(4u)(1-u)}$$

$$F^{-1}(u) = \frac{|2u-1|}{2\sqrt{u(1-u)}}$$

$$U \rightarrow 0.75$$

$$y = \frac{2(3/4)-1}{\sqrt{4(3/4)(1/4)}} = \frac{1}{\sqrt{3}} \Rightarrow F^{-1}(0.75) = \frac{1}{\sqrt{3}}$$

Q7: We want to estimate $\theta = \mathbb{E}[X^2]$ where X is an exponential random variable with rate $\lambda = 1$ ($X \sim \text{Exp}(1)$). Using importance sampling, formulate an estimator for θ by drawing N samples, $Y_1, \dots, Y_N$, from a uniform distribution $Y \sim U[0, 5]$.

$$\theta = \mathbb{E}[X^2]$$

$$X \sim \text{exp}(1) \quad [\lambda=1]$$

$$Y_1, Y_2, \dots, Y_N \text{ from } Y \sim U[0, 5]$$

$$f_X(x) = f(x) = \begin{cases} \lambda e^{-\lambda x} = e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases} \quad \left| \quad f_Y(y) = g(y) = \begin{cases} 1/5, & y \in [0, 5] \\ 0, & \text{otherwise} \end{cases}$$

$$h(X) = X^2$$

$$\mathbb{E}[h(X)] = \mathbb{E}_Y \left[h(Y) \cdot \frac{f(Y)}{g(Y)} \right]$$

$$\Rightarrow \theta = \mathbb{E}_Y \left[Y^2 \cdot \frac{e^{-Y}}{1/5} \right] = \mathbb{E}_Y [5Y^2 \cdot e^{-Y}]$$

$$\theta_N = \frac{1}{N} \cdot \sum_{i=1}^N 5Y_i^2 e^{-Y_i} \quad \leftarrow \text{Monte Carlo estimator}$$

True expectation :

$$\mathbb{E}[X^2] = \int_0^\infty x^2 e^{-x} dx$$

$$= \int_0^\infty x^2 e^{-x} dx = - \left(x^2 e^{-x} + 2x e^{-x} + 2e^{-x} \right) \Big|_0^\infty = 2$$

$$\begin{array}{rcl} 0 & & 1 \\ + x^2 & \searrow & e^{-x} \\ - 2x & \searrow & -e^{-x} \\ + 2 & \searrow & e^{-x} \\ - 0 & \searrow & -e^{-x} \end{array}$$

$$\text{But } Y \rightarrow [0, 5]$$

$$f_X(5) = e^{-5} = 0.067 \quad \leftarrow \text{Negligible}$$

Q8: Let X and Y be two independent random variables with respective moment generating functions

$$M_X(t) = \frac{1}{1-3t}, \quad M_Y(t) = \frac{1}{(1-3t)^2}, \quad t < \frac{1}{3}.$$

Find $\mathbb{E}[(X+Y)^2]$.

$$\text{Let } Z = X + Y$$

$$\begin{aligned} \text{Now, } M_Z(t) &= M_{X+Y}(t) \\ &= M_X(t) \cdot M_Y(t) \end{aligned}$$

$$\mathbb{E}[Z^2] = M_Z''(0)$$

$$M_Z(t) = \frac{1}{(1-3t)} \cdot \frac{1}{(1-3t)^2} = \frac{1}{(1-3t)^3} \quad (t < 1/3)$$

$$M_Z'(t) = \frac{-3(-3)}{(1-3t)^4} = \frac{9}{(1-3t)^4}$$

$$\begin{aligned} M_Z''(t) &= \frac{108}{(1-3t)^5} \Rightarrow M_Z''(0) = 108 \\ \therefore \mathbb{E}[(X+Y)^2] &= 108 \end{aligned}$$

Homework 6

Q1: Let $X \sim \mathcal{N}(\mu, \sigma^2)$ be a Gaussian random variable.

- (a) Find the moment generating function (MGF) of X .
 (b) Using the MGF, compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

$$f_X(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\begin{aligned}
 (a) \quad M_X(t) &= \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) \, dx \\
 &= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \, dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \, dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right)} \, dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right)} \, dx \\
 &= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{(x-(\mu+\sigma^2 t))^2}{2\sigma^2}} \cdot e^{\mu t + \frac{\sigma^2 t^2}{2}} \, dx \\
 &= \frac{e^{\mu t + \frac{\sigma^2 t^2}{2}}}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2\sigma^2}} \, du \\
 &\quad \left[\begin{array}{l} u = x - (\mu + \sigma^2 t) \\ du = \frac{1}{\sigma} \, dx \end{array} \right] \\
 &= \frac{e^{\mu t + \frac{\sigma^2 t^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} \, du \\
 &\quad \left[\begin{array}{l} \frac{2\sigma^2 tx - (x-\mu)^2}{2\sigma^2} \\ = \frac{2\sigma^2 tx - x^2 + 2x\mu - \mu^2}{2\sigma^2} \\ = -\frac{x^2 - 2x(\mu + \sigma^2 t) + \mu^2}{2\sigma^2} \\ = -\frac{x^2 - 2x(\mu + \sigma^2 t) + (\mu + \sigma^2 t)^2}{2\sigma^2} \\ \quad + \frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2} \\ = \frac{(x - (\mu + \sigma^2 t))^2}{2\sigma^2} + \frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2} \end{array} \right] \\
 &\quad \left[\begin{array}{l} \therefore M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}} \leftarrow \mathcal{N}(\mu, \sigma^2) \end{array} \right]
 \end{aligned}$$

$$(b) \quad \mathbb{E}[X] = M_X'(0) = e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot (\mu + \sigma^2 t) \Big|_{t=0} = \mu$$

$$\begin{aligned}
 \text{Var}[X] &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = M_X''(0) - (M_X'(0))^2 \\
 &= e^{\mu t + \frac{\sigma^2 t^2}{2}} (\sigma^2) + e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot (\mu + \sigma^2 t)^2 \Big|_{t=0} - (\mu)^2 \\
 &= \sigma^2 + \mu^2 - \mu^2 = \sigma^2
 \end{aligned}$$

Q2: Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with common MGF $M_X(t)$.

- (a) Show that the MGF of $Z = X_1 + X_2 + \dots + X_n$ is $M_Z(t) = (M_X(t))^n$.
 (b) Now suppose N is a positive integer-valued random variable, independent of the X_i 's, and define

$$Z = X_1 + X_2 + \dots + X_N.$$

Show that the MGF of Z is

$$M_Z(t) = \mathbb{E}[(M_X(t))^N].$$

- (c) Express $M_Z(t)$ in terms of the MGF of N , $M_N(s)$, and prove that

$$M_Z(t) = M_N(\log M_X(t)).$$

(a) If X and Y are independent,

$$\begin{aligned} \text{MGF of } Z &= M_Z(t) \\ &= \mathbb{E}[e^{tZ}] \\ &= \mathbb{E}[e^{t(X+Y)}] \\ &= \mathbb{E}[e^{tX+tY}] \\ &= \mathbb{E}[e^{tX} \cdot e^{tY}] \\ &= \mathbb{E}[e^{tX}] \cdot \mathbb{E}[e^{tY}] \\ \therefore M_Z(t) &= M_X(t) \cdot M_Y(t) \end{aligned}$$

Now, When $Z = X_1 + X_2 + \dots + X_n$

(where X_i 's are i.i.d)

$$\begin{aligned} \text{Then, } M_Z(t) &= \prod_i M_{X_i}(t) = (M_X(t))^n \\ &\hookrightarrow \mathbb{E}[e^{tZ}] \end{aligned}$$

(b) Now, if $Z = X_1 + X_2 + \dots + X_N$

$$\begin{aligned} \text{As } \mathbb{E}[Z] &= \mathbb{E}_N[\mathbb{E}[Z|N=n]] \\ &= \mathbb{E}_N[n \mathbb{E}[X]] \\ &= \mathbb{E}[N] \cdot \mathbb{E}[X] \end{aligned}$$

Similarly, for $M_Z(t)$:

$$\begin{aligned} M_Z(t) &= \mathbb{E}[e^{tZ}] \\ &= \mathbb{E}_N[\mathbb{E}[e^{tZ}|N=n]] \\ &= \mathbb{E}_N[(M_X(t))^n] \\ &= \mathbb{E}[(M_X(t))^N] \end{aligned}$$

$$\left\{ \begin{aligned} \mathbb{E}[e^{tZ}|N=n] &= \mathbb{E}[e^{t(X_1+X_2+\dots+X_n)}] \\ &= \mathbb{E}[e^{tX_1} \cdot e^{tX_2} \cdot \dots \cdot e^{tX_n}] \end{aligned} \right\}$$

(c) LHS: $M_Z(t) = \mathbb{E}[e^{tZ}]$

$$\begin{aligned} &= \mathbb{E}_N[\mathbb{E}[e^{tZ}|N=n]] \\ &= \mathbb{E}_N[(M_X(t))^n] \end{aligned}$$

$$= \mathbb{E}_N[e^{\ln(M_X(t))^n}]$$

$$= \mathbb{E}_N[e^{\ln(M_X(t)) \cdot n}]$$

$$= M_N(\log(M_X(t)))$$

$$= \text{RHS}$$

$$\left\{ \text{As } M_N(s) = \mathbb{E}_N[e^{sN}] \right\}$$

$$\therefore \text{LHS} = \text{RHS}$$

Hence Proved

Q3: The moment-generating function of a discrete random variable X is given by:

$$M_X(t) = \frac{1}{6}e^{-2t} + \frac{2}{6}e^t + \frac{3}{6}e^{4t}$$

Determine the probability mass function (PMF) of X .

$$M_X(t) = E[e^{tx}]$$

$$= \sum_x e^{tx} \cdot P_X(x)$$

Given :

$$M_X(t) = e^{t(-2)} \cdot \frac{1}{6} + e^{t(1)} \cdot \frac{1}{3} + e^{t(4)} \cdot \frac{1}{2}$$

$$\therefore P_X(x = -2) = \frac{1}{6}$$

$$P_X(x = 1) = \frac{1}{3}$$

$$P_X(x = 4) = \frac{1}{2}$$

$$\therefore P_X(x) = \begin{cases} \frac{1}{6}, & \text{if } x = -2 \\ \frac{1}{3}, & \text{if } x = 1 \\ \frac{1}{2}, & \text{if } x = 4 \end{cases}$$

Q4: The lifetime X of an electronic component follows a Weibull distribution, whose Cumulative Distribution Function (CDF) is given by:

$$F_X(x) = 1 - e^{-x^2}, \quad \text{for } x \geq 0$$

Derive a formula to generate a random sample from this distribution using the inverse transform method with a uniform random variable $U \sim \text{Uniform}[0, 1]$.

$$F_X(x) = 1 - e^{-x^2}, \quad x \geq 0$$

$$\text{Step - I : } U = F_X(x)$$

$$\Rightarrow U = 1 - e^{-x^2} \quad (x \geq 0)$$

$$\text{Step - II : } U = 1 - e^{-x^2}$$

$$\Rightarrow e^{-x^2} = 1 - U$$

$$\Rightarrow -x^2 = \ln(1 - U)$$

$$\therefore x = \sqrt{-\ln(1 - U)}, \quad U \sim \text{Uniform}(0, 1)$$

$$\left\{ \begin{aligned} f_X(x) &= \frac{d}{dx} F_X(x) \\ &= \frac{d}{dx} (1 - e^{-x^2}) \\ &= 2xe^{-x^2} \end{aligned} \right\}$$

Q5: Let X be an exponential random variable with parameter $\lambda = 2$, and let Y be a Bernoulli random variable with parameter $p = 0.5$. Assume X and Y are independent.

(a) Let $Z = X + Y$. Find the MGF of Z , denoted $M_Z(t)$.

(b) Use $M_Z(t)$ to find the expected value $E[Z]$ and the variance $Var(Z)$.

$$\left. \begin{aligned} X &\sim \exp(\lambda = 2) \\ Y &\sim \text{Bernoulli}(p = \frac{1}{2}) \end{aligned} \right\} \text{Independent}$$

$$(a) Z = X + Y$$

$$M_Z(t) = M_X(t) \cdot M_Y(t)$$

$$f_X(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} M_X(t) &= E[e^{tx}] \\ &= \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx \\ &= \int_{-\infty}^0 e^{tx} \cdot f_X(x) dx + \int_0^{\infty} e^{tx} \cdot f_X(x) dx \\ &= \int_0^{\infty} e^{tx} \cdot 2e^{-2x} dx \\ &= 2 \cdot \int_0^{\infty} e^{-(2-t)x} dx \\ &= 2 \cdot \frac{-1}{2-t} \cdot \left(e^{-(2-t)x} \right)_0^{\infty} \\ &= \frac{-2}{2-t} (0 - (1)) \quad (\text{for } t < 2) \end{aligned}$$

$$\therefore M_X(t) = \frac{2}{2-t}$$

Only for $t < 2$,
 ∞ otherwise

$$P_Y(y) = \begin{cases} \left(\frac{1}{2}\right)^y \left(\frac{1}{2}\right)^{1-y}, & \text{if } y=0 \text{ or } 1 \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{2}, & \text{if } y=0 \\ \frac{1}{2}, & \text{if } y=1 \end{cases}$$

$$\begin{aligned} M_Y(t) &= E[e^{ty}] \\ &= \sum_y e^{ty} \cdot P_Y(y) \\ &= e^0 P_Y(0) + e^t P_Y(1) \\ &= (1 + e^t)/2 \end{aligned}$$

$$\therefore M_Y(t) = \frac{1+e^t}{2}$$

$$\begin{aligned} \therefore M_Z(t) &= M_X(t) \cdot M_Y(t) \\ &= \frac{2}{2-t} \cdot \frac{1+e^t}{2} \quad (t < 2) \end{aligned}$$

$$M_Z(t) = \frac{1+e^t}{2-t} \quad \forall t < 2$$

$$(b) E[Z] = M'_Z(0)$$

$$M'_Z(t) = \frac{(2-t)e^t + (1+e^t)}{(2-t)^2} = \frac{(3-t)e^t + 1}{(2-t)^2} \Rightarrow M'_Z(0) = \frac{4}{4} = 1 \Rightarrow E[Z] = 1$$

$$M''_Z(t) = \frac{(2-t)^2 \cdot (2-t)e^t - ((3-t)e^t + 1) \cdot (-2)(2-t)}{(2-t)^4}$$

$$\Rightarrow M''_Z(0) = \frac{8 + 16}{16} = \frac{24}{16} = \frac{3}{2} \Rightarrow E[Z^2] = \frac{3}{2}$$

$$\therefore \text{Var}[Z] = E[Z^2] - (E[Z])^2$$

$$= \frac{3}{2} - 1 = \frac{1}{2} \Rightarrow \text{Var}[Z] = \frac{1}{2}$$

Q6: A Geiger counter measures background radiation. The machine's true average rate of detection, λ , is known to be either low ($\lambda_1 = 0.1$ counts/sec) or high ($\lambda_2 = 0.5$ counts/sec). The prior probability of the rate being low is $P(\lambda = \lambda_1) = 0.8$. The number of counts K in a given second follows a Poisson distribution with parameter λ . You turn on the counter for one second and observe exactly $k = 2$ counts.

What is the updated (posterior) probability that the rate is low, given your observation? That is, find $P(\lambda = \lambda_1 | K = 2)$.

$$\lambda_1 = 0.1 \text{ counts/sec} \quad \& \quad P(\lambda = \lambda_1) = 0.8$$

$$\lambda_2 = 0.5 \text{ counts/sec} \quad \therefore P(\lambda = \lambda_2) = 0.2$$

$$P(\lambda = \lambda_1 | K = 2) = \frac{P_{\lambda K}(\lambda_1, 2)}{P(K = 2)}$$

$$= \frac{P(\lambda = \lambda_1) \cdot P(K = 2 | \lambda = \lambda_1)}{P(K = 2)}$$

$$= \frac{P(\lambda_1)}{P(K = 2)} \cdot \frac{e^{-\lambda_1} \cdot \lambda_1^2}{2}$$

$$P(K | \lambda) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$$\left\{ \begin{array}{l} P(A|B) = \frac{P(A \cap B)}{P(B)} \\ P(B|A) = \frac{P(A \cap B)}{P(A)} \\ \Rightarrow P(A \cap B) = P(A) \cdot P(B|A) \\ \text{(Baye's Theorem)} \end{array} \right.$$

$$P(K = 2) = P(K = 2 | \lambda_1) \cdot P(\lambda_1) + P(K = 2 | \lambda_2) \cdot P(\lambda_2)$$

$$= \frac{e^{-\lambda_1} \cdot \lambda_1^2}{2} \cdot P(\lambda_1) + \frac{e^{-\lambda_2} \cdot \lambda_2^2}{2} \cdot \underbrace{(1 - P(\lambda_1))}_{P(\lambda_2)}$$

$$\therefore P(\lambda = \lambda_1 | K = 2) = \frac{P(\lambda_1) \cdot e^{-\lambda_1} \cdot \lambda_1^2}{P(\lambda_1) \cdot e^{-\lambda_1} \cdot \lambda_1^2 + (1 - P(\lambda_1)) \cdot e^{-\lambda_2} \cdot \lambda_2^2}$$

$$P(\lambda_1) = 4/5, \quad \lambda_1 = 1/10$$

$$P(\lambda_2) = 1/5, \quad \lambda_2 = 1/2$$

$$\Rightarrow P(\lambda = \lambda_1 | K = 2) = \frac{4/5 \cdot e^{-1/10} \cdot 1/100}{(4/5 \cdot e^{-1/10} \cdot 1/100) + (1/5 \cdot e^{-1/2} \cdot 1/4)}$$

$$= 0.193$$

$$\therefore P(\lambda = \lambda_1 | K = 2) = 0.193 \quad \&$$

$$P(\lambda = \lambda_2 | K = 2) = 0.807$$

Q7: Let $X_1, \dots, X_n$ be independent $\text{Exp}(\lambda)$ random variables (rate $\lambda > 0$). Using MGFs, show that the sum

$$Y = X_1 + X_2 + \dots + X_n$$

has the Gamma distribution $\text{Gamma}(n, \lambda)$.

$$M_Y(t) = M_{X_1}(t) \cdot M_{X_2}(t) \cdot \dots \cdot M_{X_n}(t)$$

$$\Rightarrow M_Y(t) = \left(\frac{\lambda}{\lambda - t} \right)^n$$

$$\therefore M_X(t) = \frac{\lambda}{\lambda - t}, \text{ for } \lambda > t, \infty \text{ otherwise } \quad [\text{PS-69}]$$

Note :

If a RV follows Gamma distribution,

$$f_Z(z) = \begin{cases} \frac{e^{-z} \cdot z^{\lambda-1}}{\Gamma(\lambda)}, & \lambda > 0, 0 < z < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} M_Z(t) &= E[e^{tZ}] = \int_0^{\infty} e^{tz} \cdot f_Z(z) dz \\ &= \int_0^{\infty} e^{tz} \cdot \frac{\lambda^K z^{K-1} \cdot e^{-\lambda z}}{\Gamma(K)} dz \\ &= \frac{\lambda^K}{\Gamma(K)} \int_0^{\infty} z^{K-1} \cdot e^{-(\lambda-t)z} dz \quad [\lambda > t] \\ &= \frac{\lambda^K}{\Gamma(K)} \int_0^{\infty} \left(\frac{u}{\lambda-t} \right)^{K-1} \cdot e^{-u} \cdot \frac{1}{\lambda-t} du \quad [u = (\lambda-t)z] \\ &= \frac{\lambda^K}{(\lambda-t)^K \Gamma(K)} \int_0^{\infty} u^{K-1} e^{-u} du \\ \therefore M_Z(t) &= \frac{\lambda^K}{(\lambda-t)^K} \quad [t < \lambda], \text{ for Gamma}(K, \lambda) \end{aligned}$$

$$\therefore M_Y(t) = \text{Gamma}(n, \lambda)$$

Hence Proved

Note : Gamma distribution :

For $\text{Gamma}(K, \lambda)$,

$$f_Z(z) = \frac{\lambda^K z^{K-1} \cdot e^{-\lambda z}}{\Gamma(K)} \quad \begin{cases} z \geq 0 \\ \alpha \geq 0 \text{ \& } \beta \geq 0 \end{cases}$$

$$\text{where } \Gamma(K) = \int_0^{\infty} t^{K-1} \cdot e^{-t} dt$$

Q8: Estimate the integral

$$I_2 = \int_0^{\infty} e^{-x} dx$$

using Monte Carlo integration with N random samples from $U[0, 1]$.

$$I = \int_0^{\infty} e^{-x} dx \quad U \sim \text{Uniform}[0, 1] \Rightarrow f_U(u) = \begin{cases} 1, & \text{if } u \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

Map $[0, \infty)$ to $[0, 1]$ with

$$x = \frac{u}{1-u} \Leftrightarrow u = \frac{x}{1+x} \quad \Bigg| \quad dx = \frac{du}{(1-u)^2}$$

$$I = \int_0^{\infty} e^{-x} dx = \int_0^1 \frac{e^{-u/(1-u)}}{(1-u)^2} du$$

Monte - Carlo estimator with $U_i \sim \text{Uniform}[0, 1]$

$$\hat{I} = \frac{1}{N} \sum_{i=1}^N g(U_i)$$

$$\text{where } g(u) = \frac{e^{-u/(1-u)}}{(1-u)^2}$$

Q9: Let X and Y be two random variables for which $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, and the covariance is $\text{Cov}(X, Y) = -2$.

- (a) Find the variance of the random variable $Z = 3X - 2Y + 5$.
- (b) If X and Y were independent, what would $\text{Var}(Z)$ be?

$$\begin{aligned} \text{(a) } \text{Var}[Z] &= \text{Var}[3X - 2Y + 5] \\ &= E[(3X - 2Y + 5)^2] - (E[3X - 2Y + 5])^2 \\ &= E[9X^2 + 4Y^2 + 25 - 12XY - 20Y + 30X] - (3E[X] - 2E[Y] + E[5])^2 \\ &= 9E[X^2] + 4E[Y^2] + E[25] - 12E[XY] - 20E[Y] + 30E[X] \\ &\quad - (9E^2[X] + 4E^2[Y] + E^2[5] - 12E[X] \cdot E[Y] - 4E[Y]E[5] + 6E[X]E[5]) \\ &= 9\text{Var}[X] + 4\text{Var}[Y] - 12\text{Cov}(X, Y) \\ &= 9(4) + 4(9) - 12(-2) \\ &= 96 \end{aligned}$$

$$\therefore \text{Var}[Z] = 96$$

$$\text{(b) If } X \text{ and } Y \text{ are independent} \Rightarrow E[XY] = E[X] \cdot E[Y]$$

$$\begin{aligned} \therefore \text{Var}[Z] &= 9\text{Var}[X] + 4\text{Var}[Y] \\ &= 72 \end{aligned}$$

$$\therefore \text{Cov}(X, Y) = 0$$

$$\left. \begin{array}{l} \text{Independence} \Rightarrow \text{Uncorrelation} \\ \text{Uncorrelated} \nRightarrow \text{Independence} \end{array} \right\}$$

Q10: Let X be a discrete random variable with the following probability mass function (PMF):

$$p_X(-1) = 0.2, \quad p_X(0) = 0.5, \quad p_X(1) = 0.3.$$

- Find the Moment Generating Function (MGF) of X , denoted by $M_X(t)$.
- Using the MGF from part (a), compute the first and second moments of X , i.e., $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.
- Use the results from part (b) to find the variance of X , $\text{Var}(X)$.

$$\begin{aligned} \text{(a)} \quad M_X(t) &= \sum_x e^{tx} p_X(x) \\ &= 0.2 \cdot e^{-t} + 0.5 + 0.3 \cdot e^t \\ \therefore M_X(t) &= \frac{3}{10}(e^t) + \frac{1}{5}(e^{-t}) + \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \mathbb{E}[X] &= M_X'(0) \\ M_X'(t) &= \frac{3}{10}(e^t) - \frac{1}{5}(e^{-t}) \\ \therefore \mathbb{E}[X] &= \frac{3}{10} - \frac{1}{5} = \frac{1}{10} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[X^2] &= M_X''(0) \\ M_X''(t) &= \frac{3}{10}(e^t) + \frac{1}{5}(e^{-t}) \\ \therefore \mathbb{E}[X^2] &= \frac{3}{10} + \frac{1}{5} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \text{Var}[X] &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{1}{2} - \frac{1}{100} \\ &= \frac{49}{100} \end{aligned}$$

Q11: Let X be a continuous random variable with the Gamma(α, λ) PDF

$$f_X(x) = \begin{cases} \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, & x > 0, \\ 0, & \text{otherwise,} \end{cases} \quad \alpha > 0, \lambda > 0.$$

Find the moment generating function $M_X(t) = \mathbb{E}[e^{tX}]$ (specify the region of t for which it is finite).

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} \cdot f_X(x) \, dx \\ &= \int_0^\infty e^{tx} \cdot \frac{\lambda^\alpha x^{\alpha-1} \cdot e^{-\lambda x}}{\Gamma(\alpha)} \, dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} \cdot e^{-(\lambda-t)x} \, dx \quad [\lambda > t] \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{u}{\lambda-t}\right)^{\alpha-1} \cdot e^{-u} \cdot \frac{1}{\lambda-t} \, du \quad [u = (\lambda-t)x] \\ &= \frac{\lambda^\alpha}{(\lambda-t)^\alpha \Gamma(\alpha)} \cdot \int_0^\infty u^{\alpha-1} e^{-u} \, du \\ \therefore M_X(t) &= \frac{\lambda^\alpha}{(\lambda-t)^\alpha} \quad [t < \lambda] \quad \left[\because \Gamma(\alpha) = \int_0^\infty u^{\alpha-1} \cdot e^{-u} \, du \right] \end{aligned}$$

Q12: Let $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$.

(a) Find the moment generating function $M_X(t)$.

(b) Use $M_X(t)$ to compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

$$(a) \quad P_X(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} \quad (\lambda > 0, x \geq 0)$$

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tx}] \\ &= \sum_x e^{tx} \cdot P_X(x) \\ &= \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} \\ &= \sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot (\lambda e^t)^x}{x!} \\ &= e^{-\lambda} \cdot e^{\lambda e^t} \end{aligned}$$

$$\boxed{\therefore M_X(t) = e^{\lambda(e^t - 1)}}$$

$$(b) \quad \mathbb{E}[X] = M_X'(0)$$

$$M_X'(t) = e^{\lambda(e^t - 1)} \cdot \lambda e^t$$

$$\Rightarrow M_X'(0) = \lambda = \mathbb{E}[X]$$

$$M_X''(t)$$

$$= (\lambda e^t)^2 e^{\lambda(e^t - 1)} + e^{\lambda(e^t - 1)} \cdot \lambda e^t$$

$$\Rightarrow M_X''(0) = \lambda^2 + \lambda = \mathbb{E}[X^2]$$

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \end{aligned}$$

Q13: Let X and Y be independent random variables with PMFs:

$$p_X(x) = \begin{cases} 1/3, & \text{if } x \in \{1, 2, 3\}, \\ 0, & \text{otherwise,} \end{cases} \quad p_Y(y) = \begin{cases} 1/2, & \text{if } y = 0, \\ 1/3, & \text{if } y = 1, \\ 1/6, & \text{if } y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the PMF of $Z = X + Y$, using the convolution formula.

$$Z = X + Y$$

$$P_Z(z) = \sum_{\substack{x, y \\ (x+y=z)}} P_X(x) \cdot P_Y(y) = \sum_x P_X(x) \cdot P_Y(z-x)$$

$$P_X(x) = \begin{cases} 1/3, & \text{if } x \in \{1, 2, 3\} \\ 0, & \text{otherwise} \end{cases}$$

$$P_Y(z-x) = \begin{cases} 1/2, & \text{if } z-x = 0 \\ 1/3, & \text{if } z-x = 1 \\ 1/6, & \text{if } z-x = 2 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{If } z = 1: P_Z(z) = \sum_x P_X(x) \cdot P_Y(z-x)$$

$$= \sum_{x=1}^3 \left(\frac{1}{3}\right) \cdot P_Y(z-x)$$

$$= \left(\frac{1}{3}\right) P_Y(z-1) + \left(\frac{1}{3}\right) P_Y(z-2) + \left(\frac{1}{3}\right) P_Y(z-3)$$

$$= \left(\frac{1}{3}\right) \left(\frac{1}{2}\right) = 1/6$$

If $z = 2$:

$$\begin{aligned} P_z(z) &= \sum_{x=1}^3 \left(\frac{1}{3}\right) \cdot P_Y(z-x) \\ &= \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right) + 0 \\ &= \frac{1}{9} + \frac{1}{6} = \frac{5}{18} \end{aligned}$$

If $z = 3$:

$$\begin{aligned} P_z(z) &= \left(\frac{1}{3}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right) \\ &= \frac{1}{18} + \frac{1}{9} + \frac{1}{6} \\ &= \frac{1}{3} \end{aligned}$$

$$\text{If } z = 4 : P_z(z) = 0 + \left(\frac{1}{3}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = \frac{1}{18} + \frac{1}{9} = \frac{1}{6}$$

$$\text{If } z = 5 : P_z(z) = 0 + 0 + \left(\frac{1}{3}\right)\left(\frac{1}{6}\right) = \frac{1}{18}$$

$$P_z(z) = \begin{cases} \frac{1}{6}, & \text{if } z \in \{1, 4\} \\ \frac{5}{18}, & \text{if } z = 2 \\ \frac{1}{3}, & \text{if } z = 3 \\ \frac{1}{18}, & \text{if } z = 5 \end{cases}$$

Q14: Use the convolution formula to establish that the sum of two independent Poisson random variables with parameters μ and λ respectively, is Poisson with parameter $\mu + \lambda$.

$$\left. \begin{array}{l} X \sim \text{Poisson}(\mu) \\ Y \sim \text{Poisson}(\lambda) \end{array} \right\} \text{Independent}$$

$$Z = X + Y$$

$$P_X(x) = \frac{e^{-\mu} \cdot \mu^x}{x!} \quad \text{for } x \geq 0 \quad (0, \text{ otherwise})$$

$$P_Y(y) = \frac{e^{-\lambda} \cdot \lambda^y}{y!} \quad \text{for } y \geq 0 \quad (0, \text{ otherwise})$$

Now :

$$\begin{aligned} P_z(z) &= \sum_{\substack{x, y \\ (x+y=z)}} P_X(x) \cdot P_Y(y) \\ &= \sum_x P_X(x) \cdot P_Y(z-x) \\ &= \sum_x \frac{e^{-\mu} \cdot \mu^x}{x!} \cdot \frac{e^{-\lambda} \cdot \lambda^{z-x}}{(z-x)!} \end{aligned}$$

$$\begin{aligned}
&= \sum_x \frac{e^{-(\mu+\lambda)} \cdot \mu^x \cdot \lambda^{3-x}}{x! \cdot (3-x)!} \\
&= e^{-(\mu+\lambda)} \cdot \sum_x \frac{\mu^x \cdot \lambda^{3-x}}{x! \cdot (3-x)!} \\
&= e^{-(\mu+\lambda)} \cdot \sum_x \left[\binom{3}{x} \mu^x \lambda^{3-x} \cdot \frac{1}{3!} \right] \\
&= e^{-(\mu+\lambda)} \cdot \frac{1}{3!} \sum_x \binom{3}{x} \mu^x \lambda^{3-x} \\
&= e^{-(\mu+\lambda)} \cdot \frac{1}{3!} \cdot (\mu+\lambda)^3 \\
\therefore P_2(3) &= \frac{e^{-(\mu+\lambda)} \cdot (\mu+\lambda)^3}{3!}
\end{aligned}$$

$$\therefore Z \rightarrow \text{Poisson}(\mu+\lambda)$$

Q15: We want to estimate $\theta = E[\sqrt{X}]$ where X is a Normal random variable ($X \sim \mathcal{N}(0, 1)$). Using importance sampling, formulate an estimator for θ by drawing N samples, $Y_1, \dots, Y_N$, from a uniform distribution $Y \sim U[0, 5]$.

$$\theta = E[\sqrt{X}]$$

$$X \sim \mathcal{N}(0, 1)$$

$$Y_1, Y_2, \dots, Y_N \text{ from } Y \sim U[0, 5]$$

$$f_X(x) = f(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}}, -\infty < x < \infty \quad \left| \quad f_Y(y) = g(y) = \begin{cases} 1/5, & y \in [0, 5] \\ 0, & \text{otherwise} \end{cases}$$

$$h(x) = \sqrt{x}$$

$$\Rightarrow E[h(X)] = E_Y \left[h(Y) \cdot \frac{f(Y)}{g(Y)} \right]$$

$$\Rightarrow \theta = E_Y \left[\sqrt{Y} \cdot \frac{1}{\sqrt{2\pi}} \cdot \frac{e^{-Y/2}}{1/5} \right] = E \left[\frac{5}{\sqrt{2\pi}} \sqrt{Y} \cdot e^{-Y/2} \right]$$

$$\theta_N = \frac{1}{N} \cdot \sum_{i=1}^N \frac{5}{\sqrt{2\pi}} \sqrt{Y_i} \cdot e^{-Y_i/2} \leftarrow \text{Monte Carlo estimator}$$

True expectation:

$$E[\sqrt{X}] = \int_{-\infty}^{\infty} \sqrt{x} \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{x} e^{-x^2/2} dx$$

$$Y \rightarrow [0, 5]$$

$$f_X(5) = \frac{1}{\sqrt{2\pi}} \cdot e^{-25/2}$$

$$\hookrightarrow 0.0000015 \leftarrow \text{Negligible}$$

$\hookrightarrow$ Difficult, $\therefore$ Estimator is useful

Q17: Estimate the integral

$$I_1 = \int_0^a \sqrt{x} dx$$

using Monte Carlo integration with N random samples from the uniform distribution $U[0, 1]$. Let $a > 0$ be a given constant. Also write the Monte Carlo estimate assuming X_i 's are sampled according to some pdf $g(X_i)$ in the same domain.

$$I = \int_0^a \sqrt{x} dx \quad [a > 0]$$

$$\text{Let } x = au$$

$$dx = a du$$

$$\begin{aligned} I &= \int_0^1 \sqrt{a \cdot u} \cdot a du = a^{3/2} \cdot \int_0^1 \sqrt{u} du \\ &= a^{3/2} E_U[\sqrt{U}] \end{aligned}$$

$$U_1, U_2, \dots, U_N \xrightarrow{\text{i.i.d.}} U[0, 1]$$

$$\therefore \hat{I}_N = a^{3/2} \cdot \frac{1}{N} \cdot \sum_{i=1}^N \sqrt{U_i}$$

$$\left\{ \begin{array}{l} X_i = a U_i \\ \hat{I}_N = \frac{a}{N} \sum \sqrt{X_i} \end{array} \right\}$$

As we know,

$$E[\sqrt{U}] = \int_0^1 \sqrt{u} \cdot 1 \cdot du = \frac{2}{3}$$

$$\& E[U] = 1/2$$

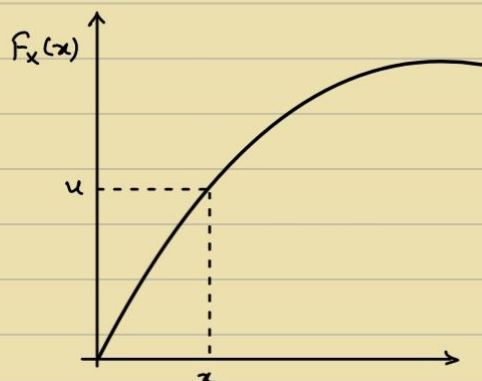
$$\begin{aligned} \text{Var}[\sqrt{U}] &= E[U] - \left(\frac{2}{3}\right)^2 \\ &= 1/2 - 4/9 \\ &= 1/18 \end{aligned}$$

Goal:

$$I = \int_0^a \sqrt{x} dx = \frac{2}{3} a^{3/2}$$

$$\text{Var}(\hat{I}_N) = a^3 \cdot \frac{1}{N} \cdot \frac{1}{18} = \frac{a^3}{18 \cdot N}$$

Note: Q16, 18 $\rightarrow$ ACCEPT - REJECT (Next class)



$$Z = 1 - e^{-\lambda x}$$

$$E[h(X)]$$

$$= \int h(x) \cdot f_X(x) dx \quad \begin{array}{l} Y \sim g_Y(y) \\ X \sim f_X(x) \end{array}$$

$$= \int \frac{h(x) \cdot f_X(x)}{g(x)} \cdot g(x) dx$$

Important Sampling $\rightarrow$ Estimation of rare events

$$\bullet P(X > x) = E[?]$$

Hint : Indicator R.V.

$$\begin{aligned} E[1_A] &= 1 \cdot P_A + 0 \cdot P(A') \\ &= P_A \end{aligned}$$

where $A : X > x$

$$\Rightarrow P(X > x) = E[1_{X>x}]$$

But, what if x is too large?

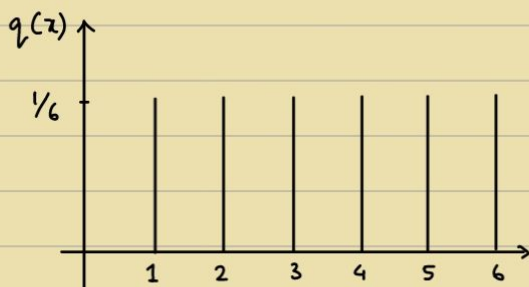
$$E[Z] = \frac{1}{100} \sum z_i$$

Importance Sampling $\rightarrow$ Better than Monte-Carlo Approximation

$\rightarrow$ Decreases variance

$\rightarrow$ Needs more samples

$\rightarrow$ ACCEPT - REJECT :



$$D = \{r_1, r_2, \dots, r_{1000}\}$$



$$D' = \{r'_1, r'_2, \dots, r'_{1000}\}$$

Goal : Generate X with $p(\cdot)$ using Y with $q(\cdot)$

Suppose : $\frac{p(y)}{q(y)} \leq c \quad \forall y \leftarrow \text{Requirement}$

Needs to be as tight as possible

Now, generate $y \sim q(\cdot)$

Also, generate $u \sim U[0,1]$

If $u \leq \frac{p(y)}{c \cdot q(y)}$, ACCEPT y as a sample from X .

If not, REJECT y and generate another y .

• $Y \sim q(y)$

y_i

$P(y | \text{ACCEPT}) \leftarrow$ Probability that the accepted sample takes value y .

$\hookrightarrow P(y)$

i.e. $P(\text{accepted sample takes value } y \mid \text{give sample accepted}) = P(y)$



$$P(y | \text{ACCEPT}) = \frac{P(\text{ACCEPT}/y) \cdot q(y)}{P(\text{ACCEPT})}$$

$\underbrace{\hspace{10em}}_{\hookrightarrow \sum P(\text{ACCEPT}/y) \cdot q(y)}$

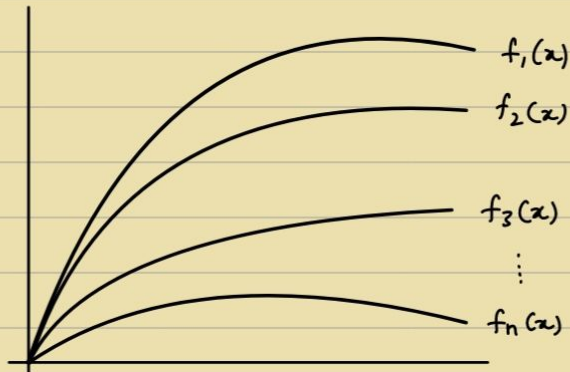
$$P(\text{ACCEPT}/y) = P\left(u \leq \frac{p(y)}{c \cdot q(y)}\right) = P\left(\frac{p(y)}{c \cdot q(y)}\right) \quad \because u \sim U[0,1]$$

$$P(y \& \text{ACCEPT}) = q(y) \cdot \frac{p(y)}{c \cdot q(y)} = \frac{p(y)}{c}$$

→ Convergence of R.V:

- $\{x_n\}$ converges to $x \in \mathbb{R}$, denoted by $x_n \rightarrow x$ if $\forall \epsilon > 0$, we find $N(\epsilon) \in \mathbb{N}$ s.t. $|x_n - x| < \epsilon$ for $n > N(\epsilon)$

- Convergence of functions:



Fix x :

$$f_1(x_1), f_2(x_1), \dots, f_n(x_1)$$

should converge

& Should converge $\forall x$ in domain

Converges to $f(x) = 0 \forall x$

$f_n(\cdot)$ converges to $F(\cdot)$ on domain $\mathbb{R}$

- Uniform convergence:

Take $N(\epsilon, x)$

We prove $N(\epsilon, x) = N(\epsilon)$

i.e. doesn't depend on x .

} → i.e. only those $f_n(x)$ are ϵ close to $F(x)$ for which $n > N(\epsilon, x)$

$\forall x$, the sequence $\{f_n(x)\} \rightarrow F(x)$

Experiment: $X(\omega) \rightarrow X_1(\omega), X_2(\omega), \dots, X_{100}(\omega), \dots, X(\omega)$

$\exists$ for some $\omega \rightarrow$ Not converges
but $P(\omega) \rightarrow$ Negligible

Sometimes $X_n \rightarrow X$

- Pointwise / Sure convergence:

$$\{X_n, n > 0\} \rightarrow X \text{ if } \forall \omega \in \Omega, \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$$

Consider $\Omega = \{H, T\}$

$$X_n = \begin{cases} 1/n, & \text{if } \omega = H \\ 1 + 1/n, & \text{if } \omega = T \end{cases}$$

$$\lim_{n \rightarrow \infty} 1/n = 0 \quad \& \quad \lim_{n \rightarrow \infty} (1 + 1/n) = 1$$

$$\text{let's take } X = \begin{cases} 0 & \text{if } \omega = H \\ 1 & \text{if } \omega = T \end{cases}$$

$$[X_n \rightarrow X]$$

Q9: You want to generate samples from a random variable with the PDF given by:

$$p(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Use the Accept-Reject method with a proposal distribution $q(x) = U(0, 1)$.

- (a) Find the smallest constant c such that $p(x) \leq c \cdot q(x)$.
- (b) Outline the algorithm to generate one sample.

- Tutorial 6

(a) $q(x) = U(0, 1)$

$$\frac{p(x)}{q(x)} \leq c \quad \forall x$$

$$q(x) = \begin{cases} 1, & \text{if } x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

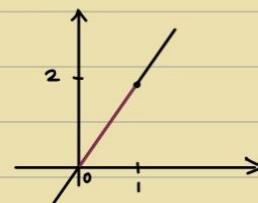
$$\Rightarrow \frac{2x}{1} \leq c \quad \forall x$$

$$\Rightarrow c \geq 2x \quad \forall x$$

$$\max(x) = 1$$

$$\therefore c \geq 2$$

$$\therefore \text{Tight bound} \Rightarrow \boxed{c = 2}$$



(b) Now, generate $x \sim q(\cdot)$

Also, generate $u \sim U[0, 1]$

$$\text{If } u \leq \frac{p(x)}{c \cdot q(x)} \Rightarrow u \leq \frac{2x}{2} \Rightarrow u \leq x$$

If $u \leq x$, ACCEPT x as a sample from X .

If not, REJECT x and generate another x .

$$E[X] = \int_0^1 x \, dx = \frac{1}{2} \quad \therefore \text{Consistent with } 1/c \text{ rule.}$$

Q16: Suppose, you want to generate samples from a Gaussian random variable with PDF $p(x)$ for $\mathcal{N}(\mu = 0.5, \sigma^2 = 0.04)$. Use the Accept-Reject method with a proposal distribution $q(x) = U(0, 1)$.

- Find the smallest constant c such that $p(x) \leq c \cdot q(x)$ for all $x \in [0, 1]$.
- Outline the algorithm to generate one sample.

- Homework 6

(a) $q(x) = U(0, 1)$

$$\frac{p(x)}{q(x)} \leq c \quad \forall x$$

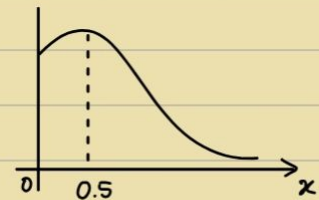
$$\Rightarrow \frac{5}{\sqrt{2\pi}} e^{-\frac{(2x-1)^2}{0.32}} \leq c \quad \forall x$$

@ $x = 1/2 \Rightarrow c = \frac{5}{\sqrt{2\pi}} \approx 1.995$

$\therefore c = \frac{5}{\sqrt{2\pi}}$
Tight bound

$$q(x) = \begin{cases} 1, & \text{if } x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$p(x) = \frac{1}{\sqrt{2\pi} \cdot 1/5} e^{-\frac{(x-1/2)^2}{0.08}}$$



(b) Now, generate $x \sim q(\cdot)$

Also, generate $u \sim U[0, 1]$

$$\text{If } u \leq \frac{p(x)}{c \cdot q(x)} \Rightarrow u \leq e^{-\frac{(x-1/2)^2}{0.08}}$$

$$\text{If } u \leq e^{-\frac{(x-1/2)^2}{0.08}}, \text{ ACCEPT } x \text{ as a sample from } X.$$

If not, REJECT x and generate another x .

Q18: Suppose we want to generate a random variable X with a target pdf $f(x)$. We have access to a method for generating a random variable Y from a simpler proposal density function $g(x)$.

Let c be a constant such that the "envelope condition" $\frac{f(y)}{g(y)} \leq c$ holds for all y for which $g(y) > 0$. The rejection method algorithm is as follows:

- Generate a candidate sample Y from the proposal density $g(y)$.
- Generate a random number U from a Uniform(0, 1) distribution.
- Acceptance Condition:** If $U \leq \frac{f(Y)}{cg(Y)}$, then set $X = Y$. Otherwise, reject the sample and return to step 1.

Prove the validity of this algorithm.

- Homework 6

Proof of ACCEPT - REJECT

for any measurable set B :

$$\begin{aligned} P(Y \in B, A) &= \int_B P(A | Y=y) g(y) dy \\ &= \int_B \frac{f(y)}{c \cdot g(y)} \cdot g(y) dy \end{aligned}$$

Taking $B = \mathbb{R}$

$$\begin{aligned} P(A) &= P(Y \in \mathbb{R}, A) = \frac{1}{c} \int f(y) dy \\ &= \frac{1}{c} \end{aligned} \quad \left[\int f(y) dy = 1 \right]$$

Let X denote the output when accepted ($X = Y | A$).

For any measurable set B ,

$$\begin{aligned} P(X \in B) &= P(Y \in B | A) = \frac{P(Y \in B, A)}{P(A)} = \frac{\frac{1}{c} \int_B f(y) dy}{\frac{1}{c}} \\ &= \int_B f(y) dy \end{aligned}$$

$\therefore X$ has PDF f .

Acceptance probability: $P(A) = \frac{1}{c}$

Expected trials per acceptance = c

If f is not normalized,

i.e. let $f' = f/z$

$$\therefore P(Y \in B, A) = \frac{1}{c} \int_B f'(y) dy$$

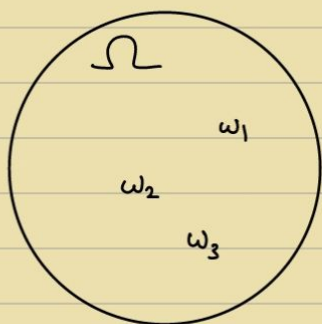
$$P(A) = \frac{1}{c} \int f'(y) dy$$

$$\Rightarrow P(X \in B) = \frac{\int_B f'(y) dy}{\int f'(y) dy}$$

$\therefore X'$ has normalized density proportional to f' .

11/10

$X_1, X_2, X_3, \dots, X_n$



$\longrightarrow X_1(\omega), X_2(\omega), \dots, X_n(\omega)$

Sometimes,

$X_1(\omega), X_2(\omega), X_3(\omega), \dots, X_{1000}(\omega) \not\rightarrow X(\omega)$

These are ω^* 's

$P(A^*) = 0 \Rightarrow$ Almost Sure convergence

$\hookrightarrow A$: Set where convergence doesn't happen

i.e. Almost Sure convergence

$$P(\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)) = 1$$

$\hookrightarrow$ [For sure convergence $\Rightarrow \forall \omega$]

$$P\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\} = 0$$

$\hookrightarrow A^* = \emptyset$

e.g. consider $\Omega = [0, 1]$

Let $X_n = \omega^n \quad \forall \omega \in \Omega$

$X(\omega) = 0 \quad \forall \omega$

$\therefore X_n(\omega) \rightarrow X(\omega) \quad \forall \omega \in [0, 1)$

$X_n(\omega) \not\rightarrow X(\omega) \quad \text{for } \omega = 1 \quad \& \quad P\{\omega = 1\} = 0$

e.g. Let $\{X_n, n > 0\} \rightarrow$ i.i.d R.Vs

mean = $\mu = E[X_i]$

If $S_n = \sum_{i=1}^n X_i$, Then $\frac{S_n}{n} \rightarrow \mu$

e.g. Toss a biased coin

Probability of head = μ

What are ω and Ω ?

$\omega = \{H, T\}$

$\Omega = \{H, T\}^\infty = \{(x_1, x_2, \dots) \mid x_i \in \{H, T\} \quad \forall i\}$

$$X_i(\omega) = \begin{cases} 1, & \text{if the } i^{\text{th}} \text{ toss is Head (H)} \\ 0, & \text{if the } i^{\text{th}} \text{ toss is Tails (T)} \end{cases}$$

→ Estimator :

$$\mu = E[X]$$

$$Z_n = \frac{S_n}{n} \rightarrow \mu$$

$$\hat{\mu}_n = \frac{S_n}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

$\{^{\wedge}$ indicates it being an estimator}

$$E[\hat{\mu}] = \mu \quad \& \quad \text{Var}[\hat{\mu}] = \frac{\text{Var}[X]}{n}$$

Now if we have $n+1$:

$$\hat{\mu}_{n+1} = \frac{\sum_{i=1}^{n+1} X_i}{n+1}$$

$$= \hat{\mu}_n + \frac{1}{n+1} [X_{n+1} - \hat{\mu}_n]$$

$$\therefore \hat{\mu}_{n+1} = \hat{\mu}_n + \frac{1}{n+1} [X_{n+1} - \hat{\mu}_n]$$

- Gradient descent:

$$X_{n+1} = X_n - \alpha \nabla f(X_n)$$

$$\mu_{n+1} = \frac{X_1 + X_2 + \dots + X_n + X_{n+1}}{n+1}$$

$$\begin{aligned} \frac{(n+1)\mu_{n+1}}{n} &= \frac{X_1 + X_2 + \dots + X_n}{n} + \frac{X_{n+1}}{n} \\ &= \hat{\mu}_n + \frac{X_{n+1}}{n} \end{aligned}$$

$$\Rightarrow \mu_{n+1} = \frac{n}{n+1} \left(\hat{\mu}_n + \frac{X_{n+1}}{n} \right)$$

$$= \frac{n}{n+1} \hat{\mu}_n + \frac{1}{n+1} (X_{n+1})$$

$$= \hat{\mu}_n - \frac{1}{n+1} \hat{\mu}_n + \frac{1}{n+1} (X_{n+1})$$

$$= \hat{\mu}_n + \frac{1}{n+1} [X_{n+1} - \hat{\mu}_n]$$

$\left\{ \begin{array}{l} \text{Stochastic Gradient descent} \\ \text{Stochastic Approximation} \end{array} \right\}$

- Q Learning (DL) formula:

$$Q_m(s, a) = Q_n(s, a) + \alpha [r + \gamma \max_{a'} Q_n(s', a') - Q_n(s, a)]$$

→ Borel - Cantelli Lemma:

Proof: Out of Syllabus

if $\forall \epsilon, \sum_{n=1}^{\infty} P(|X_n - X| > \epsilon) < \infty$, then $X_n \rightarrow X$ (almost surely) {Th. 7.5, pc.com}

↳ Only Sufficient condition

• H.W: Th. 7.6

(probabilitycourse.com)

e.g. Consider uniform R.V. U

$$X_n = n \cdot 1_{\{U \leq 1/n\}}$$

Sol: $X_n = n$ when $U \leq 1/n$

$X_n = 0$ otherwise

$$P(U=0) = 0$$

$\therefore$ Once an $X_n = 0$, then rest $X_{n+1}, X_{n+2}, \dots$ also $= 0$.

when U is fixed

Happens for $U \neq 0 \leftarrow$ No convergence

$$E[X_n] = n(1/n) + 0 = 1 ?$$

$$\Rightarrow E[X_n] \not\rightarrow 0$$

Almost Sure convergence $\not\Rightarrow$ Sure convergence

If U is not fixed,

$\{X_n\}$ may not converge almost surely.

$\therefore X_n \rightarrow X$ only if $\lim_{n \rightarrow \infty} P(|X_n - X| > \varepsilon) = 0 \quad \forall \varepsilon > 0$
(in probability)

e.g. $X_n = n$ with probability $1/n$

$X_n = 0$ otherwise

$$P(|X_n - X| > \varepsilon) = P(X_n > \varepsilon) = \frac{1}{n} \quad \text{when } n > \varepsilon$$

$$P(|X_n - X| > \varepsilon) = 0 \quad \text{when } n < \varepsilon$$

Once $n > \varepsilon$,

$$\lim_{n \rightarrow \infty} P(X_n > \varepsilon) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$X_n \rightarrow 0$ in probability, but not almost surely.

Tutorial - 7

Q1: Let $(\Omega, \mathcal{F}, P)$ be a probability space on which the random variable U is defined, where $U \sim \text{Uniform}[0, 1]$. Define a sequence of random variables $(X_n)_{n \geq 1}$ by

$$X_n(\omega) = \begin{cases} n, & \text{if } U(\omega) \leq 1/n, \\ 0, & \text{if } U(\omega) > 1/n. \end{cases}$$

Investigate the convergence of the sequence X_n to 0 in the following modes.

- (a) Does $X_n \xrightarrow{d} 0$?
- (b) Does $X_n \xrightarrow{P} 0$?
- (c) Does $X_n \xrightarrow{\text{a.s.}} 0$. Can Borel-Cantelli Lemma be applied?
- (d) Does $X_n \xrightarrow{L1} 0$ (Convergence in mean)?

$$X_n(\omega) = \begin{cases} n, & U(\omega) \leq 1/n \\ 0, & \text{otherwise} \end{cases}$$

$$X_n = \begin{cases} n, & \text{with probability } 1/n \\ 0, & \text{with probability } 1 - 1/n \end{cases}$$

$$\lim_{n \rightarrow \infty} P(|X_n - 0| > \varepsilon) = 0 \quad \forall \varepsilon > 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(\{\omega : |X_n(\omega) - 0| > \varepsilon\}) = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(|X_n| > \varepsilon) = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \begin{cases} P(|X_n| = \varepsilon), & n > \varepsilon \\ 0, & \text{otherwise} \end{cases}$$

$$\Rightarrow P(|X_n| = n) = \lim_{n \rightarrow \infty} 1/n = 0$$

$$\therefore X_n \xrightarrow{P} 0$$

Q3: Let X be a discrete random variable with support

$$R_X = \{0, 1\}$$

and probability mass function:

$$p_X(x) = \begin{cases} 1/3, & \text{if } x = 1, \\ 2/3, & \text{if } x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Consider a sequence of random variables $\{X_n\}$ whose generic term is:

$$X_n = \left(1 + \frac{1}{n}\right) X.$$

Show that $\{X_n\}$ converges in probability to X .

$$|X_n - X| = \begin{cases} 0, & X = 0 \\ 1/n, & X = 1 \end{cases}$$

For any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(X = 1 \text{ and } 1/n > \varepsilon)$$

$$= \begin{cases} 1/3, & \varepsilon < 1/n \\ 0, & \varepsilon \geq 1/n \end{cases}$$

For every $\varepsilon > 0$, $\exists N$ with $1/N \leq \varepsilon$, follows $\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0$

$$\therefore X_n \xrightarrow{P} X \quad (\text{Almost Surely})$$

$$\begin{aligned} & \begin{cases} X = 0, & X_n = 0 \\ X = 1, & (1 + 1/n) \rightarrow 1 \end{cases} \\ & \mathbb{E}[|X_n - X|] = \frac{1}{n} \cdot \frac{1}{3} \rightarrow 0 \\ & \therefore X_n \rightarrow X \end{aligned}$$

Q6: Let X be an exponential random variable with parameter λ and let Y be a random variable with the Gamma distribution $Y \sim \text{Gamma}(k, \theta)$.

- Show how to generate X using a uniform random variable U drawn from the interval $[0, 1]$.
- Show how to generate Y using k uniform random variables drawn from $[0, 1]$.

$$\begin{aligned} \text{(a)} \quad f_X(x) &= \lambda e^{-\lambda x}, \text{ for } x \geq 0 \quad (0 \text{ otherwise}) \\ F_X(x) &= 1 - e^{-\lambda x}, \text{ for } x \geq 0 \end{aligned}$$

$$F_X(X) = U$$

By Inverse Sampling,

$$U = 1 - e^{-\lambda x}$$

$$\Rightarrow e^{-\lambda x} = 1 - U$$

$$\Rightarrow -\lambda x = \ln(1 - U)$$

$$\Rightarrow x = -\frac{1}{\lambda} \ln(1 - U)$$

$$\because 1 - U \sim \text{Uniform}(0, 1)$$

$$X = -\frac{1}{\lambda} \ln(1 - U) = -\frac{1}{\lambda} \ln(U)$$

(b) Gamma distribution $\rightarrow$ Sum of k independent exponential R.V.s with rate $1/\theta$, i.e. mean θ

$$Y = \sum_{i=1}^k X_i, \text{ where } X_i \sim \text{exponential}(1/\theta)$$

$$\text{From (a)} \Rightarrow X_i = -\theta \ln(U_i)$$

$$\Rightarrow Y = -\theta \sum_{i=1}^k \ln(U_i)$$

where $U_i \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1)$

Extra practice problems (recommended).

- (1) Let $X_n \sim \text{Poisson}(n)$. Define $Y_n = X_n/n$. Show $Y_n \xrightarrow{p} 1$. (*Hint: compute mean and variance of Y_n and use Chebyshev.*)
- (2) Let Y_n satisfy $\mathbb{E}[Y_n] = 1/n$ and $\text{Var}(Y_n) = 1/n^2$. Show $Y_n \xrightarrow{p} 0$. (*Hint: Chebyshev.*)

Resources for practice:

- Weak Law of Large Numbers (proof and discussion)
- Convergence in Probability (see Example 7.9)
- **From the textbook:** Try solving Example 5.6, Example 5.7, and Problem 6 and 7 in Chapter "Limit Theorems."

Q9: Suppose a sequence of random variables X_n satisfies

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X_n - c|^\alpha] = 0 \text{ for } \alpha > 0.$$

Show that $X_n \xrightarrow{p} c$.

$$\mathbb{E}[g(x)] = \int_{\mathcal{X}} g(x) \cdot f_x(x) dx$$

$$\begin{aligned} \mathbb{E}[|X_n - c|^\alpha] &= \int_{\mathcal{X}} |x_n - c|^\alpha \cdot f_x(x) dx \\ &= \int_{|x_n - c| \leq \varepsilon} |x_n - c|^\alpha + \int_{|x_n - c| \geq \varepsilon} |x_n - c|^\alpha \end{aligned}$$

$$\begin{aligned} \mathbb{E}[] &\geq \int_{|x_n - c| \geq \varepsilon} |x_n - c|^\alpha \cdot f_x(x) dx \\ &\geq \int_{|x_n - c| \geq \varepsilon} \varepsilon^\alpha f_x(x) dx \end{aligned}$$

$$\geq \varepsilon^\alpha \int_{|x_n - c| \geq \varepsilon} f_x(x) dx$$

$$\mathbb{E}[|X_n - c|] \geq \varepsilon^\alpha \cdot P(|X_n - c| \geq \varepsilon)$$

$$\Rightarrow 0 \leq \varepsilon^\alpha P(|X_n - c| \leq \varepsilon) \leq \mathbb{E}[|X_n - c|^\alpha]$$

$$\leq \frac{\mathbb{E}[|X_n - c|^\alpha]}{\varepsilon^\alpha}$$

$$\leq \left(\frac{\mathbb{E}[|X_n - c|]}{\varepsilon} \right)^\alpha$$

$$\leq \frac{0}{\varepsilon^2}$$

• $X_1, X_2, X_3, \dots, X_n \mid X_{n+1} \dots X$

$$Z_t = \frac{S_t}{t} = \frac{\sum_{i=1}^t X_i}{t} \quad \{ E[X_i] = \mu \}$$

$Z_1, Z_2, Z_3, \dots, Z_t, \dots, Z_n \rightarrow Z$

• $P(|X_n - X| > \epsilon)$
 if $A_n = |X_n - X| > \epsilon$
 $\therefore 1_{A_n} = 1_{(|X_n - X| > \epsilon)}$

$X_1, X_2, X_3, \dots, X_n \mid X_{n+1} \dots X$
 $1_{A_1} \quad 1_{A_2} \quad 1_{A_3} \quad \dots \quad 1_{A_n}$
 $\therefore E\left[\sum_{i=1}^{\infty} 1_{A_i}\right]$

• Almost sure $\Rightarrow$ Convergence in probability

$\rightarrow$ Convergence in r^{th} mean :

X_n converges to X in r^{th} mean if
 $\lim_{n \rightarrow \infty} E[|X_n - X|^r] = 0$

$\left\{ \begin{array}{l} \text{when } X_n \xrightarrow{a.s} X \\ E[X_n] \not\rightarrow E[X] \end{array} \right\}$

$r = 2 \Rightarrow$ convergence in mean-squared sense
 & if $X = 0 \Rightarrow$ Second moments converge to 0.

Convergence in r^{th} mean $\Rightarrow$ Convergence in probability [For any r]
 Proof : H.W

Note : $r \leq s ; X_n \xrightarrow{L_s} X \Rightarrow X_n \xrightarrow{L_r} X$ [7.3]
 Proof : H.W

7.10 :

$X_n \sim U[0, 1/n]$

To show : $X_n \xrightarrow{L_r} 0$ for any $r \geq 1$

$E[|X_n - X|^r]$
 $X = 0$ (given)

$\Rightarrow E[|X_n|^r]$ As X_n is always non-negative
 $\Rightarrow E[X_n^r]$

$$f_{X_n}(x_n) = \begin{cases} n, & \text{if } x_n \in [0, \frac{1}{n}] \\ 0, & \text{otherwise} \end{cases}$$

$$\Rightarrow \int_0^{\frac{1}{n}} x_n^r \cdot n \, dx_n$$

$$\Rightarrow n \cdot \left(\frac{x_n^{r+1}}{r+1} \right)_0^{\frac{1}{n}} \Rightarrow n \left(\frac{\frac{1}{n}^{r+1}}{r+1} - 0 \right)$$

$$\Rightarrow \frac{1}{r+1} \cdot n \cdot \left(\frac{1}{n} \right)^{r+1}$$

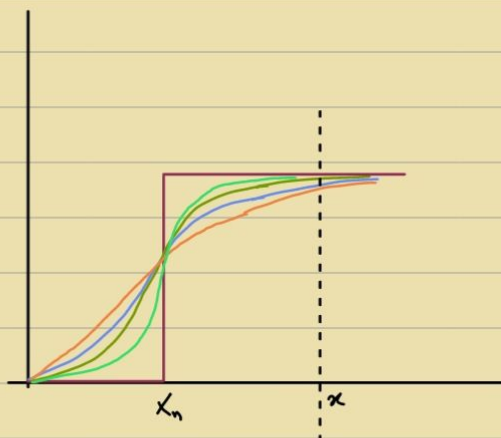
$$\text{Now, } \lim_{n \rightarrow \infty} \frac{1}{r+1} \cdot n \left(\frac{1}{n} \right)^{r+1}$$

$$\Rightarrow \frac{1}{r+1} \cdot \lim_{n \rightarrow \infty} n^{1-(r+1)}$$

$$\Rightarrow \frac{1}{r+1} \cdot \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^r \Rightarrow 0$$

→ Weak convergence (in distribution)

X_n converges to X if $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$
for all continuity points of $F_X(\cdot)$



$$P(X_n \leq x)$$

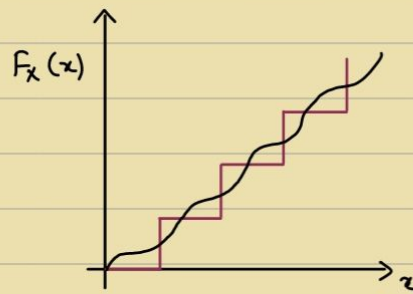
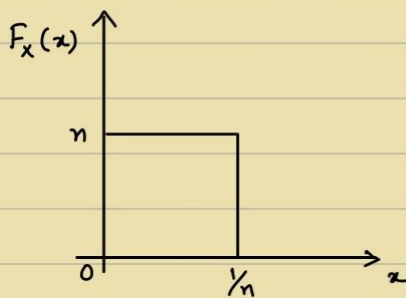
$$F_{X_n}(x)$$

e.g. If $X_n \rightarrow \exp(\lambda)$

$$F_{X_n}(x) = 1 - e^{-n\lambda x} \quad \& \quad F_X(x) = 1 \quad \forall x$$

As $x = 0 \rightarrow$ Discontinuity point

$$F_{X_n}(x) = 0 \quad \& \quad F_X(0) = 1 \quad \leftarrow \text{@ discontinuity}$$



H.W : $X_n \rightarrow$ i.i.d Binomial($n, \lambda/n$) converges in distribution to Poisson

- Mean square convergence $\Rightarrow$ Convergence in Probability

Almost Sure convergence $\nearrow$ $\Downarrow$ Convergence in distribution

- Point-wise convergence : $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \quad \forall \omega$
- Almost Sure convergence : $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$ for some ω
- Convergence in Probability : $\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$ for any $\epsilon > 0$
- Convergence in Distribution : $\lim_{n \rightarrow \infty} F_n(x) = F(x)$ for any continuity point x
- Mean - Square convergence : $\lim_{n \rightarrow \infty} E[|X_n - X|^2] = 0$

$\rightarrow$ Central Limit Theorem (CLT) :

$$\hat{\mu}_n = \frac{S_n}{n} \quad \text{where} \quad S_n = \sum_{i=1}^n X_i$$

where X_i : i.i.d with mean μ & Variance σ^2

$$E[\hat{\mu}_n] = \mu$$

$$\text{Var}[\hat{\mu}_n] = \frac{\sigma^2}{n}$$

Now, Let $Y_n = \frac{\hat{\mu}_n - \mu}{\sigma/\sqrt{n}}$ $\nearrow$ Centering & Scaling

$$E[Y_n] = E\left[\frac{\hat{\mu}_n - \mu}{\sigma/\sqrt{n}}\right]$$

$$= \frac{E[\hat{\mu}_n] - \mu}{\sigma/\sqrt{n}} = 0$$

$$\text{Var}[Y_n] = \frac{\text{Var}[\hat{\mu}_n - \mu]}{\sigma^2/n}$$

$$= \frac{n}{\sigma^2} \cdot \frac{\sigma^2}{n} = 1$$

$$F_Y(\cdot) = ?$$

$$\lim_{n \rightarrow \infty} F_Y(\cdot) = \Phi(\cdot) = F_{N(0,1)}(\cdot)$$

$$F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$P \left\{ \frac{X_1 + X_2 + X_3 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq a \right\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^a e^{-x^2/2} dx \quad (\text{as } n \rightarrow \infty)$$

Now, if $E[X_i] = 0$

$$\text{Var}[X_i] = 1$$

in this case, $Y_n = \frac{S_n}{\sqrt{n}}$ & converges in distribution to $N(0,1)$

$\therefore \frac{S_n}{n}$ converges almost surely to 0.

$\frac{S_n}{\sqrt{n}}$ converges to $N(0,1)$

- CLT gives approximation distribution of $\hat{\mu}_n$.

Take large enough n ,

$$Y_n \sim N(0,1) \leftarrow \text{approximate distribution}$$

$$\Rightarrow \mu_n \sim N(\mu, \sigma^2/n) \quad \left\{ \begin{array}{l} \because \text{Affine Transformations} \\ \text{Linear combination} \end{array} \right\}$$

i.e.

if X_i 's are bernoulli (i.i.d)
 $\searrow (p)$

$$Z_n = \frac{X_1 + X_2 + \dots + X_n - np}{\sqrt{np(1-p)}}$$

$$\left[\begin{array}{l} Y_n \rightarrow N(0,1) \\ Y_n + \mu \rightarrow N(\mu, 1) \\ \frac{\sigma}{\sqrt{n}} (Y_n + \mu) \rightarrow N(\mu, \sigma^2/n) \end{array} \right]$$

- $P(S_n < c) \approx \Phi(z)$, where $z = \frac{c - n\mu}{\sigma\sqrt{n}} \leftarrow \text{Approximation}$
if $n \rightarrow \text{large}$

$$S_n = X_1 + X_2 + \dots + X_n ; X_i : \text{i.i.d}$$

[Mean μ and Variance σ^2]